When Do Local Income Taxes Cause Residential Sorting?

Evidence from Pennsylvania's Piecewise EIT System

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Abstract

Do local income taxes cause residential sorting across sub-county boundaries? I use LEHD Origin-Destination data linking each worker's residence block to their workplace block at annual frequency, covering approximately 2,500 Pennsylvania municipalities and 500 school districts over 2002–2019.

Pennsylvania's Earned Income Tax operates a piecewise max-rule: each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where τ^{home} stacks the municipal and school-district resident rates and $\tau^{\text{nr,work}}$ is the work-location nonresident rate. A statutory rate change only moves the bill when the changed side of the max binds. I exploit this structure with a continuous stacked difference-in-differences design that decomposes each rate event by direction and binding-state transition, generating a within-design placebo: rate hikes whose post-event home stack stays below the nonresident rate (Inc-nobind) do not change the bill and should not move residents.

The preferred causal estimate is at the census block, where both rate-setting jurisdictions enter at their native geography. A 1 percentage-point municipal resident-rate

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increase in the binding cell reduces the log count of workers living in the block by -0.21 log points (approximately 19%), with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The Inc-nobind within-design placebo is statistically null on the same count outcome.

The municipality-level Inc-bind coefficient is larger (-0.40 log points on residents working in their own municipality) but pre-trended; I report it as policy-jurisdiction magnitude calibration rather than as the causal headline. School-district rate changes do not produce robust evidence of sorting and are treated as an aggregation diagnostic. Industry decomposition shows consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, consistent with a local-services demand spillover from departing residents rather than direct nonresident-tax incidence.

1 Introduction

Do local income taxes cause residential sorting? The Tiebout hypothesis predicts that households "vote with their feet" by sorting across jurisdictions offering different bundles of taxes and public services (Tiebout, 1956), and a large literature documents tax-driven migration at the state level for high earners (Kleven et al., 2020; Moretti & Wilson, 2017; Young et al., 2016) and at sub-national borders for sales taxes (Agrawal, 2015). Yet sub-state income-tax sorting—the canonical Tiebout margin, where moving costs are small and tax boundaries are crossed by a short relocation—is essentially untested in the United States. Recent work on Indiana local income taxes (Yang & Heim, 2017) uses county-level IRS data and finds county-level migration responses, but cannot resolve the muni-level variation that defines the actual tax boundary; the canonical U.S. sub-state place-of-taxation studies (Coomes & Hoyt, 2008; Rohlin et al., 2014) likewise operate at the state-border / county level. A literature survey by Agrawal (2026) explicitly identifies sub-state local-income-tax sorting as an open question, and Agrawal et al. (2022) characterize the empirical evidence as "thin."

The gap persists for two reasons. First, the jurisdictions that levy local income taxes are sub-county, so standard migration data sources do not observe the correct margins: the IRS Statistics of Income tracks county-to-county migration, and the American Community Survey measures place-level population without linking workers to employers. Second, local income-tax systems feature credit mechanisms, overlapping school-district taxes, and uneven enforcement that complicate the mapping from statutory rates to effective tax burden. A naive comparison of rates and population changes conflates tax incidence with amenity valuation, fiscal capacity, and enforcement quality.

I use Pennsylvania (2002–2019) and the LEHD Origin-Destination Employment Statistics (LODES), the only administrative data source that links each worker's residence to their workplace at annual frequency and at sub-municipal geography. Pennsylvania's Earned Income Tax (EIT) is unusual along two dimensions that the empirical design exploits. First, the EIT operates a piecewise max-rule: each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where

τ^{home} is the municipal-plus-school-district stack at the worker's residence and $\tau^{\text{nr,work}}$ is the nonresident rate at the work location. A statutory rate change only moves the worker's bill when the changed side binds. Second, Act 32 of 2008 (effective 2012) consolidated EIT collection into county-based Tax Collection Districts and mandated employer withholding at the higher of the two rates, which made the max-rule mechanically enforced and provides an enforcement-break heterogeneity test.

The max-rule turns a single statutory rate change into heterogeneous predicted effects depending on the binding-state transition. I classify each rate event by direction (Increase / Decrease) and binding-state transition into six cells per side. The increase cells carry the identification because the decrease cells are thin (eleven Dec-bind, four Dec-becomes-non, fifteen Dec-nobind munis in total) and most contain no cities; the discussion below focuses on the three Inc cells, with decrease results reported alongside for completeness. In the Inc-bind cell the home stack was binding before and after, so the bill rises by the full $\Delta\tau$ and the predicted response is strongest. In the Inc-becomes cell the home stack flips from non-binding to binding, so the bill rises only partially (the portion above the nonresident rate) and the predicted response is intermediate. In the Inc-nobind cell the home stack stays below the nonresident rate throughout, so the bill does not move and the predicted response is zero. Inc-nobind is therefore a within-design placebo: if the same statutory increase produces a response in Inc-bind and no response in Inc-nobind, the mechanism interpretation is credible; if Inc-nobind moves, the mechanism interpretation fails.

At the block level, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by -0.21 log points, approximately 19%, with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The Inc-nobind within-design placebo is statistically null on the same count outcome. The block-level decomposition is the preferred causal estimate because both rate-setting jurisdictions—the municipality and the school district—enter at their native geography and the pre-trend is clean. The municipality-level Inc-bind coefficient is larger in absolute value (-0.40 log points on the

count of residents who both live and work in the municipality, implying a $\approx 33\%$ decline per pp rate hike), with a mild positive pre-event drift in the joint Wald test ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$) that is driven primarily by a single pre-period coefficient at $k = -3$ rather than a sharp pre-trend failure; the block-level diagnostic is cleaner. I therefore report the municipality-level estimate as policy-jurisdiction magnitude calibration rather than as the causal headline. School-district rate changes do not produce robust evidence of sorting and are treated as an aggregation diagnostic. Industry decomposition shows that consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, consistent with a local-services demand spillover from departing residents rather than with direct nonresident-tax incidence on jobs.

The narrow, credible claim of the paper is that local income taxes cause residential sorting when the institutional rule makes them mechanically change the worker's bill—that is, in the Inc-bind cell, and at the muni level also in the Inc-becomes cell where the hike flips the home stack from non-binding to binding ($\beta = -0.181^{**}$ on the count of residents who both live and work in the muni). The pooled Inc/Dec specification that a conventional local-tax design would estimate is reported as a benchmark, but it mixes binding and non-binding events. The binding-state decomposition is the paper's contribution.

2 Institutional Background

Among U.S. states that authorize sub-state income taxation, Pennsylvania has the deepest cross-section of jurisdictions setting their own rate: approximately 2,500 municipalities and 500 school districts each levy a separate Earned Income Tax (EIT) on essentially all earned income earned in or by Pennsylvania residents (Agrawal, 2026; Bagchi, 2022). Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries within the same metropolitan area, creating sharp spatial discontinuities in tax burden that can influence where workers

choose to live and work.

2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act of 1965 (Act 511; 53 P.S. § 6924.101 et seq.). It applies to all earned income with no minimum threshold; retirees are exempt. The standard combined cap is 1%, typically split as 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap under the Home Rule Charter and Optional Plans Law (53 Pa. C.S. § 2961 et seq.), and Philadelphia operates separately under the Sterling Act of 1932 (Act of August 5, 1932, P.L. 45) at approximately 3.9% (Bagchi, 2022). For most non-Home-Rule boroughs and townships the EIT is the primary discretionary revenue lever, because property-tax millage limits and reassessment lags constrain the property side; quantifying that constraint would require a 2002–2019 muni-level millage panel that is not currently available from DCED, and I leave the comparison to property-tax sorting as a separate exercise. Across 2002–2019, the panel contains 210 muni resident-rate increases, 119 muni decreases, 611 school-district increases, and 52 school-district decreases; the binding-state breakdown of these events is in Table 13.

Each Pennsylvania municipality has three EIT rates relevant to a worker: the muni resident rate $\tau^{\text{res,muni}}$ levied on residents of the muni; the muni nonresident rate $\tau^{\text{nr,muni}}$ levied on workers employed in the muni who live elsewhere; and the school-district rate τ^{SD} levied on residents only (school districts cannot tax nonresidents). Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (muni, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill

is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where j is the worker's home muni and i is the work muni. The work jurisdiction always collects its full nonresident rate first; the home muni and school district then split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate exceeds the home stack, the home jurisdictions get nothing. The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\right). \tag{2}$$

Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate muni or school district. When the work nonresident rate binds, the worker can lower the bill only by changing job to a lower-nonres-rate work location; moving home without changing job has no effect. For workers who live and work in the same muni, only the home stack applies—they pay $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$.

A worker who lives in muni j and moves to a lower-rate muni j' while keeping the same job at i saves at least the SD rate at j: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD}. Residence moves therefore do save tax under Pennsylvania's piecewise structure, because the home-side stack includes the SD rate, which the work jurisdiction cannot collect from nonresidents. This savings logic is built directly into the empirical design: the treatment $\Delta\tau^{\text{muni}}$ and the binding-state classification are constructed from the worker's home stack, so the regressions identify sorting in response to changes in the same quantity a relocating worker would optimize against.

Philadelphia is the one exception to the Act 32 framework. It operates under the Ster-

ling Act of 1932: Philadelphia residents pay the city resident rate regardless of work location with no offset, and Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania munis. My main estimates retain Philadelphia in the sample for completeness; results are robust to dropping it.

Act 32 of 2008 is central to identification in this paper. Before its 2012 effective date, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The piecewise rule described in Equation 1 existed on paper but was not effectively enforced.

Act 32 transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *mechanically binding*: rate changes now appear in every worker's paycheck through employer withholding, regardless of whether the worker resides in the taxing municipality. Second, it eliminated the possibility of *de facto* avoidance of the work-location nonresident claim through weak enforcement. After Act 32, a worker could no longer effectively duck the nonresident portion owed to the work municipality.

Importantly, Act 32 did not eliminate all tax savings from residential relocation. The piecewise rule in Equation (1) continues to imply that a worker whose original home stack binds can reduce the bill by moving to a sufficiently lower-stack home jurisdiction: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which falls below the old home stack τ_j^{home} whenever $\tau_{j'}^{\text{home}} < \tau_j^{\text{home}}$ and the original home stack was the binding side. Act 32 thus changed enforcement and made the max-rule mechanically binding through withholding; it did not eliminate the residential-relocation margin that the mechanism in this paper exploits.

School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across my panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework: Binding-State Decomposition

The piecewise tax allocation in Equation (1) is the analytical engine of the empirical design. A worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side of the max binds. This generates a sharp prediction: the same statutory rate change can produce a strong behavioral response, a moderate response, or no response at all depending on the binding state pre- and post-event.

3.1 Six Cells per Side

For each muni i and each rate variable $\tau \in \{\tau^{\text{res,muni}}, \tau^{\text{SD}}, \tau^{\text{nr,muni}}\}$, classify the muni's first observed rate event by direction (Increase / Decrease) and binding-state transition relative to the muni's own home-stack vs nonresident rate comparison: An *Inc-bind* event is a rate increase in which the home stack was binding both pre and post, so the bill rises by the full $\Delta\tau$ for in-muni residents; this is the cell with the predicted strongest behavioral response. An *Inc-becomes* event is a rate increase that flips the home stack from non-binding to binding, so the bill rises by less than the full $\Delta\tau$ (only the portion above the work nonres rate),

generating an intermediate predicted response. An *Inc-nobind* event is a rate increase in which the home stack stays below the nonres rate throughout, so the bill never moves; the predicted response is zero, and this cell serves as the within-design placebo.

The decrease cells mirror these definitions. A *Dec-bind* event is a rate decrease in which the home stack stays binding, so the bill falls by the full $\Delta\tau$—the predicted mirror of Inc-bind under symmetry. A *Dec-becomes-non* event is a rate decrease that flips the home stack from binding to non-binding, so the bill falls only down to the nonres rate. A *Dec-nobind* event is a rate decrease in which the home stack stays below the nonres rate throughout, so the bill never moves; this is the second placebo cell. A seventh category, NC (*never-changed*), comprises munis whose τ never changes across the panel. NC serves as the control pool.

The fundamental testable prediction of the institutional rule is an ordered ranking of cell magnitudes: Inc-bind effects exceed Inc-becomes effects, which in turn exceed Inc-nobind effects, with the last close to zero in absolute value and serving as the within-design placebo. The central empirical comparison of the paper is between Inc-bind and Inc-nobind: if statutory rates matter mechanically through tax liability, the residential response should appear in Inc-bind and not in Inc-nobind. If Inc-nobind moves, the mechanism interpretation fails—the response is something other than the binding rule, because Inc-nobind events do not change the worker's bill. This makes the design self-falsifying in a way that a conventional pooled Inc/Dec specification is not: in the pooled specification, every increase is a treated event, while here the institutional rule predicts that only a subset of statutory increases should generate a response.

3.2 Three Treatment Margins

Pennsylvania's institutional structure separates three rate variables that change independently across muni-years. The muni resident rate $\tau^{\text{res,muni}}$ directly affects the bill of in-muni residents through the home-stack channel. The school district rate τ^{SD} also affects the home stack, but is set by school districts rather than municipalities, so the predicted mech-

anism is the same binding-rule channel as the muni rate even though the SD's institutional decision-making differs from the muni's. The muni nonresident rate $\tau^{\text{nr,muni}}$ affects nonresident workers commuting in, since the piecewise rule means it moves the bill of workers from low-home-stack origins.

Two predictions follow. First, each rate's binding-cell decomposition should show the same ranking of Inc-bind above Inc-becomes above Inc-nobind. Second, channel separation should hold: the muni resident rate moves residents (live-and-work, total residents); the muni nonresident rate moves nonresident workers (in-commuters); and the SD rate's effect depends on whether SD changes induce similar residential responses.

4 Data

4.1 Data Construction Pipeline

The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The build sequence scrapes the DCED MunStats portal for the rate panel, scrapes the DCED Annual Financial Reports for muni fiscal controls, downloads block-level LODES files for 2002–2019 (a ∼5 GB raw OD database), crosswalks PA municipalities to TIGER COUSUBFP codes by normalized name and county (99.6% match), builds a block-to-municipality crosswalk by point-in-polygon assignment of block centroids to 2020 cousub geometries (Census blocks nest in cousubs by construction), and aggregates block-pair-year LODES flows to the municipality, school-district, and block levels of observation. Step-by-step pipeline detail (scraper specs, crosswalks, imputation rules, and the auxiliary LODES file for cross-state inflows) is in the Companion Appendix (`appendix.tex`).

4.2 Data Sources

The residence–workplace data come from LEHD LODES Version 8, 2002–2019 (U.S. Census Bureau, 2024c): block-level origin–destination links derived from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages (Abowd et al., 2009). I trim 2020–2021 due to COVID-related disruptions in both employment patterns and LODES coverage. The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality; the mechanism is calibrated so that aggregation to larger geographies produces estimates that approximate ground truth and have been validated against external sources at the tract level and above (Graham et al., n.d.). I aggregate blocks to county subdivisions (municipalities) using the TIGER 2020 block-to-municipality crosswalk (U.S. Census Bureau, 2020) described in Section 4.1, item 3.

Municipal and school-district EIT rates come from the Pennsylvania Department of Community and Economic Development (DCED) MunStats portal (Pennsylvania Department of Community and Economic Development, 2024a), scraped annually. The panel covers resident and nonresident municipal EIT rates plus school district EIT rates for all ∼2,500 PA municipalities, matched to Census COUSUBFP via a name-based crosswalk (99.6% match rate). Pre-2012 nonresident rates, which the registry recorded only sporadically before Act 32 took effect, are backfilled from the earliest observed post-2012 value per muni.

Municipal fiscal controls come from the PA DCED Statewide Municipal Annual Financial Report database (Pennsylvania Department of Community and Economic Development, 2024b), 2002–2019 (trimmed from the available 2002–2023 to exclude COVID/ARPA fiscal distortion). Each muni–year record contains 71 columns covering EIT revenue, real estate tax revenue, police expenditure, streets expenditure, total debt, state intergovernmental aid, population, and assessed market value of real estate. Crime data come from FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest, in the concatenated agency-year files compiled by Kaplan (2024), 2002–2019: monthly agency-level counts of Part I index crimes (murder, robbery, aggravated assault, burglary, larceny-theft, motor vehicle theft),

aggregated to annual totals and matched to PA municipalities via `fips_place_code` (93.5% agency match rate). Housing-supply indicators come from the Census Bureau Building Permits Survey (U.S. Census Bureau, 2024b), place-level annual files, 2002–2019: total housing units permitted by structure type (1-unit, 2-unit, 3–4 unit, 5+ unit), matched via `fips_mcd_code` = COUSUBFP (93.3% MCD match rate).

4.3 Sample Restrictions and Transformations

The sample period runs from 2002 to 2019. LODES coverage begins in 2002, and I trim at 2019 because COVID disrupted both commuting patterns (LODES 2020–2021) and municipal finances (ARPA transfers distort AFR data 2020–2023); including these years would conflate pandemic-driven relocation with tax-motivated sorting.

I impose a minimum employment threshold of 50 total jobs at the municipality level because small municipalities have share_live_in values dominated by sampling noise (a single worker moving changes the ratio by $1/N$). The 50-job threshold balances precision against sample size, and sensitivity analysis (Section 7) confirms stability across thresholds from 10 to 500. I also apply rate outlier filters: PA DCED reports the Local Services Tax (a flat dollar amount) in the same field as the EIT rate (a percentage) in some years, so dropping rates $> 5\%$ removes these data entry errors without affecting any real EIT rate (the highest standard EIT is $\sim$1%; only Philadelphia exceeds 3%). For April 1 alignment, LODES records employment as of April 1, and a rate change that takes effect mid-year could be misattributed if not aligned, so I assign each municipality-year the rate in effect on April 1, ensuring the tax rate a worker faces and the OD flow recording their location refer to the same date.

The primary outcome variable for the headline causal analysis is the block-level log count of workers living in the block. The count directly measures the residential location of active workers and avoids the denominator fragility of block-level shares, which are mechanically sensitive at sparse geography because both the numerator and denominator move. At the municipality and school-district aggregations I also report resident-worker counts, workplace

counts, in-commuter flows, and share outcomes (*share_live_in*, *share_work_outside*). Share outcomes at coarser geographies retain the interpretation of earlier literature; at the block level I treat them as supporting evidence rather than as the primary estimand. Fiscal outcomes include log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, and log market value of real estate from the PA DCED Annual Financial Reports. Pre-treatment balance tests cover eleven observables including crime and building permits.

For treatment definitions I use both a continuous and a binary specification. The continuous treatment is the current rate level (resident municipal, school district, and the home stack as defined in Equation 1), capturing the intensive margin via TWFE and DCDH. The binary cohorts equal 1 after a municipality's first rate increase (or first decrease), enabling Sun-Abraham estimation of staggered-DiD-robust dynamic effects. The control group reflects the fact that Pennsylvania's EIT is universal, so the relevant control pool is municipalities with a stable rate over 2002–2019 (1,264 never-changed municipalities). For the Q3 clean-groups specification I further partition treated municipalities into increase-only (97) and decrease-only (24), dropping the flip-floppers that change in both directions.

Finally, fiscal controls address the concern that rate increases may coincide with other fiscal changes (police hiring, debt issuance) that independently affect residential attractiveness. I include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid as time-varying controls, plus revenue shares (EIT, real-estate, intergovernmental) from the PA DCED AFR. The main estimates are unchanged with and without these controls (the robustness section (Section 7)).

4.4 Descriptive Statistics

This section reports summary statistics and key distributions of the estimation panels. Table 1 reports the municipality-year panel; Table 2 reports the block-year panel that underlies the headline causal estimate. Table 3 reports the school-district aggregation. Table 4 sum-

marizes the binding-state cell classification on the muni resident-rate side, including the dose magnitude in each cell.

Table 1: Muni-year summary statistics (2002–2019)

Variable	N	Mean	SD	p25	p50	p75	Min	Max
Resident rate τ^M (%)	43,374	0.525	0.163	0.500	0.500	0.500	0.000	4.331
Nonresident rate $\tau^{nr,M}$ (%)	34,080	0.883	0.237	1.000	1.000	1.000	0.000	3.820
SD rate τ^{SD} (%)	43,374	0.622	0.289	0.500	0.500	0.500	0.000	2.050
Home stack $\tau^M + \tau^{SD}$ (%)	43,374	1.147	0.316	1.000	1.000	1.000	0.000	4.331
log live-and-work	40,748	3.638	2.011	2.079	3.714	5.088	0.000	12.877
log total jobs	45,291	5.819	2.006	4.489	5.883	7.234	0.000	13.353
Share live-and-work in muni	45,291	0.095	0.081	0.037	0.080	0.133	0.000	1.000
Total jobs	45,291	2076.691	13031.730	89.000	359.000	1385.500	1.000	629698.000

Notes: Muni-year level; one observation per (municipality, year). Rates in percentage points. $\tau^{nr,M}$ has reporting gaps before Act 32; the N for that row is smaller than the full panel.

Table 2: Block-year summary statistics (2002–2019)

Variable	N	Mean	SD	p25	p50	p75	Min	Max
Home count (workers living)	5,030,958	18.822	31.677	4.000	10.000	22.000	0.000	1572.000
log home count	4,795,213	2.293	1.198	1.386	2.398	3.135	0.000	7.360
Work count (jobs in block)	5,030,958	18.826	138.957	0.000	0.000	4.000	0.000	20945.000
log work count	1,894,432	2.256	1.663	1.099	2.079	3.367	0.000	9.950
Share live-and-work in muni	4,795,213	0.154	0.222	0.000	0.067	0.203	0.000	1.000
Share work outside muni	4,795,213	0.846	0.222	0.797	0.933	1.000	0.000	1.000
Resident rate τ^M (%)	4,782,928	0.762	0.799	0.500	0.500	0.500	0.000	4.331
SD rate τ^{SD} (%)	4,782,928	0.607	0.376	0.500	0.500	0.500	0.000	2.050
Nonresident rate $\tau^{nr,M}$ (%)	4,181,214	1.063	0.657	1.000	1.000	1.000	0.000	3.820
Blocks per muni	2,561	120.922	384.585	41.000	73.000	132.000	1.000	16550.000

Notes: Block-year level; one observation per (census block, year). Panel covers 309,680 blocks in 2,561 municipalities.

Table 3: School-district-year summary statistics (2002–2019)

Variable	N	Mean	SD	p25	p50	p75	Min	Max
SD rate τ^{SD} (%)	49,226	0.610	0.278	0.500	0.500	0.500	0.000	2.050
Resident rate τ^{M} (%) (muni-year)	49,226	0.523	0.162	0.500	0.500	0.500	0.000	4.564
Nonres rate $\tau^{nr,M}$ (%) (muni-year)	38,536	0.881	0.237	1.000	1.000	1.000	0.000	3.967
share_state_rev_sd	9,278	0.416	0.166	0.279	0.414	0.555	0.000	0.821
share_local_rev_sd	9,278	0.536	0.186	0.376	0.536	0.691	0.103	1.000
share_federal_rev_sd	9,278	0.048	0.033	0.024	0.040	0.062	0.000	0.247
share_property_tax_sd	9,278	0.420	0.176	0.270	0.407	0.562	0.000	0.853
share_instructional_sd	9,278	0.522	0.070	0.492	0.536	0.567	0.000	1.000
share_support_sd	9,278	0.279	0.039	0.259	0.284	0.305	0.000	0.438

Notes: SD-year level. F-33 share controls are constructed from Census F-33 revenue/expenditure totals.

Table 4: Cell counts: muni resident-rate side, 2002–2019

Cell	Munis	Muni-years	Mean dose (pp)	Max dose (pp)
NA	1	18	0.000	0.000
NC	1,793	32,266	0.000	0.000
Inc-bind	40	720	0.191	1.500
Inc-becomes	58	1,044	0.339	4.331
Inc-nobind	26	468	0.273	0.975
Dec-bind	11	198	-0.104	0.350
Dec-becomes-non	4	72	-0.217	0.000
Dec-nobind	15	270	-0.283	0.700

Notes: Cells classify each muni's first resident-rate event by direction (Inc/Dec) and the binding-state transition relative to the muni's nonresident rate. NC is the never-changed control pool. Dose is the cumulative $\Delta\tau$ at event-time $k = +5$.

Figure 1 plots the joint distribution of the three rate variables that enter the piecewise max-rule: the muni resident rate τ^{M}, the muni nonresident rate $\tau^{nr,M}$, and the home stack

$\tau^M + \tau^{SD}$. The large mass points at $\tau^M = 0.5\%$ and at $\tau^M + \tau^{SD} = 1\%$ are mechanical, not artifacts of the panel. Under Act 511 of 1965, the combined home-side EIT is capped at 1% for ordinary boroughs and townships and is conventionally split 0.5% muni plus 0.5% SD; the vast majority of PA municipalities and school districts simply sit at this statutory default and never adjust. Approximately 121 of the $\sim$ 2,500 munis change their resident rate at any point in the sample, and most school districts likewise hold at 0.5%. The plot is therefore read as a small treated tail embedded in a large institutional control pool. The long right tail of τ^M and $\tau^M + \tau^{SD}$ corresponds to two policy carve-outs: Act 47 distressed cities (Pittsburgh-area munis, Wilkes-Barre, Hazleton, Johnstown, and others) that exceeded the muni half of the 1% cap, and Act 1 school districts that converted property-tax authority into EIT authority. Philadelphia operates under the Sterling Act of 1932 rather than Act 511 and accordingly sits at the far right of the resident-rate distribution. These tail munis and SDs generate the identifying variation that the binding-state decomposition then splits by whether each statutory rate change moved the worker's bill.

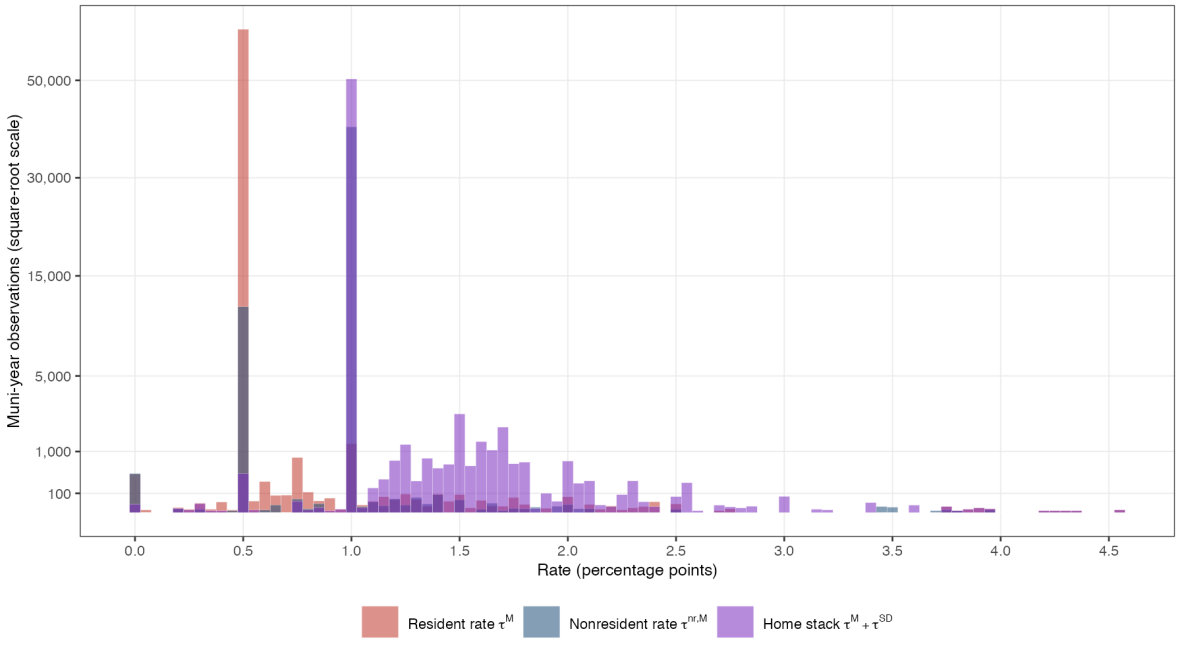


Figure 1: Distribution of PA EIT rates, 2002–2019. Resident rate τ^M, nonresident rate $\tau^{nr,M}$, and the home stack $\tau^M + \tau^{SD}$.

Figure 2 reports the timing of muni resident-rate changes across the panel. Resident-rate increases (red, above zero) and decreases (blue, below zero) occur in roughly similar numbers in any given year, with increases dominating after Act 32 takes effect in 2012.

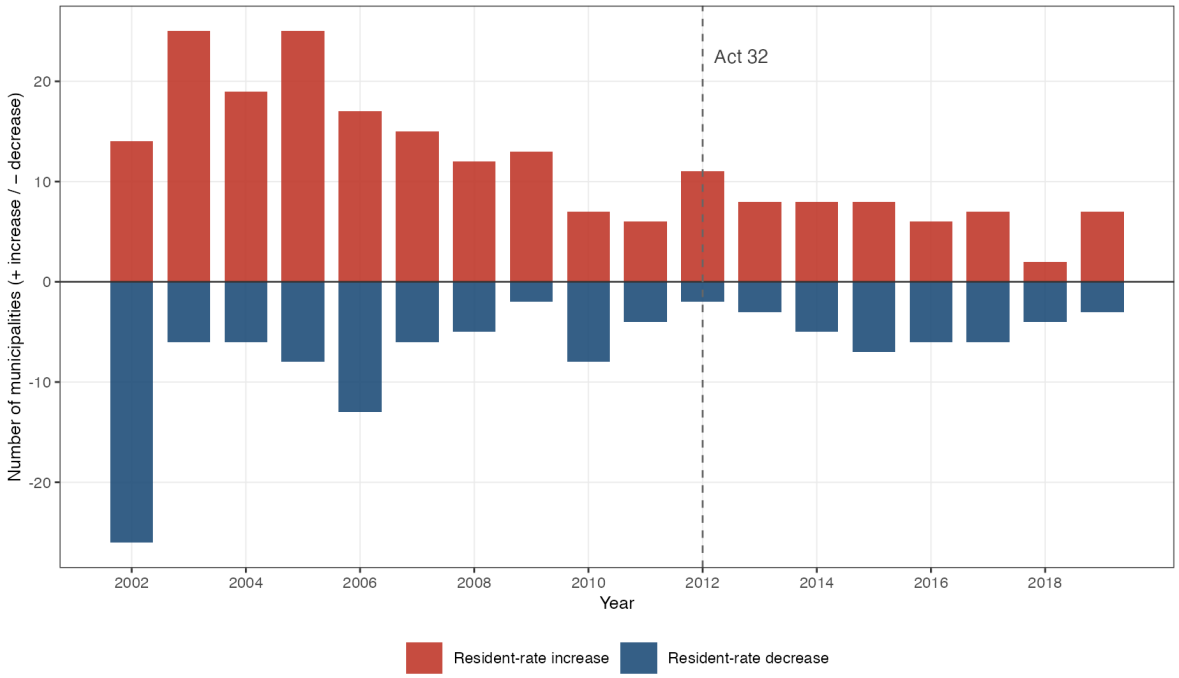


Figure 2: Number of PA municipalities changing their resident rate, by year. Increases shown above zero; decreases shown below. Dashed line: 2012, when Act 32 consolidated withholding.

Figure 3 documents the institutional break at Act 32. Before 2012, fewer than 60% of PA munis reported a nonresident rate to DCED; after 2012, when consolidated withholding made the max-rule mechanically binding, coverage rises sharply. This is the pre-/post-Act-32 enforcement break used in the heterogeneity test of Section 7.2.

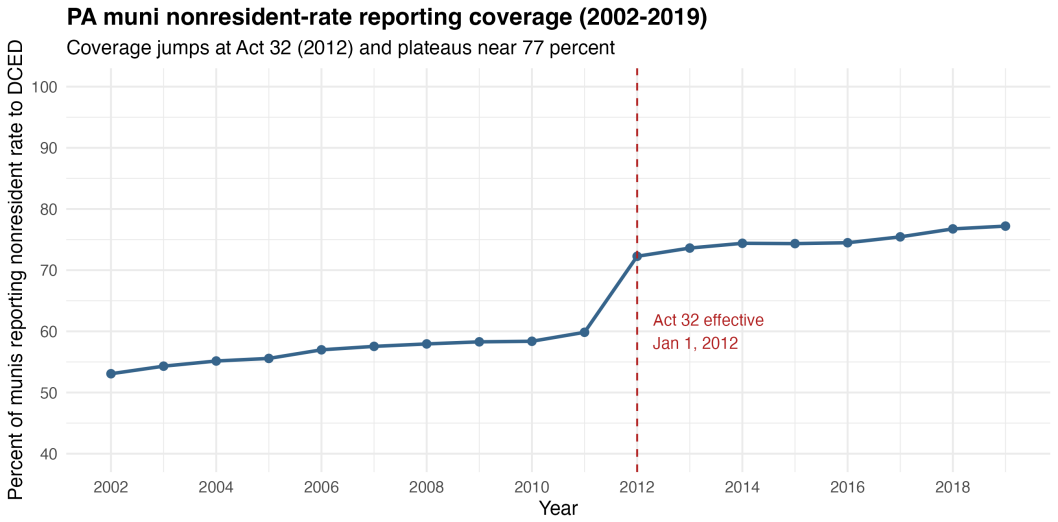


Figure 3: Share of PA municipalities reporting a nonresident rate to DCED, by year. The break at 2012 reflects the Act 32 requirement that employers withhold at the higher of the home resident and work nonresident rates.

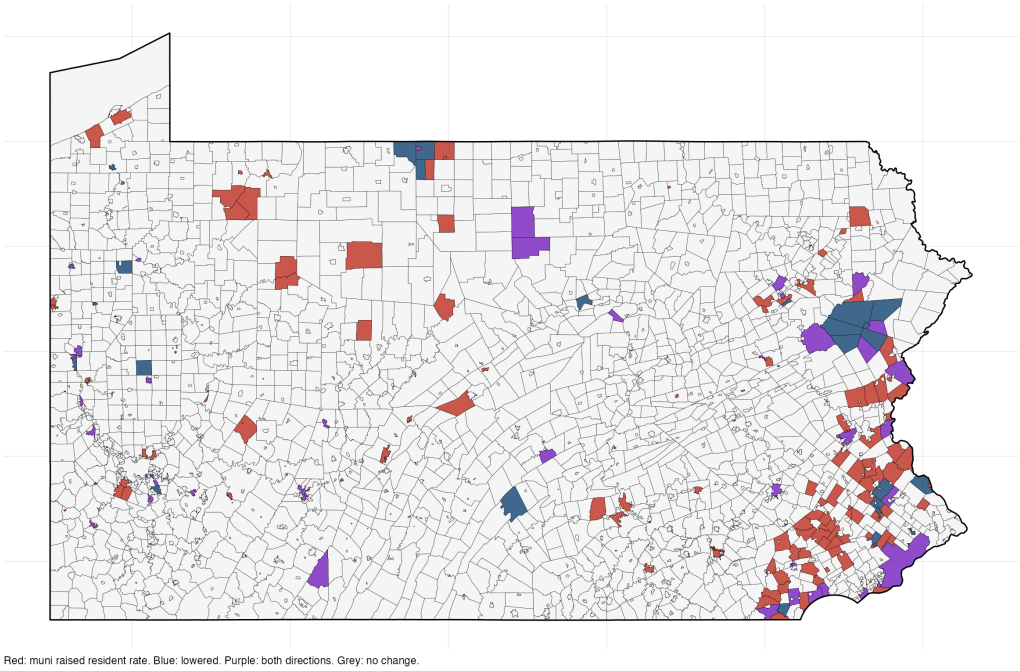


Figure 4: Pennsylvania EIT rate changes, 2000–2019. Red = increased; blue = decreased; purple = both; grey = no change.

5 Empirical Strategy

The empirical framework is a continuous stacked difference-in-differences design with binding-state cell decomposition, estimated on a panel spanning 2002–2019 at three units of observation: municipality, school district, and census block. Pre-2002 data are not consistently available in LODES; post-2019 years are excluded because COVID disrupted commuting patterns and ARPA fiscal transfers distorted municipal budgets. Each estimating unit plays a different role. The block-level estimate is the preferred causal specification because both rate-setting jurisdictions enter the regression at their native geography and the realized-dose pre-trend on the binding cell is clean. The municipality-level estimate is retained as policy-jurisdiction magnitude calibration, and the school-district aggregation is used as a diagnostic. The first-difference dose-response specification estimates the post-event effect of a one percentage-point change in the relevant tax rate; a separate realized-dose event-study diagnostic tests for differential pre-event trajectories across units of varying eventual treatment intensity.

The resident-side specification includes two simultaneous rate-setter treatments: $\Delta\tau_{it}^{\text{muni}}$, the cumulative change in the muni resident rate from baseline, and $\Delta\tau_{it}^{\text{SD}}$, the cumulative change in the school-district rate. Baseline for each is the rate at year $c_i^s - 1$, where c_i^s is unit i's first event year on side $s \in \{\text{muni, SD}\}$, and cumulative dose accumulates with subsequent rate changes within i. The work-side nonresident rate τ^{nr} enters the binding-state classification of cells but is not itself a treatment in the headline regression: I focus on the resident-side mechanism and defer the work-side behavioral response to a separate analysis.

Each rate-change event is classified by direction $\in \{\text{Inc, Dec}\}$ and binding-state transition $\in \{\text{bind, becomes, nobind}\}$ to yield the six non-NC cells introduced in Section 3.1. For each unit i and each rate-setter side s, I locate the first rate event year c_i^s and assign i to one of the six cells based on the direction of the change and the binding state of the home stack relative to $\tau^{\text{nr,muni}}$ in the pre- and post-event years. The NC reference category is restricted strictly to units whose rate τ^s never changes over 2002–2019. Units with rate changes whose

first event has an undefined binding state—typically because the muni's nonresident rate is missing in the pre-event year due to DCED reporting gaps before Act 32—are categorized as "Uncertain" and dropped: pooling them with NC would contaminate the control trend with treated-but-unclassified units, while reporting them in their own cell would mix true binding states with measurement gaps.

Outcomes are run in parallel across counts and shares. The municipality-level workplace and residence outcomes are the logs of `live_and_work` (residents employed in the muni), `in_commuters` (workers in the muni from elsewhere), `out_commuters`, `total_jobs`, and `total_residents`, plus the shares of workers living in their own muni, of workers who are nonresidents, and of residents working outside the muni. The school-district aggregation uses the same outcomes aggregated to the SD via the muni–SD assignment. At the block level, outcomes are the log count of workers living in the block, the log count of workers working in the block, the share of block residents working in their own muni, and the share working outside; geometry and aggregation details are in Section 5.3.

Fiscal controls are share-based and time-varying, with the exact set keyed to the estimating unit. Municipality-level regressions use muni AFR shares (police, streets, and debt service over total expenditures; EIT, real estate, and intergovernmental over total revenues). SD-level regressions use Census F-33 SD shares (U.S. Census Bureau, 2024a) (state, local, federal, and property-tax shares of revenue; instructional, support, capital, and debt-interest shares of expenditure). Block-level regressions use both, since each block is exposed to both rate-setters' fiscal posture.

5.1 Pooled Aggregate Specification

Before the binding-state decomposition I report an aggregate specification that pools all six binding-state cells per side into two direction tags (Inc for any rate increase, Dec for any rate decrease). The pooled regression is the literature-comparable baseline: it answers "what is the average per-pp effect of a muni-rate hike on residents, ignoring binding state?", which

is the elasticity reported in prior local-tax-incidence work (Bagchi, 2022; Yang & Heim, 2017). The binding-state decomposition that follows in Equation (4) then explains *which* sub-population of the treated muni cohort drives the pooled coefficient.

For each unit-time observation (i, t) on the stacked panel, define the per-pp pooled spec

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} + \beta^{M}_{\text{Inc}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Inc}\} \;+\; \beta^{M}_{\text{Dec}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Dec}\} \\ + \beta^{SD}_{\text{Inc}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Inc}\} \;+\; \beta^{SD}_{\text{Dec}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Dec}\} \\ + \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{3}$$

where $d^{s}_{i} \in \{\text{NC}, \text{Inc}, \text{Dec}\}$ is the direction tag on side $s \in \{M, SD\}$ (NC is the omitted reference), $\Delta\tau^{s}_{it}$ is the cumulative dose defined in Section 5.2, and the fixed effects and clustering follow the rule in Section 5.3: unit-by-stack and year-by-stack fixed effects, with clustering at the policy-jurisdiction level.

The pooled coefficient β^{M}_{Inc} is a weighted average of the six binding-state cell coefficients on the muni side (Inc-bind, Inc-becomes, Inc-nobind; with equivalent decomposition for Dec). It is interpretable as the elasticity a prior-literature specification would recover. The decomposition in Section 6.3 then reveals that the pooled coefficient is dominated by the Inc-bind cell at the block level and attenuated toward zero in the Inc-nobind placebo cell—direct evidence that the institutional binding rule, not the statutory rate *per se*, drives the response.

I also report a *binary* analog of Equation (3) in which the four per-pp interactions are replaced by four post-treatment indicators $\{\text{everInc}^{M}_{it}, \text{everDec}^{M}_{it}, \text{everInc}^{SD}_{it}, \text{everDec}^{SD}_{it}\}$, each equal to 1 in the event year and after for the relevant direction on the relevant side. The binary coefficients are comparable to the event-study coefficients in Bagchi (2022); both versions appear together in Section 6.3.

5.2 Binding-State Decomposition: Stacked Continuous DiD

The binding-state decomposition extends the pooled spec by splitting each direction tag into its three binding-state cells, and is the main estimator of the paper. The construction is stacked continuous difference-in-differences. For each side s and each cohort year c at which any muni's first event on side s occurs, build a sub-experiment $\mathcal{S}_c^s$:

The treated set in sub-experiment $\mathcal{S}_c^s$ consists of munis i with $c_i^s = c$ and $\text{cell}_i^s \neq \text{NC}$; the control set consists of all NC^s munis (never-changed on side s). The window covers years $t \in [c - K, c + K]$ with $K = 5$. Each muni-year observation in the stack carries a stack indicator $s_i = c$, relative event time $\ell_{it} = t - c$, and stack-muni and stack-year identifiers $\tilde{i} = (s_i, i)$ and $\tilde{t} = (s_i, t)$.

Stacking all sub-experiments yields a single panel where each NC muni appears once per cohort and each treated muni appears in exactly one sub-experiment. Within each sub-experiment, controls are clean (no treated muni from another cohort serves as control), avoiding the negative-weighting problem in vanilla TWFE under staggered treatment (Goodman-Bacon, 2021).

The stacked-DiD construction is in the lineage of Cengiz et al. (2019) (138 minimum-wage events, stacked event-by-event with continuous treatment), Fadlon and Nielsen (2021) (severe health shocks, stacked with future-treated households as controls), and Deshpande and Li (2019) (SSA field-office closings, stacked with future-closing ZIP codes as controls). Three differences are worth flagging.

The three designs differ first in the choice of control group. Fadlon and Nielsen (2021) and Deshpande and Li (2019) use *future-treated* units (those that will be treated at $\tau + \Delta$) as controls, because never-treated units are scarce or selected in their settings (no never-dying households; selection into closing eligibility). Cengiz et al. (2019) uses *clean controls* (states with no minimum-wage events in the 8-year window around the focal event), which permits states with events outside the window to serve as controls. I use the strictest specification: *never-changed* (NC) munis, defined as those whose rate τ^s never changes across 2002–2019.

Since 95% of PA municipalities are NC on at least one rate side, the never-changed pool is abundant and I do not need to relax to future-treated or clean-within-window controls.

On the second dimension—continuous treatment—Cengiz et al. (2019) is the closest analogue: their treatment is continuous (log minimum wage) within each stacked sub-experiment. Their identification logic—"aligning events by event-time, not calendar time, equivalent to a setting where the events happen all at once" (Cengiz et al., 2019, Online Appendix C)—applies identically in my setup. I extend their continuous-treatment design by extracting *cell-specific* dose-response coefficients via the $i(\text{cell}, \Delta\tau)$ interaction, which their pooled event-by-event spec does not produce.

The third difference is the binding-state cell partitioning, which is new relative to all three antecedents. Cengiz et al. (2019) pools all minimum-wage events without a within-design taxonomy of treated units; Fadlon and Nielsen (2021) and Deshpande and Li (2019) have homogeneous treatments (death, closure). My decomposition exploits the piecewise PA EIT rule to partition treated munis into cells whose ex ante predicted response differs by binding state, generating a within-design placebo (Inc-nobind) that the standard stacked DiD cannot produce.

Some treated munis change rates more than once in 2002–2019, raising a coding question about which event defines the cohort. A treated muni i may have more than one rate-change event over the panel. I classify each such muni by the binding-state transition of its *first* event and assign i to the corresponding cell on side s. The muni then appears in exactly one sub-experiment (its first-event cohort c_i^s), not in subsequent cohorts triggered by later events.

Subsequent rate changes within the same muni accumulate into $\Delta\tau_{it}^s$, the *cumulative* dose from the muni's pre-event baseline. Concretely: suppose muni i has $\tau_i^s = 0.50$ in years 2005–2009, $\tau_i^s = 0.70$ in 2010 (first event, $c_i^s = 2010$, baseline = 0.50), and $\tau_i^s = 1.00$ from 2014 onward (second event). Then $\Delta\tau_{it}^s = 0$ for $t < 2010$, $\Delta\tau_{it}^s = 0.20$ for $t \in [2010, 2013]$, and $\Delta\tau_{it}^s = 0.50$ for $t \geq 2014$. The stacked-continuous specification absorbs this trajectory

through the dose-stack interaction in a single regression on muni i's 2005–2015 window (its first-event cohort 2010 with $K = 5$). Muni i is *not* duplicated as a 2014 cohort treated muni; it stays in its 2010 cohort. The cumulative dose path encodes the second event by raising $\Delta\tau_{it}^{s}$ from 0.20 to 0.50 at year 2014.

This first-event classification has three deliberate consequences. First, no treated muni serves as control to itself; each muni appears in exactly one stack. Second, multi-event treated munis do not receive disproportionate weight from appearing in multiple stacks. Third, the per-pp coefficient β_k^s is interpretable as a dose-response elasticity over the cumulative rate change, which is the welfare-relevant magnitude for fiscal calibration.

For each treated muni i in stack $\mathcal{S}_c^s$, define the cumulative dose-stack as $\Delta\tau_{it}^{s} = \tau_{it}^{s} - \tau_{i,c-1}^{s}$, with $\Delta\tau_{it}^{s} = 0$ for $t < c$ (pre-event) and for NC munis at all t. The stacked continuous DiD specification, for outcome Y, runs the regression on the stacked panel:

$$
\begin{aligned}
Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta_k^{\mathrm{muni}}\, \Delta\tau_{it}^{\mathrm{muni}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{muni}} = k\} \\
+ \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta_k^{\mathrm{SD}}\, \Delta\tau_{it}^{\mathrm{SD}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{SD}} = k\} \qquad (4) \\
+ \mathbf{X}_{it}'\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}},
\end{aligned}
$$

where $\mathcal{C}_{\setminus \mathrm{NC}} = \{$Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind$\}$ is the set of non-NC cells; NC is the omitted reference category on each side, so for each of the two rate-setter sides (muni and SD) the regression estimates six β_k coefficients, each interpreted as the dose-response coefficient *relative to the NC trend on that side*, holding the other side's cell $\times$ dose interaction fixed; each unit i appears in exactly one sub-experiment for each side where it is in a non-NC cell, while NC units appear in every sub-experiment on the side(s) where they are stable; $\alpha_{\tilde{i}}$ absorbs all time-invariant features of the muni-stack pair (so each muni has a separate FE for each cohort it appears in) and $\gamma_{\tilde{t}}$ absorbs cohort-specific time effects; $\mathbf{X}_{it}$ is the share-based control vector specified above; and standard errors are

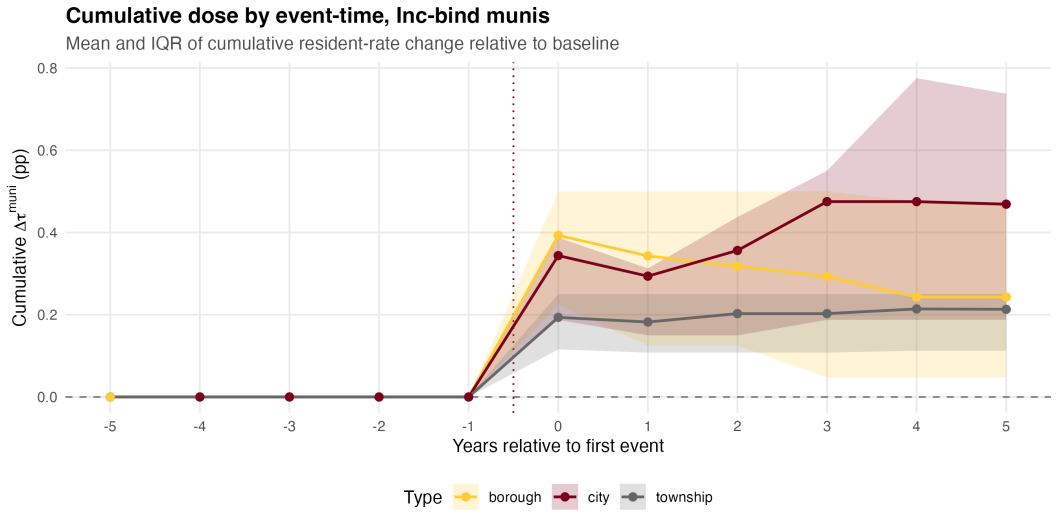
clustered at the muni level (not the muni-stack level), accommodating within-muni serial correlation across sub-experiments.

The cell coefficient β_k^s is the per-pp dose-response of outcome Y to a 1 percentage-point change in rate s, restricted to munis whose first event on side s falls in cell k, with NC munis on side s as controls. Specifically, $\beta^{\text{muni}}_{\text{Inc-bind}}$ is the dose-response of Y to muni-rate hikes among munis whose home stack was binding both pre- and post-event, identified by comparing their post-event trajectory to NC munis' contemporaneous trajectory. The placebo within the design is $\beta^{\text{muni}}_{\text{Inc-nobind}}$: under the institutional rule, this should be zero (the rate hike does not move the bill).

Within each sub-experiment, the comparison is treated cohort vs NC controls in the same calendar-year window. The cell decomposition partitions the treated set by whether the rate change moved the bill. The reference for each cell coefficient is the NC trend on the same side. The triple decomposition (muni $\times$ SD $\times$ nr) lets us read off, in a single regression: which rate moves which outcome, conditional on the muni's binding state on that side.

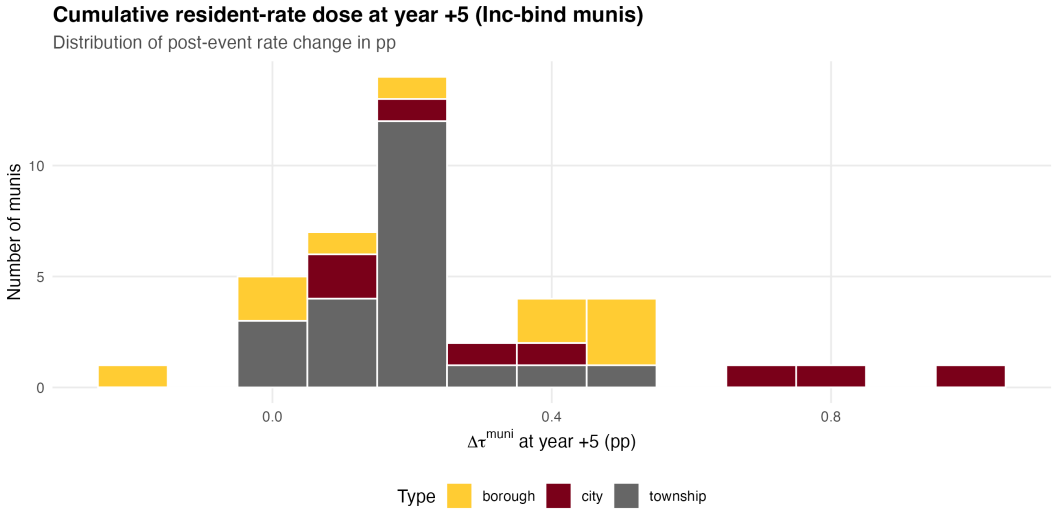
The identifying variation in this design comes from the cross-unit distribution of cumulative rate changes after each unit's first event. Figure 5 traces the mean cumulative muni-resident-rate change (with IQR) of Inc-bind units relative to their pre-event baseline. The pre-event window is mechanically flat at zero by construction (dose is measured relative to the unit's own $c-1$ rate); the post-event window shows the realized cumulative $\Delta\tau^{\text{muni}}$ rising to roughly 0.2 percentage points for townships, 0.3 for boroughs, and 0.4 for cities by $k=+5$. Figure 6 reports the cross-unit distribution of the cumulative dose at $k=+5$: most Inc-bind munis experience cumulative changes of 0.2–0.4 pp, with a small right tail (Wilkes-Barre and Coatesville) at 0.8–0.9 pp. The realized-dose ES diagnostic exploits this across-unit variation to test pre-trends conditional on eventual treatment intensity.

Figure 5: Cumulative muni-resident-rate dose by event-time, Inc-bind units, by muni type.



Notes: Mean and inter-quartile range of cumulative $\Delta\tau_{it}^{\text{muni}}$ (cumulative resident-rate change from baseline $\tau_{i,c-1}^{\text{muni}}$) by relative event-time k, separately for cities, boroughs, and townships. Pre-event values are zero by construction. Post-event cumulative dose averages ≈ 0.4 pp for cities and ≈ 0.2 pp for townships by $k = +5$.

Figure 6: Distribution of cumulative muni-resident-rate dose at $k = +5$, Inc-bind units, by muni type.



Notes: Cross-unit distribution of cumulative resident-rate change at the end of the post-event window ($k = +5$). The mass at 0.2–0.4 pp reflects the standard half-point increment; the right tail at 0.8–0.9 pp is driven by a small set of Home Rule cities.

5.3 Three Estimating Units

The headline equation (4) is estimated at three units of observation. The choice of unit interacts directly with how the two rate-setting bodies (muni and school district) enter the regression.

At the municipality level, the unit of observation is municipality × year. Workplace-panel and residence-panel outcomes are constructed by aggregating LODES origin–destination flows over blocks within the municipality. The municipal resident rate enters the regression at its native level. The school district resident rate is assigned to each muni-year as $\max(\tau^{SD})$ across the school districts the muni reports to in DCED records; for the 97.6% of munis that report to a single SD this is exact, and for the 2.4% that span multiple SDs the max-aggregation is conservative. Fiscal share controls are taken from the muni AFR. Fixed effects: muni-by-stack and year-by-stack. Standard errors clustered at the muni level.

At the school district level, the unit is SD × year. Outcomes are constructed by aggregating LODES OD flows over blocks within the SD. The SD resident rate enters at its native level. The municipal resident rate is assigned to each SD-year as the within-SD aggregation across the SD's constituent munis. Fiscal share controls are taken from Census F-33 SD-level revenue and expenditure data, mirroring the muni AFR shares. Fixed effects: SD-by-stack and year-by-stack. Standard errors clustered at the SD level.

At the block level, the unit is census block × year. Each block is assigned its own muni and its own SD, so both rate-setters' policy variation enters the regression at its native level with no within-jurisdiction aggregation. Outcomes are block-defined: log of workers living in the block, log of workers working in the block, the share of block residents who work in the block's own municipality, and the share of block residents who work outside the municipality. Fiscal share controls include both muni AFR shares and F-33 SD shares, since each block is exposed to both governments. Fixed effects: block-by-stack and year-by-stack. Standard errors two-way clustered at the muni and SD levels.

The block level is the preferred causal estimate for two reasons. First, the block is

the only level where both rate-setting jurisdictions enter at their native geography, with no within-jurisdiction aggregation step the reader has to defend. Second, the realized-dose pre-trend diagnostic is cleanest at the block level (Table 9): all four pre-event coefficients are individually $p > 0.10$, and the joint Wald $\beta_{\text{pre}} = -0.027$ ($|t| < 1$). The municipality-level analogue on the log count of residents working in their own municipality has a mild positive joint pre-trend ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$) driven primarily by the $k = -3$ coefficient, with the other three pre-periods not individually distinguishable from zero. The muni-level diagnostic is therefore noisier than the block-level analogue rather than a sharp failure, but it is enough that I treat the block-level estimate as the causal headline and the muni-level as policy-jurisdiction magnitude calibration: the typical PA municipality has $\sim$40 live-and-work workers, a meaningful per-jurisdiction benchmark, but the noisier pre-period in the diagnostic means I do not promote the muni-level Inc-bind coefficient to a second causal headline. The school-district aggregation produces direction-mixed coefficients that do not survive the pre-trend diagnostic and is reported as an aggregation diagnostic in the Companion Appendix.

5.4 Pre-Trend Diagnostics

Because the first-difference dose $\Delta\tau_{it}$ is mechanically zero in all pre-event periods by construction, the headline specification cannot estimate conventional pre-treatment lead coefficients. I report two complementary event-study diagnostics designed for this setting. The first tests *strong* parallel trends across treatment-intensity magnitudes (recommended by the recent continuous-DiD literature for designs with a continuous dose). The second tests *average* parallel trends under a binary cohort treatment and complements the first by capturing pre-event mean differences regardless of dose.

5.4.1 Realized-Dose Sun–Abraham Event Study

To assess parallel trends in the cross-dose sense (Callaway & Sant'Anna, 2021; de Chaisemartin et al., 2024), I estimate a separate event-study specification using each unit's realized cumulative dose, $\widetilde{D}_i = \tau^M_{i,c+5} - \tau^M_{i,c-1}$, which is constant within unit and varies across units in proportion to eventual treatment intensity. For each muni-side cell c, restrict to the subset of units in $\{\text{NC}, c\}$ and run

$$Y_{it} = \alpha_{\tilde{\imath}} + \gamma_{\tilde{t}} + \sum_{k \neq -1} \beta^M_k \, \mathbf{1}\{\ell_{it} = k\} \, \widetilde{D}_i + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{5}$$

where $\ell_{it} = t - c^M_i$ is event-time relative to the unit's first rate event. The β^M_k at pre-event leads ($k \leq -2$) test whether units with larger eventual rate changes were on differentially sloped pre-event trajectories; under parallel trends in the strong sense, these coefficients are zero. The post-event coefficients (β^M_k for $k \geq 0$) recover the per-pp dose-response at each event time and average to the corresponding cell coefficient in the headline specification when the unit's rate path is a single step at the event year.

5.4.2 Binary Sun–Abraham IW Event Study

The realized-dose diagnostic above tests whether *larger* eventual rate changes are associated with differential pre-trends. A complementary diagnostic asks the simpler question of whether ever-treated units differ in mean trajectory from never-treated units, ignoring dose magnitude. For this I use the Sun–Abraham interaction-weighted event study (Sun & Abraham, 2021) implemented in `fixest::sunab()`:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{e} \sum_{k \neq -1} \delta_{e,k} \, \mathbf{1}\{c_i = e\} \, \mathbf{1}\{\ell_{it} = k\} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{6}$$

estimating a separate dynamic ATT $\delta_{e,k}$ per cohort e and event-time k, then interaction-weighting the $\delta_{e,k}$ across cohorts to deliver a single cohort-robust path of pre- and post-event

coefficients. Binary cohort assignment is constructed per binding-state cell: ever-Inc-bind munis are coded **1** in the year of their first Inc-bind event and thereafter, NC munis are coded **0** throughout, and other cells are excluded from the cell-specific subset.

The two diagnostics are complementary: failure of the realized-dose test indicates dose-magnitude selection (high-dose units were trending differently), while failure of the binary test indicates *any* mean differential trajectory between treated and control units. Passing both is required for clean identification under the strong parallel-trends assumption. Block-level binary SA-IW is unstable (thin cohort $\times$ event-time cells under stacking) and is reported at the municipality and SD levels only.

I do not impose a uniform pre-trend standard across every outcome in the paper, because not every outcome carries the causal claim. The primary residential-sorting outcome is the block-level log count of workers living in the block, and the realized-dose pre-trend on the binding cell must be clean here for the headline to go through; this is the outcome the paper asks the reader to believe. Municipality-level resident counts—log residents working in their own municipality, log employed residents—are retained as policy-jurisdiction magnitude calibration. I acknowledge and caveat imperfect pre-trends at the municipality scale, but these outcomes do not carry the causal claim. Secondary equilibrium outcomes such as log in-commuters, log total jobs, and the industry decomposition are reported as suggestive evidence of a local-services demand channel; imperfect pre-trends are acceptable there because these outcomes are downstream of the residential-sorting margin and are not the identification object. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are sensitive to sparse denominators and to endogenous denominator dynamics at block geography, so I report them in the Appendix and do not lean on them. The headline block-level log home count in the Inc-bind cell passes the strong standard; other outcomes are retained with the caveats appropriate to their role.

5.5 Sample Variants

The DCED nonresident-rate panel has reporting gaps—55–60% coverage pre-Act 32, 72–80% post—which affect the binding-state classification when the nonresident rate is missing. I report three sample variants. The strict variant drops muni-years with no recorded nonresident rate. The filled variant imputes `muni_nonres_eit` as the contemporaneous resident rate when missing, which is statute-anchored: under Act 511 the nonresident rate is bounded above by the resident rate. The drop-incomplete variant keeps all muni-years and drops NA only at regression time per outcome, maximizing N. The headline tables use the drop-incomplete sample; the other two are reported as robustness columns, and coefficients are stable across them.

Table 5 summarizes the role of each specification that follows—the headline binding-state decomposition at each estimating unit, the pooled Inc/Dec literature benchmark, and the auxiliary diagnostics (industry decomposition, pre/post Act 32 split, dose-magnitude terciles, alternative control pools, and sample variants) reported either later in the main text or in the Appendix.

Table 5: Specification roadmap

Specification	Role in paper	Causal status	Where reported
Block-level binding-state decomposition	Main causal estimate (block log home count, Inc-bind)	Preferred	Main text, §6.3
Pooled Inc/Dec (aggregate)	Conventional literature benchmark; motivates decomposition	Descriptive / benchmark	Main text, §6.1
Municipality-level binding-state decomposition	Policy-jurisdiction magnitude calibration	Secondary, caveated (joint pre-trend marginal)	Main text, §6.4
Industry decomposition (muni level)	Mechanism: local-services demand spillover	Suggestive	Main text, §6.5
School-district aggregation	Aggregation diagnostic (non-result)	Not load-bearing	Brief §6.6; full tables in Appendix
Pre/post Act 32 split	Enforcement-break heterogeneity	Robustness	Brief main text; full in Appendix
Block share outcomes	Sparse-denominator caveat	Supporting	Appendix only
Binary Sun–Abraham IW ES	Average parallel-trends diagnostic	Diagnostic	Appendix
Dose-magnitude tercile	Linearity check	Robustness	Appendix
Clean-in-window control	Alternative control pool	Robustness	Appendix
Sample-variant comparison	Strict / filled / drop-incomplete	Robustness	Appendix

Notes: The block-level binding-state decomposition is the preferred causal estimate because the block observes both rate-setting jurisdictions at their native level and its realized-dose pre-trend is clean. The pooled Inc/Dec model is a literature-comparable benchmark but mixes binding and non-binding events. The municipality-level estimate is retained as policy-jurisdiction calibration; its Inc-bind realized-dose joint pre-trend is marginal ($|t| = 2.66$) and driven by a single pre-period coefficient. The school-district aggregation does not produce robust evidence of sorting and is treated as a diagnostic non-result.

6 Results

I report results in three layers. Section 6.1 presents the pooled Inc/Dec benchmark—what a conventional local-tax design would recover if it ignored the Pennsylvania max-rule. Sections 6.3–6.6 then report the binding-state decomposition at three levels of aggregation. The block-level decomposition (§6.3) is the preferred causal estimate: both rate-setting jurisdictions enter at their native geography and the realized-dose pre-trend is clean. The municipality-level decomposition (§6.4) is reported as policy-jurisdiction magnitude calibration; the Inc-bind pre-trend fails there, so it is not the causal headline. The school-district aggregation (§6.6) is a diagnostic non-result and is summarized briefly, with full tables in the Appendix. The industry mechanism, most naturally read at the municipality level, appears in §6.5. The pre-/post-Act-32 split, where the block-level binding-cell effect strengthens after consolidated withholding made the EIT mechanically binding, appears in §7.2. The central empirical comparison throughout is between the Inc-bind cell (where the statutory rate change moves the worker's bill) and the Inc-nobind cell (where it does not); Inc-nobind is the within-design placebo.

6.1 Pooled Inc/Dec Benchmark

I first report the pooled Inc/Dec specification from Equation (3), which collapses each rate-setter side to two direction tags. The pooled coefficients are useful because they approximate the estimand in a conventional local-tax design: the average per-pp dose-response of a muni-rate change and an SD-rate change without distinguishing whether the rate change moves the worker's bill. In Pennsylvania, however, the pooled specification is not the structural object of interest. It averages together rate changes that raise tax liability (Inc-bind) and rate changes that do not (Inc-nobind), and the binding-state decomposition in Sections 6.3–6.4 shows which sub-population of the pooled treated cohort drives each coefficient.

Tables 6 and 7 report the per-pp pooled coefficients at the block and municipality levels.

Both tables include four coefficients of interest per outcome: the muni resident-rate Inc effect β^{M}_{Inc}, the muni resident-rate Dec effect β^{M}_{Dec}, and the analogous school-district-rate coefficients $\beta^{SD}_{\text{Inc}}, \beta^{SD}_{\text{Dec}}$.

Table 6: Pooled Inc/Dec headline — block level (per pp)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Municipality rate (τ^{M})				
$\Delta\tau > 0$ (Inc)	-0.045	-0.066***	-0.041***	0.041***
	(0.034)	(0.024)	(0.009)	(0.009)
$\Delta\tau < 0$ (Dec)	0.255***	0.097	-0.025	0.025
	(0.089)	(0.229)	(0.022)	(0.022)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.035	0.028	-0.000	0.000
	(0.031)	(0.025)	(0.005)	(0.005)
$\Delta\tau < 0$ (Dec)	0.001	-0.002	-0.008***	0.008***
	(0.008)	(0.011)	(0.003)	(0.003)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^{M}_{Inc}, β^{M}_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Pooled Inc/Dec headline — municipality level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	-0.228***	-0.209***	-0.035	-0.019**	0.024***
	(0.058)	(0.056)	(0.033)	(0.008)	(0.006)
$\Delta\tau < 0$ (Dec)	0.260**	0.014	0.058	0.022	-0.005
	(0.107)	(0.143)	(0.138)	(0.019)	(0.003)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.128*	0.105	0.051	0.007	-0.002
	(0.070)	(0.067)	(0.038)	(0.005)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.310***	-0.170**	0.020	-0.003	0.005**
	(0.110)	(0.076)	(0.051)	(0.007)	(0.003)
N	218,677	239,023	242,722	239,033	242,722
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The school-district-level analogue (pooled across the SD's constituent muni rates) is in Table 8.

Table 8: Pooled Inc/Dec headline — school district level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	0.017	0.013	0.014	0.000	0.001
	(0.022)	(0.017)	(0.014)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	-0.017***	-0.010*	-0.005***	-0.000	0.001***
	(0.003)	(0.005)	(0.002)	(0.000)	(0.000)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.150**	0.154**	0.196***	0.001	0.003
	(0.070)	(0.069)	(0.054)	(0.004)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.008	0.089	0.139***	-0.012	0.011**
	(0.069)	(0.078)	(0.040)	(0.010)	(0.005)
N	51,440	51,474	51,474	51,474	51,474
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.003	0.002	0.004	0.004	0.003
Dep. mean	6.260	8.335	8.768	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: SD-level regression: τ^{SD} is the school-district's own rate; the muni-rate row uses the within-SD mean of constituent muni resident rates as an aggregated treatment. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Tables 9 and 10 report the ATT-summary statistics for the block-level pooled event study (muni- and SD-cohort stacks respectively), with cluster-robust Wald pre-trend p-values. Figure 7 then plots all four event-time paths side by side. Tables 11 and 12 report the analogous ATT summaries for the municipality- and SD-level pooled specifications; Figure 8 overlays those event studies on the same axes, with color encoding direction and line type encoding aggregation level (solid = municipality, dotted = school district). The Wald p-values are the joint pre-trend test $H_0\colon \beta_k = 0$ for all $k \leq -2$ using the model's cluster-robust variance matrix.

Table 9: Pooled realized-dose event study — block level (muni-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc (post)	-0.032	-0.053	-0.035	0.035
	(0.021)	(0.014)	(0.005)	(0.005)
Inc (pre)	0.079	0.028	0.008	-0.008
	(0.020)	(0.011)	(0.005)	(0.005)
Inc Wald p	0.0019	0.2118	0.0002	0.0002
Dec (post)	0.217	0.123	-0.045	0.045
	(0.046)	(0.100)	(0.007)	(0.007)
Dec (pre)	0.171	0.160	0.018	-0.018
	(0.052)	(0.118)	(0.010)	(0.010)
Dec Wald p	<0.0001	<0.0001	0.0021	0.0021

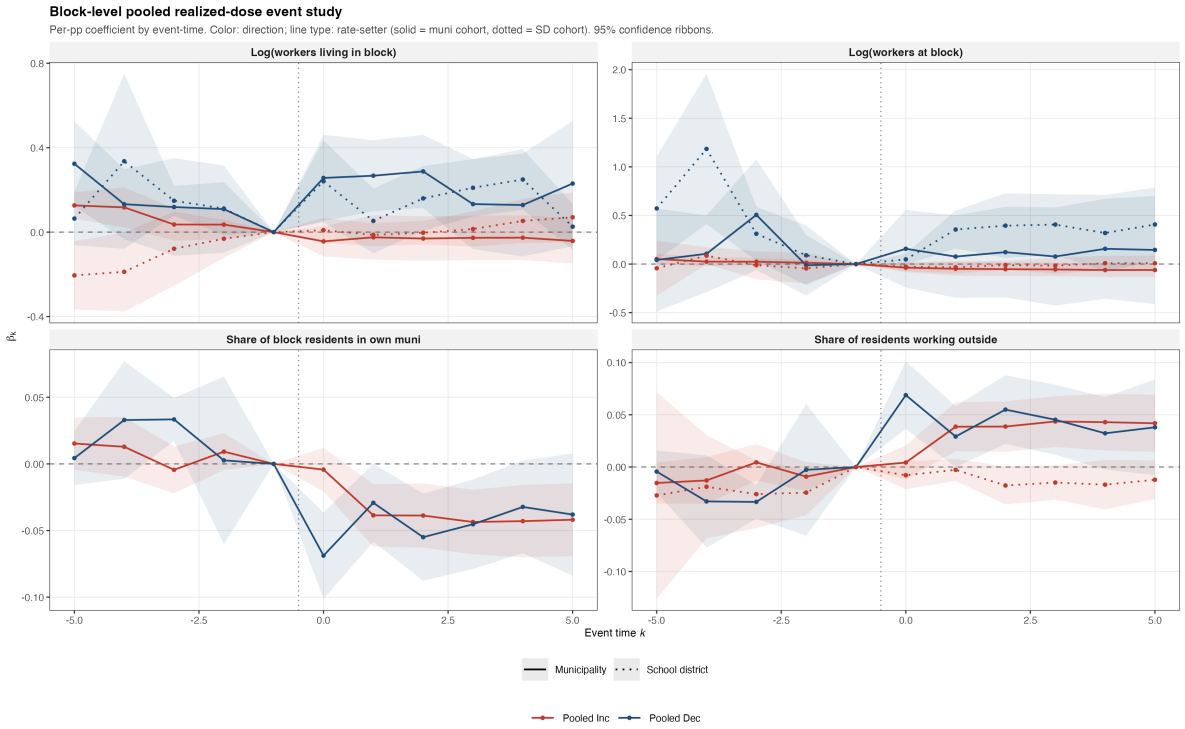
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of $H_0 \colon \beta_k = 0$ across pre-event k. Stack built on muni-cohort years; event time defined against each block's muni-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Pooled realized-dose event study — block level (SD-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc (post)	0.022	-0.012	—	-0.012
	(0.017)	(0.014)	—	(0.004)
Inc (pre)	-0.126	-0.003	—	-0.024
	(0.040)	(0.047)	—	(0.015)
Inc Wald p	0.0573	0.0151	—	0.2450
Dec (post)	—	0.322	-0.117	—
	—	(0.063)	(0.011)	—
Dec (pre)	—	0.540	-0.012	—
	—	(0.131)	(0.008)	—
Dec Wald p	—	<0.0001	0.0067	—

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Stack built on SD-cohort years; event time defined against each block's SD-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 7: Block-level pooled realized-dose event study: four pre- and post-event paths corresponding to the four pooled coefficients in Table 6.



Notes: Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects and two-way clustering at the municipality and school-district levels. Each line corresponds to one of the four coefficients in Table 6: color encodes direction (red = pooled Inc; blue = pooled Dec); line type encodes rate-setter (solid = muni-cohort stack with `realized_dose_muni` as the event-time interaction; dotted = SD-cohort stack with `realized_dose_sd`). The two stacks are parallel: each block enters the muni stack at its first muni-rate event year c_b^M and the SD stack at its first SD-rate event year c_b^{SD}; relative-event-time k is defined against the relevant rate-setter's event year. Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Table 11: Pooled realized-dose event study — municipality level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc (post)	-0.197	-0.218	-0.039	-0.015	0.018
	(0.026)	(0.028)	(0.014)	(0.003)	(0.003)
Inc (pre)	0.094	0.010	0.054	0.011	-0.014
	(0.034)	(0.058)	(0.032)	(0.006)	(0.003)
Inc Wald p	<0.0001	0.0513	0.4834	0.5071	0.0216
Dec (post)	0.674	0.264	0.175	0.020	0.000
	(0.178)	(0.091)	(0.041)	(0.008)	(0.004)
Dec (pre)	0.190	-0.139	0.169	0.004	0.006
	(0.217)	(0.100)	(0.044)	(0.007)	(0.007)
Dec Wald p	<0.0001	<0.0001	0.0003	0.0001	<0.0001

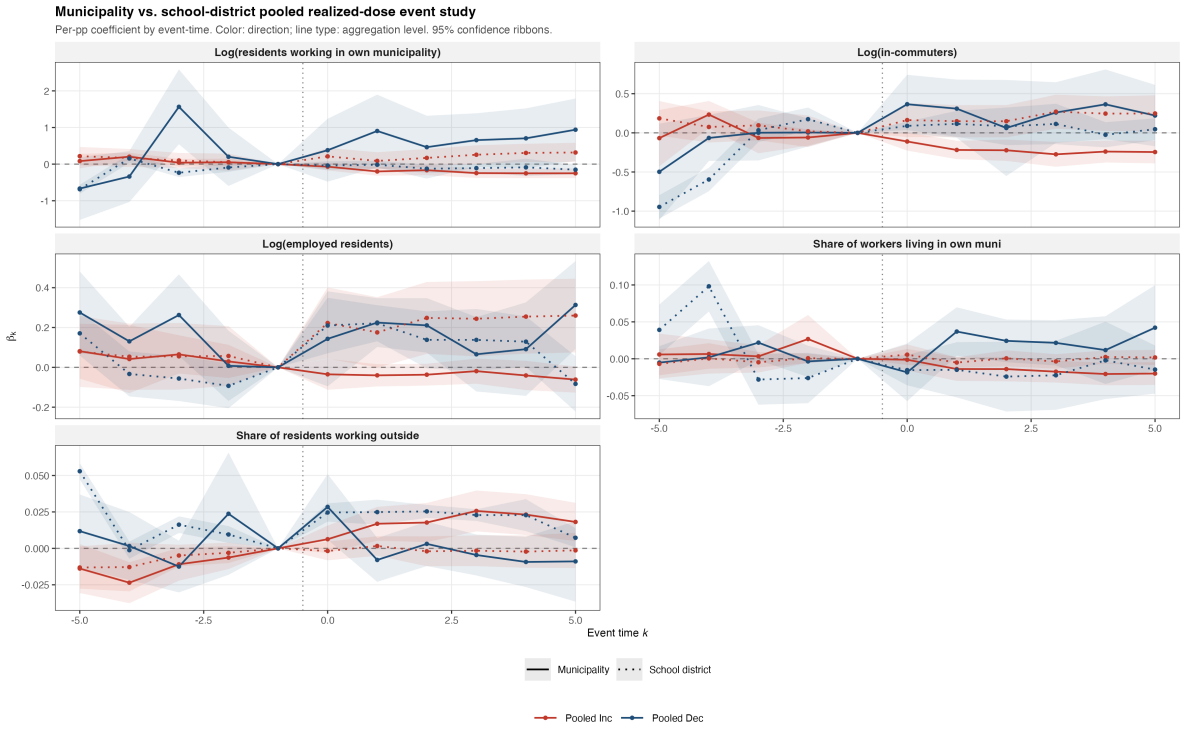
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12: Pooled realized-dose event study — school district level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc (post)	0.225	0.204	0.234	0.000	-0.001
	(0.051)	(0.045)	(0.038)	(0.002)	(0.002)
Inc (pre)	0.134	0.094	0.061	-0.003	-0.008
	(0.060)	(0.052)	(0.042)	(0.004)	(0.003)
Inc Wald p	0.4556	0.2738	0.8816	0.8688	0.4907
Dec (post)	-0.088	0.071	0.126	-0.016	0.021
	(0.030)	(0.038)	(0.028)	(0.009)	(0.001)
Dec (pre)	-0.214	-0.334	-0.003	0.021	0.019
	(0.030)	(0.038)	(0.029)	(0.009)	(0.001)
Dec Wald p	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 8: Municipality vs. school-district pooled realized-dose event study: direction (color) and aggregation level (line type).



Notes: Per-pp realized-dose coefficient at each event-time. Color: direction (red = Inc, blue = Dec). Line type: aggregation level (solid = municipality stacked-continuous DiD; dotted = school-district stacked-continuous DiD on the aggregated outcomes). Ribbons are 95% confidence bands.

The pooled coefficients in Tables 6–8 mask substantial within-direction heterogeneity relative to the cell-decomposed magnitudes. At the block level, the pooled-Inc post-event coefficient on the log count of workers living in the block is approximately −0.08, while the realized-dose ES post-event mean restricted to the Inc-bind cell alone is −0.305 (a fourfold attenuation when Inc-bind is averaged with Inc-becomes and the Inc-nobind placebo cells). At the municipality level the pooled-Inc coefficient on the log count of residents working in their own municipality is approximately −0.23, compared to an Inc-bind cell coefficient of −0.401. The pooled coefficient is a sample-share weighted average of the binding-state cell coefficients; the decomposition in Sections 6.3– 6.4 shows that this aggregate is driven almost entirely by Inc-bind, while Inc-nobind delivers the predicted null behavioral response. This is the central point of the pooled-then-decompose structure: the institutional binding rule,

not the statutory rate change itself, identifies the response.

Pooled-Dec coefficients deserve a separate caveat on pre-trends. The block-level pooled-Dec ES coefficient on the log count of workers living in the block has a post-event mean of +0.22 and a pre-event lead of +0.17; the post/pre difference is small and the pre-trend fails parallel trends. The Dec direction reflects selection of a small set of declining or politically distinct munis into rate cuts, not a clean binding-rule mirror of the Inc effect, and is reported here for completeness. The Inc direction, with its clean pre-trend on the count outcomes, carries the causal weight in the analysis.

A binary pooled-spec analog (Companion Appendix, "Pooled Inc/Dec Binary-Event Analog") replaces the per-pp dose interactions with four post-treatment indicators, comparable to a Bagchi (2022) binary DiD spec. Coefficients agree in sign with the per-pp pooled estimates and are deferred to the Companion Appendix in long form; I report them here only as a robustness column at the headline outcome.

6.2 Cell Counts

Table 13 reports the number of unique munis in each cell on each side after the NC-purity filter described in Section 5.

Table 13: Cell composition by side (drop sample, post-NC-fix)

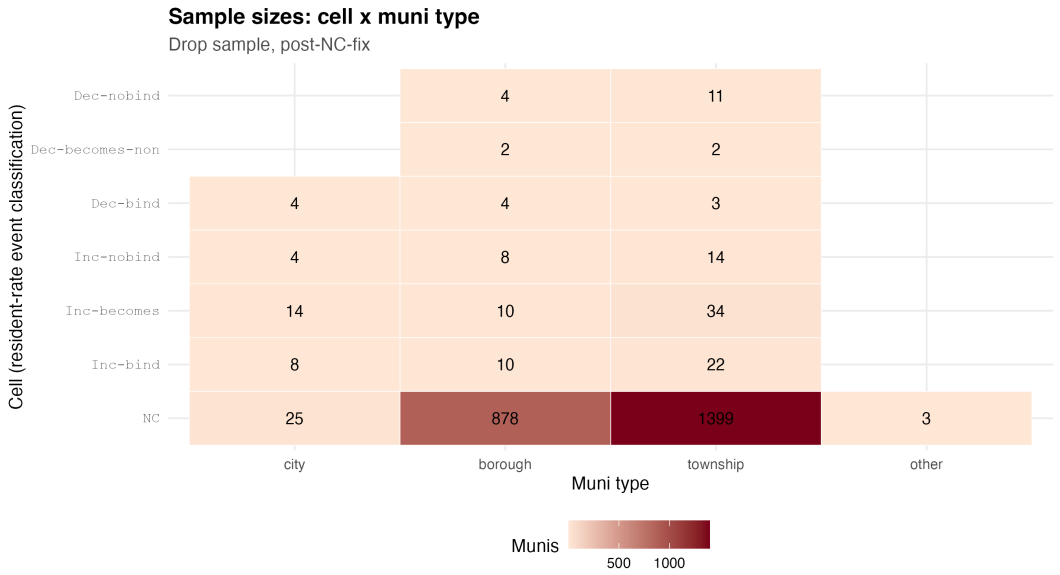
Side	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes-non	Dec-nobind	NC
Muni resident (τ^{muni})	40	58	26	11	4	15	2,406
School district (τ^{SD})	75	197	14	4	10	—	2,253
Muni nonresident (τ^{nr})	0	0	171	0	5	109	2,276

Notes: Cells classified at each muni's first observed rate event on the relevant side. Munis with rate changes whose first-event binding state cannot be determined (NA pre-event nonresident rate) are dropped (not pooled with NC). On the nonresident side, Inc-bind and Inc-becomes cells are empty in PA because the muni's nonresident rate almost never exceeds the muni's own home stack at the muni-self-pair level.

The cell counts also vary substantially by muni type. Figure 9 disaggregates the cell

composition by muni type (city, borough, township). The 40 Inc-bind munis comprise 8 cities, 10 boroughs, and 22 townships; the muni-type heterogeneity analysis in the Companion Appendix shows the headline magnitude is concentrated in the 8 post-industrial cities. Decrease cells are extremely thin: 11 Dec-bind, 4 Dec-becomes-non, 15 Dec-nobind units in total, with no cities in Dec-becomes-non or Dec-nobind. The binding-rule symmetry test between Inc and Dec is therefore not feasible under conventional standard-error inference.

Figure 9: Sample sizes by binding-state cell and muni type (drop sample, post-NC-fix).

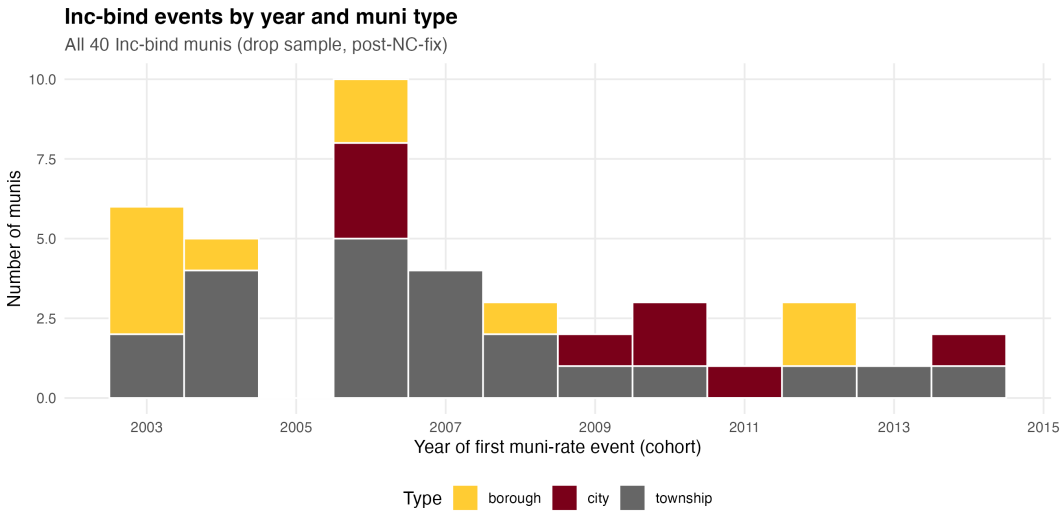


Notes: Each cell reports the count of unique municipalities in the (binding-state cell × muni type) bin. NC dominates by sample size (878 borough + 1,399 township never-changers); treated cells are small per type.

The stacked design's identifying support comes from the joint distribution of cohort timing and event-time coverage. Figure 10 reports the distribution of muni-rate first-event years for the 40 Inc-bind municipalities. The cohort distribution is bimodal: an early spike in 2003–2004 (boroughs and townships responding to baseline EIT setup) and a 2006 spike of 10 units (Act 1 of 2006 enabled SD-rate increases that mechanically flipped some munis into Inc-bind via the joint $\tau^{\text{muni}} + \tau^{\text{SD}}$ stack), with a long tail through 2014. Figure 11 shows the resulting stacked-design support: the 2006 cohort dominates the post-event window with 10 treated munis, and most other cohorts contribute 1–4 treated munis to the stack.

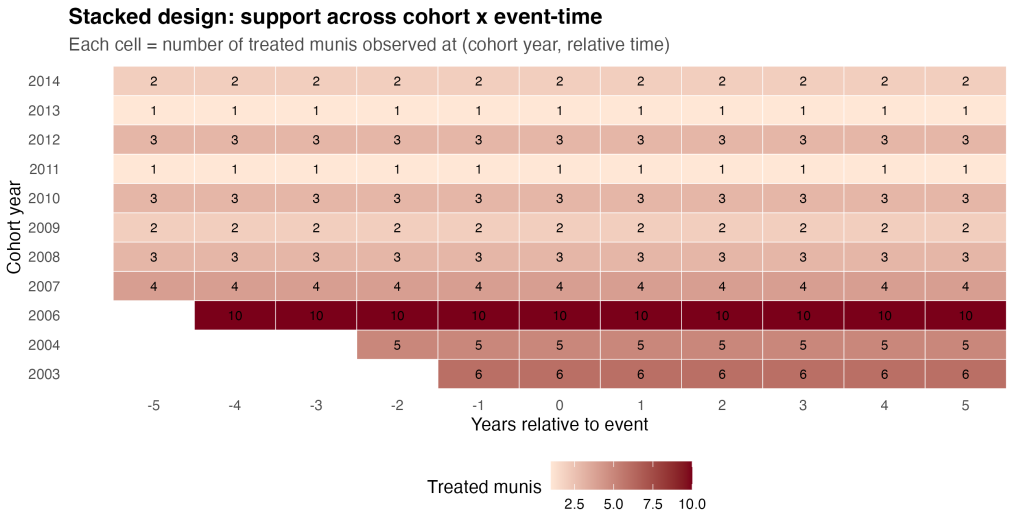
The thin tail cohorts limit the effective sample size for late-event-time coefficients in each cohort-specific stack.

Figure 10: Inc-bind muni-rate first-event timing, by muni type.



Notes: Counts of Inc-bind municipalities by year of first muni-rate event. The 2006 spike reflects Act 1 of 2006 (school-district EIT authorization) flipping some munis into the binding cell via the joint home-stack mechanism.

Figure 11: Stacked design support: treated muni count by cohort year × relative event-time.



Notes: Each cell reports the number of Inc-bind treated munis observed at the (cohort year, relative event-time k) combination. Pre-event cells before 2002 are empty because LODES coverage begins in 2002 (cohorts at $c = 2003$ and $c = 2004$ have truncated pre-event windows).

6.3 Block-Level Headline Results (Binding-State Decomposition)

The preferred causal estimate is the block-level municipal Inc-bind coefficient on the log count of workers living in the block. This is the cleanest test of the EIT mechanism for two reasons. First, both rate-setting jurisdictions enter the regression at their native geography—each block is exposed to a single municipality and a single school district, so no within-jurisdiction aggregation is required to assign policy variation. Second, the realized-dose pre-trend on the binding cell is near zero (Table 9; $\beta_{\text{pre}} = -0.027$, $|t| < 1$). Together these features make the block-level decomposition the level at which the paper asks the reader to believe a causal claim.

Table 14 reports the block-level dose-response on the muni-rate cells, and Table 15 reports the SD-rate cells from the same unified regression. The main residential-sorting outcome is the log count of workers living in the block; the log count of workers working in the block is reported alongside as a secondary count outcome. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are mechanically fragile at block geography because both the numerator and denominator can move and many blocks contain small worker counts. I report them in Appendix **??** as supporting estimates and do not use them to identify the headline causal effect.

Table 14: Block-level dose-response: muni resident-rate cells (count outcomes)

Outcome	Log(workers live)	Log(workers work)
Inc-bind	-0.208***	-0.097*
	(0.054)	(0.052)
Inc-becomes	-0.013	-0.053**
	(0.032)	(0.022)
Inc-nobind	-0.056	-0.113
	(0.098)	(0.075)
Dec-bind	0.269*	0.167
	(0.153)	(0.353)
Dec-becomes-non	0.193	0.413***
	(0.299)	(0.137)
Dec-nobind	0.242***	-0.053
	(0.086)	(0.254)
N	23,567,655	8,751,581
R^2	0.858	0.865
Within R^2	0.000	0.000
Dep. mean	2.238	2.318
Dep. SD	1.186	1.655

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix; sparse block-level denominators make share estimates mechanically fragile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: Block-level dose-response: SD resident-rate cells (count outcomes)

Outcome	Log(workers live)	Log(workers work)
Inc-bind	0.097	0.099**
	(0.062)	(0.047)
Inc-becomes	0.016	0.009
	(0.032)	(0.025)
Dec-bind	0.207***	-0.052
	(0.021)	(0.144)
Dec-becomes-non	-0.013*	0.001
	(0.008)	(0.007)
Dec-nobind	-0.486***	0.502***
	(0.099)	(0.168)
N	23,567,655	8,751,581
R^2	0.858	0.865
Within R^2	0.000	0.000
Dep. mean	2.238	2.318
Dep. SD	1.186	1.655

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Inc-nobind cell serves as the within-design placebo: it contains munis whose home stack remained below the muni's nonresident rate after the rate hike, so the bill never moved for in-muni residents. The institutional rule predicts no behavioral response, and the Inc-nobind coefficient on the log count of residents working in their own municipality is statistically null (-0.062, SE 0.169); on log total jobs is -0.201 (SE 0.203); and on the

share of workers living in their own muni is −0.012 (SE 0.008). The placebo passes on the headline outcomes. Two outcomes show small marginally significant deviations (the log count of out-commuters $+0.217^{**}$ and log employed residents $+0.194^{**}$); with $n = 26$ in this cell these are imprecise and I disclose rather than claim a perfectly clean placebo.

The joint regression also separates the two rate-setting channels, since it includes both muni-resident-rate cells and SD-resident-rate cells. The muni resident rate moves residents (Inc-bind coefficient on log residents working in their own municipality: -0.401^{***}; on log employed residents: -0.100^{***}) and in-commuters (-0.212^{**}), consistent with the local-services demand spillover in Section 6.5. The SD resident-rate Inc-bind coefficient on the same outcomes is $+0.177^{*}$ on log residents working in their own municipality and +0.063 on log employed residents: SD rate changes do not produce a clean sorting response through the binding channel even when SD changes flip the binding state. An earlier draft using tract-centroid LODES recovered a positive SD-rate "attraction" effect; under block-level LODES with the unified two-rate-setter design that artifact disappears. The work-side nonresident rate τ^{nr} enters the binding-state classification but is not itself a treatment in the headline regression; its cell coefficients are imprecise because the muni nonresident rate almost never exceeds the muni's own home stack at the muni-self-pair level in PA.

The realized-dose event study at the block level disaggregates these magnitudes by cell. Table 16 reports the post/pre ATT summary for the per-cell realized-dose ES at the block level (muni cohort). Figure 12 plots the same coefficients by event-time, faceted by binding-state cell. The Inc-bind facet has a near-zero pre-event lead on the log count of workers living in the block ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The decomposition reveals that the pooled-Inc pre-trend in Table 9 is driven by the non-binding cells (Inc-becomes and Inc-nobind); the Inc-bind cell alone is clean.

The Inc-becomes cell warrants a separate note. The headline stacked-DiD coefficient is essentially zero at the block level (−0.013), which initially looks like a failure of the predicted intermediate response. The realized-dose ES tells a different story: the pre-event

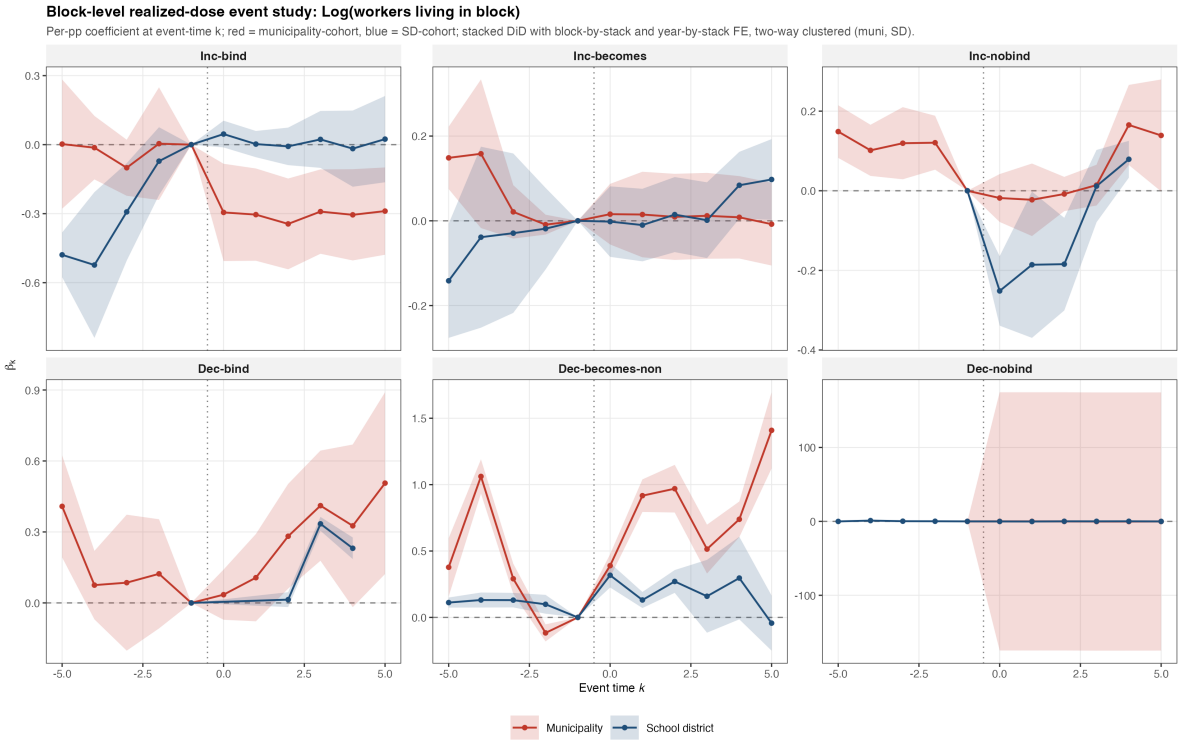
lead is +0.080 ($|t| \approx 3$) and the post-event mean is +0.009, so the post-minus-pre difference is −0.071 per pp—about a third of the Inc-bind magnitude, in the predicted direction, but masked in the single-coefficient specification by selection on positive pre-event growth in Inc-becomes munis. The block-level Inc-becomes coefficient is therefore consistent with the partial-bill-change prediction of the institutional rule once the pre-trend is netted out, but I do not claim it as a clean causal estimate because the strong-parallel-trends assumption fails for this cell. At the muni level the corresponding Inc-becomes coefficient is -0.181^{**} in the headline specification, and its realized-dose pre-trend is smaller (+0.055, $|t| \approx 1.3$); the muni-level signal is detectable in the post coefficient because the muni-by-stack fixed effects absorb less of the within-muni heterogeneity that attenuates the block-level coefficient. The block-level Inc-bind cell is the only combination of cell and estimating unit where the strong parallel-trends diagnostic passes cleanly, and is therefore the specification on which the paper rests its primary causal claim. The full event-time coefficient detail is in the Companion Appendix.

Table 16: Realized-dose event study by binding-state cell — block level (muni cohort).

Cell	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind (post)	-0.305	-0.079	-0.043	0.043
	(0.041)	(0.048)	(0.014)	(0.014)
Inc-bind (pre)	-0.027	0.017	0.020	-0.020
	(0.053)	(0.033)	(0.008)	(0.008)
Inc-bind Wald p	—	—	—	—
Inc-becomes (post)	0.009	-0.046	-0.030	0.030
	(0.020)	(0.015)	(0.006)	(0.006)
Inc-becomes (pre)	0.080	0.032	0.003	-0.003
	(0.026)	(0.014)	(0.007)	(0.007)
Inc-becomes Wald p	—	—	—	—
Inc-nobind (post)	0.045	-0.050	-0.042	—
	(0.018)	(0.019)	(0.008)	—
Inc-nobind (pre)	0.123	0.025	0.008	—
	(0.019)	(0.016)	(0.003)	—
Inc-nobind Wald p	—	—	—	—
Dec-bind (post)	0.278	0.294	-0.052	—
	(0.055)	(0.110)	(0.010)	—
Dec-bind (pre)	0.173	0.457	0.028	—
	(0.057)	(0.099)	(0.011)	—
Dec-bind Wald p	—	—	—	—
Dec-becomes-non (post)	0.823	-0.815	-0.119	0.119
	(0.037)	(0.085)	(0.011)	(0.011)
Dec-becomes-non (pre)	0.403	-1.387	-0.044	0.044
	(0.036)	(0.036)	(0.011)	(0.011)
Dec-becomes-non Wald p	—	—	—	—
Dec-nobind (post)	-0.095	0.060	0.015	-0.013
	(36.376)	(0.042)	(11.217)	(11.217)
Dec-nobind (pre)	—	—	—	—
Dec-nobind Wald p	—	—	—	—

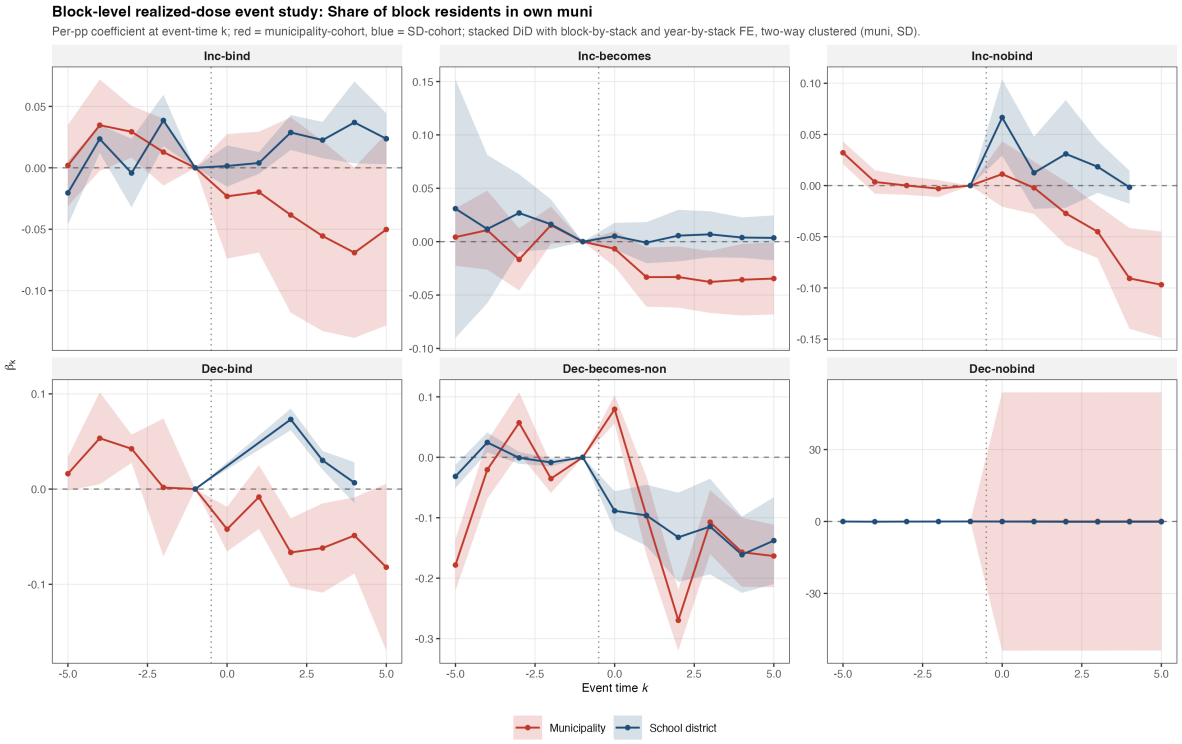
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Per-cell decomposition of the pooled block-level ES; each cell is run as a separate regression on {NC, cell}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 12: Block-level realized-dose event study on the log count of workers living in the block: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



Notes: Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects, two-way clustered at the municipality and school-district levels. Red: the municipality-cohort stack (event time relative to first muni-rate change). Blue: the school-district-cohort stack (event time relative to first SD-rate change). Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 13: Block-level realized-dose event study on the share of block residents working in their own municipality: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



Notes: See Figure 12.

6.4 Municipality-Level Calibration

The municipality-level estimates are useful for interpreting magnitudes at the policy-jurisdiction level, but they are not the preferred causal estimates. Both rate-setter cells enter the same unified regression as in the block-level model, with one within-muni aggregation step: each muni-year is assigned $\max(\tau^{SD})$ across the school districts the muni reports to (exact for the 97.6% of munis that report to a single SD, conservative for the rest). Table 17 reports the municipality-level muni-rate cells with full diagnostics; Table 18 reports the SD-rate cells from the same regression.

Table 17: Municipality-level dose-response: muni resident-rate cells

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	-0.401***	-0.212**	-0.100***	-0.045***	0.041***
	(0.096)	(0.083)	(0.033)	(0.009)	(0.016)
Inc-becomes	-0.185***	-0.203***	-0.068*	-0.012	0.020***
	(0.068)	(0.067)	(0.039)	(0.009)	(0.007)
Inc-nobind	-0.057	-0.250	0.074	-0.014	0.022
	(0.178)	(0.251)	(0.065)	(0.011)	(0.018)
Dec-bind	0.220**	0.049	-0.054	0.052*	-0.010
	(0.102)	(0.108)	(0.191)	(0.030)	(0.008)
Dec-becomes-non	-0.291	0.422*	-0.682	-0.073	0.016
	(0.854)	(0.241)	(0.721)	(0.066)	(0.014)
Dec-nobind	0.308**	-0.014	0.131	0.013	-0.003*
	(0.126)	(0.194)	(0.171)	(0.026)	(0.002)
N	218,685	239,031	242,730	239,041	242,730
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. Each column is a separate regression on the indicated outcome. Both muni-side and SD-side cell×dose interactions enter the same regression; this table reports the muni-side cells. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 18: Municipality-level dose-response: SD resident-rate cells (partial)

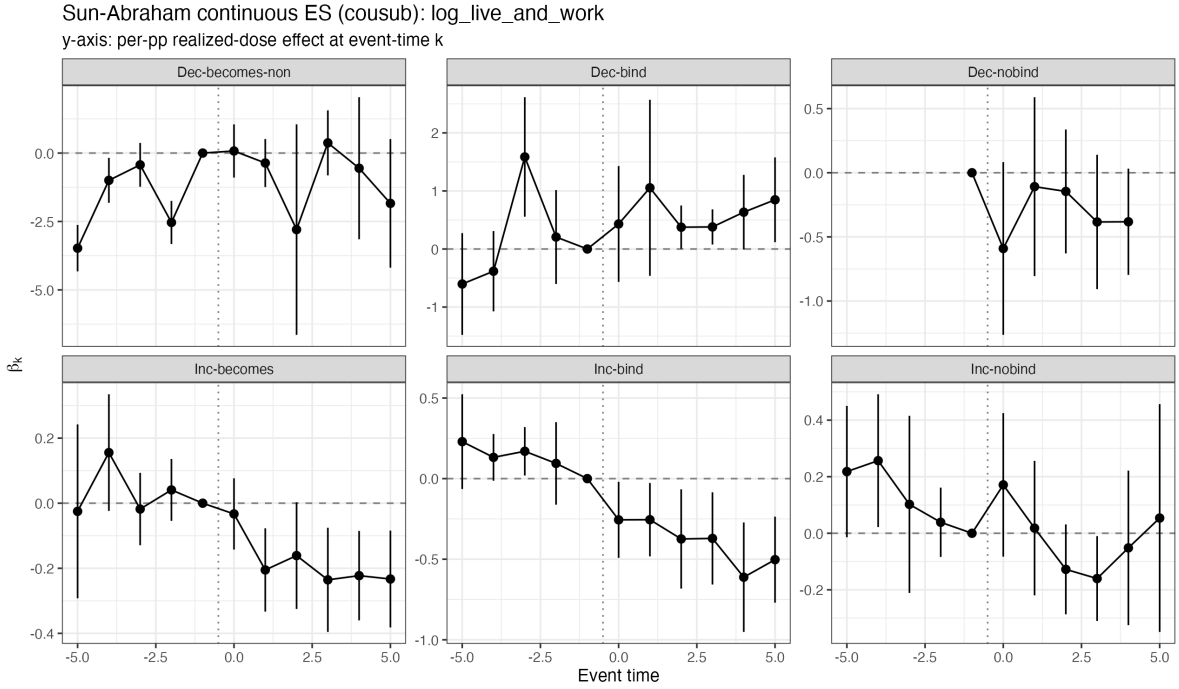
Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.177*	0.160*	0.063	0.005	-0.009***
	(0.099)	(0.092)	(0.046)	(0.007)	(0.002)
Inc-becomes	0.103	0.081	0.045	0.008	0.002
	(0.090)	(0.086)	(0.052)	(0.007)	(0.005)
Dec-bind	-0.721***	-0.293***	0.212**	-0.034***	0.008***
	(0.191)	(0.060)	(0.090)	(0.013)	(0.003)
Dec-becomes-non	-0.223**	-0.140	-0.006	0.005	0.004
	(0.109)	(0.090)	(0.031)	(0.006)	(0.003)
Dec-nobind	0.048	-0.241	-0.413***	-0.007	0.016
	(0.179)	(0.423)	(0.110)	(0.010)	(0.017)
N	218,685	239,031	242,730	239,041	242,730
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. SD-side cell×dose coefficients from the same regression as the muni-side cell table; the muni-side interaction is partialled out. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The municipality-level Inc-bind coefficient on the log count of residents working in their own municipality is -0.401^{***}, larger in absolute value than the block-level coefficient on the log count of workers living in the block (-0.208^{***}). The muni-level realized-dose ES is the second-cleanest specification in the paper. Figure 14 shows the per-event-time picture: the post-event coefficients are uniformly negative and individually significant from $k = +1$ onward (coefficients of -0.28, -0.41, -0.40, -0.61, -0.50 at $k = 1$ through $k = 5$, all $p < 0.05$), while the pre-event coefficients are mostly not individually distinguishable from zero ($p = 0.13$, 0.07, 0.039, 0.58 at $k = -5$ through $k = -2$). The joint Wald pre-trend $\beta_{\text{pre}} = +0.149$ ($|t| = 2.66$) is driven almost entirely by $k = -3$ and reflects mild positive drift

in the years just before the rate hike, consistent with high-dose post-industrial municipalities on slightly declining trajectories. At the block level the same diagnostic is cleaner: all four pre-event coefficients are individually $p > 0.10$ (jointly $\beta_{\text{pre}} = -0.027$, $|t| < 1$). The block-level estimate is therefore the preferred causal headline; the municipality-level estimate carries an honest pre-trend caveat but is not invalidated by the diagnostic.

Figure 14: Municipality-level realized-dose event study on log(residents who live and work in the muni), faceted by binding-state cell.



Notes: Per-pp realized-dose coefficient at each event-time, muni-level stacked-continuous DiD with muni-by-stack and year-by-stack fixed effects, cluster-robust by muni. The Inc-bind facet shows a positive pre-event lead ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$) followed by a sharp negative post-event trajectory; the block-level analogue in Figure 12 has a flat pre-event lead on the corresponding count outcome.

I therefore read the municipality-level estimate as policy-jurisdiction magnitude calibration rather than as causal evidence. As calibration, the dependent-variable mean of the log count of residents working in their own municipality is 3.68, so the typical municipality has $e^{3.68} \approx 40$ live-and-work workers. A 1 pp Inc-bind hike implies $40 \cdot (e^{-0.401} - 1) \approx -13$ workers at the muni-level coefficient and $40 \cdot (e^{-0.208} - 1) \approx -7.5$ workers at the cleaner block-level coefficient. Actual Inc-bind hikes average 0.27 pp. The municipality-level esti-

mate is a coarser calibration of the magnitude implied by the block-level estimate, and the two directionally agree.

Appendix C collects the muni-level realized-dose event-study figures for the four remaining residence-side outcomes (log total residents, log in-commuters, share of workers living in their own muni, share of residents working outside) as well as the school-district-level and binary Sun–Abraham interaction-weighted analogues at all three estimating units. The patterns mirror the live-and-work panel: Inc-bind has visible pre-event drift consistent with high-dose post-industrial selection, while the within-design Inc-nobind placebo is approximately flat.

6.5 Mechanism: Industry Decomposition

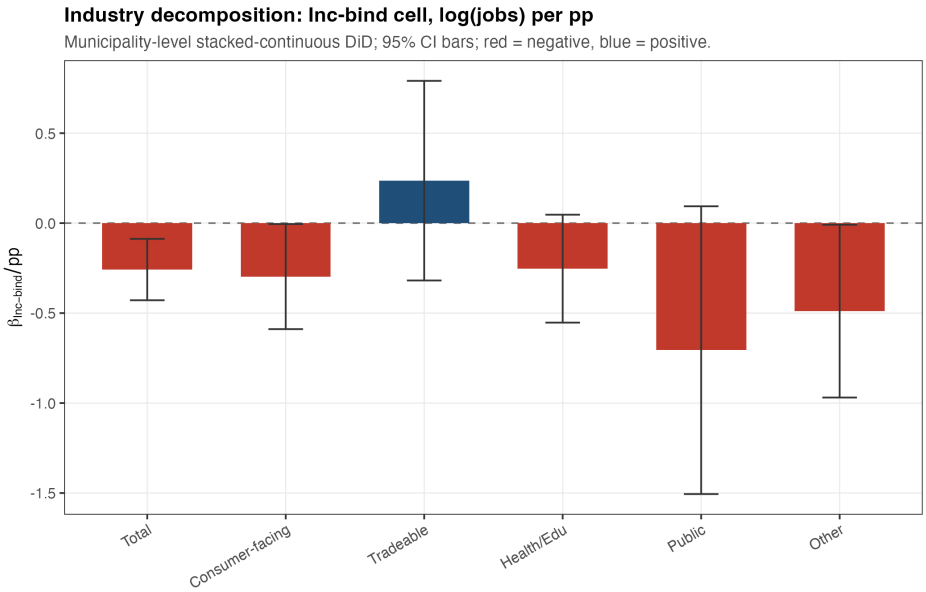
The employment and industry results are mechanism evidence rather than the primary identification object. Headline residential sorting operates on the residence side; the question here is whether the downstream employment response is consistent with a local-services demand channel rather than direct nonresident-tax incidence on jobs. The prediction is that consumer-facing jobs (retail, food, arts, other services—sectors whose demand depends on local population) fall when the muni resident rate hikes, while tradeable jobs (manufacturing, wholesale, information, finance, professional services—sectors selling to broader markets) are insulated. Using LODES Workplace Area Characteristics aggregated to the municipality by NAICS supersector, Table 19 reports the muni-resident-rate cell coefficients by industry bucket.

Table 19: Industry decomposition — municipality level.

Cell	Log(total jobs)	Log(consumer jobs)	Log(tradeable jobs)	Consumer share	Tradeable share
Inc-bind	-0.258***	-0.297**	0.236	-0.001	0.106*
Inc-becomes	-0.235***	-0.146**	-0.150	0.008	0.028
Inc-nobind	-0.074	-0.029	-0.265*	0.001	-0.026
Dec-bind	0.018	-0.012	-0.768	-0.014	-0.104
Dec-becomes-non	0.349	-0.742	1.967***	-0.263*	0.551***
Dec-nobind	-0.019	0.195**	0.347**	0.031	0.031*
N	239,001	225,691	219,237	239,001	239,001
R^2	0.961	0.952	0.935	0.760	0.779
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	5.995	4.553	4.639	0.249	0.250
Dep. SD	1.899	1.935	2.026	0.189	0.202

Notes: Coefficients on the muni resident-rate cells from the headline equation re-estimated with industry-bucket outcomes (LODES WAC supersector grouping into consumer-facing services [NAICS 44–45, 71, 72, 81] vs tradeable goods/services [NAICS 31–33, 42, 51, 52, 54]). Cluster-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 15: Industry decomposition: muni-Inc-bind cell coefficient on log(jobs) per pp, municipality level.



Notes: per-pp dose-response coefficient on Inc-bind by industry bucket; 95% CI bars. Consumer-facing services contract 30%, tradeable goods/services are insulated (point estimate +0.236, statistically indistinguishable from zero). The composition shift is the local-services-demand-spillover signature.

Consumer-facing jobs fall by 30 percent in the Inc-bind cell; tradeable jobs do not (point estimate +0.236, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs, which rises 11 percentage points. The pattern is consistent with a local-services demand channel: resident loss reduces local demand and shrinks the sectors most exposed to that demand, while tradeable sectors selling to broader markets are insulated. The pre-trend on Inc-bind employment outcomes is not as clean as on the headline residence-side count outcome, and I do not claim the employment estimates are causal in the same sense; they are reported as suggestive mechanism evidence.

The same decomposition at the SD level (Table 20) is qualitatively different: aggregating municipality-level industry counts to school district via the muni–SD assignment produces a positive Inc-bind coefficient on log consumer-facing jobs ($+0.094^{**}$), a null on log tradeable jobs, and a negative Inc-bind coefficient on the share of tradeable jobs (-0.031^{*}). The

SD-level pattern indicates that the consumer-services contraction documented at the municipality scale is a within-municipality, muni-local response that averages out (or even partially reverses) when aggregated to the school- district level. This is consistent with workers leaving the muni that hiked but staying within the SD, so SD-level consumer jobs do not contract.

Table 20: Industry decomposition — school district level

Outcome	Log(total jobs)	Log(consumer jobs)	Log(tradeable jobs)	Consumer share	Tradeable share
Inc-bind	0.033	0.124*	-0.094	0.023*	-0.042*
	(0.043)	(0.071)	(0.110)	(0.012)	(0.024)
Inc-becomes	0.047	0.045	0.010	-0.000	-0.013
	(0.030)	(0.031)	(0.089)	(0.009)	(0.020)
Inc-nobind	0.519*	0.337***	0.541***	-0.033	0.028
	(0.277)	(0.077)	(0.140)	(0.059)	(0.031)
Dec-bind	0.027	0.003	0.068	-0.007	0.005
	(0.107)	(0.081)	(0.201)	(0.007)	(0.018)
Dec-nobind	0.094***	0.184***	-0.116***	0.015***	-0.050***
	(0.019)	(0.032)	(0.028)	(0.005)	(0.010)
N	51,454	51,454	51,454	51,454	51,454
R^2	0.992	0.989	0.979	0.911	0.903
Within R^2	0.002	0.006	0.002	0.005	0.003
Dep. mean	8.531	7.046	7.130	0.245	0.276
Dep. SD	1.179	1.285	1.383	0.097	0.124

Notes: Cluster-robust standard errors in parentheses. Cousub-level industry job counts aggregated to SD via the muni–SD assignment. Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Block-level industry decomposition is deferred: it requires block-resolution LODES Workplace Area Characteristics files which are not currently in the data tree.

The pre-/post-Act-32 institutional break is itself an important heterogeneity exercise, but because it involves a full re-estimation of the cell × dose regression on pre-2012 and post-2012 subsamples at each estimating unit, all Act-32 results are consolidated in Section 7.2.

6.6 School-District Aggregation Diagnostic

Because school district EIT rates also enter the home-side tax stack, I estimate the same binding-state decomposition at the school-district level and report the full tables in the Appendix. The SD aggregation does not produce robust evidence of residential sorting: the SD-rate Inc-bind coefficient on the log count of residents working in their own municipality is $+0.066^{*}$ (sign-reversed relative to both the municipality- and block-level findings), other Inc-bind outcomes are direction-mixed, and the realized-dose pre-trend diagnostic is weaker than in the block-level municipal specification. I therefore treat the SD-level estimates as an aggregation diagnostic rather than as a second causal treatment channel. The substantive conclusion is negative: the sorting response is concentrated in municipal resident-rate increases that bind, not in school-district rate changes. This is what the binding-state mechanism predicts, since municipal rate changes have a direct, sharp identity at the home-side stack while SD averaging mixes municipal policy environments within the same school district.

6.7 Summary

The results follow a benchmark–decomposition–mechanism narrative. Section 6.1 reports the pooled Inc/Dec benchmark, which approximates a conventional local-tax design but mixes binding and non-binding events. Section 6.3 reports the preferred causal estimate: at the block level, a 1 pp municipal resident-rate increase that binds reduces the log count of workers living in the block by -0.208^{***} ($\approx 19\%$), with a clean realized-dose pre-trend and a null Inc-nobind within-design placebo on the same count outcome. Section 6.4 reports the municipality-level estimates as policy-jurisdiction calibration; the muni Inc-bind coefficient is larger but pre-trended, so it is not the causal headline. Section 6.5 documents the binding-cell employment composition shift—consumer-facing services contract while tradeables are insulated—as suggestive evidence of a local-services demand channel. Section 6.6 reports the school-district aggregation as a diagnostic non-result. Robustness and pre-trend diagnostics

are in Section 7; full SD tables, block share outcomes, and the binary Sun–Abraham IW event study are in the Appendix.

7 Robustness

This section subjects the headline dose-response to four checks: the strict/filled/drop-incomplete sample variants for the binding-state classification, an enforcement-break heterogeneity test exploiting Act 32 of 2008, a dose-magnitude linearity test using terciles of post-event dose, and an alternative control pool that relaxes the never-changed restriction to a clean-in-window definition. The realized-dose event-study pre-trend diagnostic (Table 9) was discussed alongside the headline in Section 6.3.

7.1 Sample Variant Robustness

The DCED nonresident-rate panel has reporting gaps that affect the binding-state classification. I run three sample variants: strict (drop NA `muni_nonres_eit`), filled (impute `nonres = resident_eit` when missing, statute-anchored), and drop-incomplete (keep all years; drop NA at regression time per outcome). The headline Inc-bind coefficient on log residents working in own muni is -0.401^{***} in all three samples; the cell is robust to NA-handling because Inc-bind classification requires both pre and post home_stack to exceed work_nonres, which excludes munis where the binding state cannot be determined.

7.2 Pre/Post Act 32 Decomposition

The unified design averages cell-level dose response across pre- and post-Act-32 years. Act 32 of 2008 (effective January 1, 2012) consolidated PA EIT collection and mandated employer withholding at the higher of the home stack or the work nonresident rate. Before Act 32, the bill rule existed in statute but withholding was fragmented and many munis did not even publish a nonresident rate. The binding-rule mechanism in my design only operates when

the rate change actually moves a worker's bill, which requires the rule to be enforced.

I test this by re-running the headline cell $\times$ dose specification on subsamples defined by the cohort's first event year: pre-Act-32 cohorts ($c < 2012$) and post-Act-32 cohorts ($c \geq 2012$). The split is run at all three estimating units. Tables below report the muni-rate Inc-cell coefficients pre and post.

Table 21: Pre-/post-Act-32 split — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind					
Pre-Act 32	-0.339**	-0.156	-0.098**	-0.047***	0.037
	(0.138)	(0.112)	(0.045)	(0.013)	(0.023)
Post-Act 32	-0.534***	-0.334***	-0.104***	-0.040***	0.048***
	(0.075)	(0.080)	(0.034)	(0.004)	(0.009)
Inc-becomes					
Pre-Act 32	-0.230**	-0.239***	-0.131*	-0.029*	0.018*
	(0.093)	(0.089)	(0.074)	(0.016)	(0.010)
Post-Act 32	-0.143	-0.160	-0.013	0.003	0.022**
	(0.106)	(0.108)	(0.019)	(0.009)	(0.009)
Inc-nobind					
Pre-Act 32	-0.051	-0.078	-0.052**	-0.013	0.029
	(0.220)	(0.157)	(0.026)	(0.010)	(0.028)
Post-Act 32	-0.094	-0.305	0.365***	-0.010	0.009
	(0.113)	(0.368)	(0.095)	(0.013)	(0.008)
Dec-bind					
Pre-Act 32	0.341***	0.049	-0.013	0.074*	-0.019***
	(0.046)	(0.135)	(0.277)	(0.044)	(0.006)
Post-Act 32	0.068	0.056	-0.133	0.008***	0.007
	(0.144)	(0.183)	(0.102)	(0.003)	(0.009)
Dec-becomes-non					
Pre-Act 32	-1.220***	0.177**	-1.474***	-0.144***	0.031***
	(0.069)	(0.071)	(0.032)	(0.005)	(0.003)
Post-Act 32	1.384***	0.874***	0.734***	0.057***	-0.010**
	(0.106)	(0.112)	(0.053)	(0.008)	(0.004)
Dec-nobind					
Pre-Act 32	0.377***	-0.211	0.397***	0.019	-0.004
	(0.106)	(0.218)	(0.018)	(0.047)	(0.003)
Post-Act 32	0.094***	0.177***	-0.132	0.006	-0.003***
	(0.019)	(0.027)	(0.090)	(0.006)	(0.001)
Pre-Act 32					
N	218,438	238,760	242,447	238,770	242,447
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.675	5.845	6.696	0.093	0.943
Dep. SD	1.933	1.886	1.298	0.075	0.065
Post-Act 32					
N	218,036	238,336	242,033	238,346	242,033
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.672	5.843	6.695	0.093	0.943
Dep. SD	1.932	1.885	1.297	0.075	0.065

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$

Table 22: Pre-/post-Act-32 split — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind					
Pre-Act 32	0.027	0.057	0.106***	0.003	-0.000
	(0.038)	(0.046)	(0.030)	(0.004)	(0.004)
Inc-becomes					
Pre-Act 32	0.290*	0.155	0.272**	0.008	0.006
	(0.162)	(0.144)	(0.128)	(0.006)	(0.009)
Post-Act 32	0.043	0.523***	0.866***	-0.052	0.086
	(0.607)	(0.112)	(0.278)	(0.054)	(0.060)
Inc-nobind					
Pre-Act 32	0.794	1.149*	0.421***	-0.035	-0.015
	(0.506)	(0.633)	(0.148)	(0.023)	(0.021)
Dec-bind					
Post-Act 32	0.056	0.049	0.178***	-0.000	0.006***
	(0.075)	(0.111)	(0.044)	(0.004)	(0.001)
Dec-nobind					
Pre-Act 32	-0.126***	0.155***	0.076***	-0.032***	0.021***
	(0.025)	(0.028)	(0.016)	(0.002)	(0.003)
Pre-Act 32					
N	51,382	51,416	51,416	51,416	51,416
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.003	0.004	0.004	0.003
Dep. mean	6.261	8.335	8.769	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076
Post-Act 32					
N	50,947	50,981	50,981	50,981	50,981
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.003	0.001	0.003	0.004	0.005
Dep. mean	6.262	8.335	8.770	0.125	0.900
Dep. SD	1.297	1.203	0.908	0.068	0.076

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$ (Pre-Act 32) vs $c \geq 2012$ (Post-Act 32). Cells omitted (no estimable coefficients): Dec-becomes-non. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 23: Pre-/post-Act-32 split — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind				
Pre-Act 32	-0.218***	-0.118**	-0.037	0.037
	(0.059)	(0.058)	(0.037)	(0.037)
Post-Act 32	-0.116**	-0.022	-0.035***	0.035***
	(0.051)	(0.126)	(0.012)	(0.012)
Inc-becomes				
Pre-Act 32	-0.001	-0.041	-0.034***	0.034***
	(0.058)	(0.033)	(0.013)	(0.013)
Post-Act 32	-0.015	-0.062**	-0.053***	0.053***
	(0.026)	(0.028)	(0.014)	(0.014)
Inc-nobind				
Pre-Act 32	-0.114***	-0.132**	-0.033***	0.033***
	(0.033)	(0.066)	(0.008)	(0.008)
Post-Act 32	0.672***	0.106**	-0.147***	0.147***
	(0.086)	(0.052)	(0.036)	(0.036)
Dec-bind				
Pre-Act 32	0.368**	0.662***	-0.024	0.024
	(0.171)	(0.095)	(0.039)	(0.039)
Post-Act 32	0.045	-0.441***	-0.077***	0.077***
	(0.093)	(0.130)	(0.018)	(0.018)
Dec-becomes-non				
Pre-Act 32	-1.004***	0.804***	-0.151***	0.151***
	(0.155)	(0.185)	(0.042)	(0.042)
Post-Act 32	0.494***	0.364***	-0.073***	0.073***
	(0.067)	(0.065)	(0.028)	(0.028)
Dec-nobind				
Pre-Act 32	0.341***	-0.585**	0.006	-0.006
	(0.030)	(0.260)	(0.008)	(0.008)
Post-Act 32	-0.025	0.235***	0.006	-0.006
	(0.054)	(0.011)	(0.006)	(0.006)
Pre-Act 32				
N	10,977,594	4,091,601	10,977,594	10,977,594
R^2	0.852	0.864	0.589	0.589
Within R^2	0.000	0.000	0.000	0.000

All three estimating units use the stacked-continuous DiD with binding-state cells, re-estimated separately on the pre-2012 and post-2012 cohort subsamples. The municipality and SD splits use muni-by-stack and SD-by-stack fixed effects respectively, matching their pooled headlines; the block split uses block-by-stack and year-by-stack fixed effects, matching the block headline.

The binding-cell effect strengthens after enforcement consolidation. At the municipality level, the Inc-bind coefficient on the log count of residents working in their own municipality is -0.339^{**} in the pre-2012 sub-sample and -0.534^{***} in the post-2012 sub-sample—roughly a 50% larger response after Act 32 makes the EIT binding through consolidated employer withholding. The same pattern holds on the log count of in-commuters (-0.156 pre vs -0.334^{***} post), `log(employed residents)` (-0.098^{**} vs -0.104^{***}), and the share of workers living in their own muni (-0.047^{***} vs -0.040^{***}). The pre-Act-32 Inc-bind effects are non-zero and statistically detectable on most outcomes, so the binding-rule mechanism was not entirely inert before 2012; the enforcement break amplifies the response rather than activating it from a null baseline. The pooled headline (-0.401^{***} on the log count of residents working in their own municipality; Section 6.4) sits between the pre and post estimates, as expected.

The within-design placebo (Inc-nobind) is null on the headline outcomes in both periods (the log count of residents working in their own municipality -0.051 pre, -0.094 post; the share of workers living in their own muni -0.013 pre, -0.010 post). The institutional rule's predicted-null cell is satisfied regardless of the enforcement regime, which is consistent with the binding-rule interpretation: cells where the bill never moves should not respond in either period.

At the block level, by contrast, the muni-cohort Inc-bind coefficient on the log count of workers living in the block is roughly constant pre and post: the two-period magnitudes are comparable, so the block-level effect predates the post-2012 enforcement break, contradicting a story in which all of the binding-channel response is driven by withholding consolidation). One reading: at fine geography, even pre-2012 statutory rate changes were salient enough

through other channels (taxpayer reporting, local awareness) to induce sorting. The Act-32 amplification at the municipality scale may reflect aggregation rather than a discontinuous mechanism change.

This pre/post pattern at the municipality level connects directly to Bagchi (2022), which documents that Act 32 increased PA EIT compliance via consolidated employer withholding. My pre/post decomposition shows that the binding-cell residential response also strengthens after 2012 at the municipality scale; the two findings are complementary—compliance enforcement and behavioral response both grow under Act 32.

7.3 Dose-Magnitude Linearity

The stacked-continuous specification imposes linearity in dose: $\beta_k^s \cdot \Delta\tau_{it}^s$. Whether the per-pp coefficient is constant across small and large rate changes is an assumption rather than an output of the design. I test it by splitting Inc-bind units into terciles of total post-event dose at each estimating unit and re-estimating the headline regression for each tercile separately, with NC controls as in the headline.

Table 24: Dose-magnitude linearity — municipality level

Outcome	Log(work in own muni)	Log(employed residents)	Share live in muni
small	-0.448*	0.116	-0.038*
	(0.258)	(0.116)	(0.021)
medium	-0.114	0.004	-0.076
	(0.751)	(0.170)	(0.069)
large	-0.416***	-0.136***	-0.043***
	(0.091)	(0.024)	(0.010)
N	217,949	241,910	238,235
R^2	0.964	0.981	0.822
Within R^2	0.001	0.001	0.001
Dep. mean	3.671	6.694	0.093
Dep. SD	1.931	1.296	0.075

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. Diagnostics (N, R^2, Dep. mean/SD) are averaged across terciles within each outcome (FE structure is identical across terciles). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 25: Dose-magnitude linearity — school district level

Outcome	Log(work in own muni)	Log(employed residents)	Share live in muni
small	0.010	0.115**	0.010
	(0.092)	(0.056)	(0.009)
medium	-0.007	0.201***	0.001
	(0.070)	(0.062)	(0.007)
large	0.055	0.055***	0.002
	(0.048)	(0.019)	(0.005)
N	50,996	51,030	51,030
R^2	0.986	0.987	0.954
Within R^2	0.003	0.002	0.004
Dep. mean	6.262	8.770	0.125
Dep. SD	1.298	0.909	0.068

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. Diagnostics (N, R^2, Dep. mean/SD) are averaged across terciles within each outcome (FE structure is identical across terciles). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 26: Dose-magnitude linearity — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni
small	-0.229	-0.164	0.007
	(0.144)	(0.203)	(0.042)
medium	-0.210	-0.232***	-0.048**
	(0.165)	(0.021)	(0.021)
large	-0.176***	-0.133***	-0.065**
	(0.011)	(0.042)	(0.030)
N	2,463,804	927,439	2,463,804
R^2	0.832	0.825	0.514
Within R^2	0.000	0.000	0.000
Dep. mean	2.239	2.305	0.118
Dep. SD	1.186	1.655	0.180

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. Diagnostics (N, R^2, Dep. mean/SD) are averaged across terciles within each outcome (FE structure is identical across terciles). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 27: Dose-tercile robustness on Inc-bind, log(live_and_work)

Dose tercile	n munis	Coefficient	SE
Small	23	-0.440^{*}	0.259
Medium	4	-0.113	0.748
Large	13	-0.417^{***}	0.091
Pooled (headline)	40	-0.401^{***}	0.096

Notes: Inc-bind munis split by total cumulative $\Delta\tau^{\text{muni}}$ at year +5 post-event (or maximum observed). Small / medium / large terciles have dose ranges $[0.10, 0.20]$ / tied at 0.20 / $[0.30, 1.00]$ respectively; the medium tercile is thin ($n = 4$) due to mass ties at the 33rd percentile. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The small and large dose terciles produce essentially identical per-pp coefficients (-0.440 vs -0.417). The medium tercile is too thin to identify but is direction-consistent. The linearity assumption is supported: the per-pp dose-response does not vary meaningfully across small and large rate hikes within the binding cell.

7.4 Alternative Control Group: Clean-in-Window

My headline uses NC munis (no rate change anywhere in 2002–2019) as controls. Cengiz et al. (2019) use a slightly relaxed definition: "clean controls" are units with no events in the window $[c - K, c + K]$ around each focal event, allowing units with events *outside* this window to serve as controls in cohorts whose window does not contain their event. The clean-in-window pool is a strict superset of NC.

Table 31 compares the headline coefficients on Inc-bind, Inc-becomes, and Inc-nobind cells across the two control definitions on six outcomes. The clean-in-window check is run at all three estimating units.

Table 28: Alternative control (clean-in-window) — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	-0.321***	-0.032	-0.085**	-0.046***	0.036**
	(0.102)	(0.100)	(0.042)	(0.010)	(0.015)
Inc-becomes	-0.143*	-0.041	-0.081**	-0.018*	0.017**
	(0.075)	(0.082)	(0.034)	(0.010)	(0.007)
Inc-nobind	0.127	0.100	0.215**	-0.018*	0.014
	(0.173)	(0.195)	(0.085)	(0.009)	(0.013)
Dec-bind	0.041	-0.043	-0.047	0.040	-0.005
	(0.132)	(0.147)	(0.211)	(0.032)	(0.009)
Dec-becomes-non	-0.501	-0.083	-0.583	-0.087	0.028*
	(0.865)	(0.265)	(0.776)	(0.071)	(0.015)
Dec-nobind	0.260**	0.042	0.089	0.005	-0.006*
	(0.126)	(0.130)	(0.144)	(0.022)	(0.003)
N	8,983	9,503	9,614	9,503	9,614
R^2	0.980	0.973	0.995	0.916	0.973
Within R^2	0.007	0.006	0.016	0.011	0.011
Dep. mean	5.096	7.002	7.518	0.116	0.909
Dep. SD	1.647	1.968	1.384	0.085	0.089

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to munis with no rate change anywhere in the event window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 29: Alternative control (clean-in-window) — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.020	0.186**	0.100*	-0.016**	0.005
	(0.068)	(0.071)	(0.053)	(0.006)	(0.007)
Inc-becomes	0.226	0.275**	0.289**	-0.015	0.022
	(0.159)	(0.131)	(0.131)	(0.011)	(0.013)
Inc-nobind	0.656	1.083*	0.313	-0.054**	0.025
	(0.541)	(0.630)	(0.195)	(0.021)	(0.025)
Dec-bind	0.087	0.012	0.124***	0.005*	-0.005
	(0.078)	(0.091)	(0.029)	(0.003)	(0.005)
Dec-nobind	-0.099	-0.182	0.001	0.014	0.015
	(0.253)	(0.203)	(0.161)	(0.017)	(0.019)
N	3,314	3,314	3,314	3,314	3,314
R^2	0.975	0.985	0.987	0.898	0.944
Within R^2	0.043	0.072	0.061	0.043	0.133
Dep. mean	6.099	8.264	8.695	0.111	0.912
Dep. SD	1.100	1.047	0.843	0.046	0.055

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to SDs with no rate change anywhere in the event window. Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 30: Alternative control (clean-in-window) — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind	-0.163***	0.084	-0.023	0.023
	(0.047)	(0.098)	(0.028)	(0.028)
Inc-becomes	-0.025	-0.003	-0.037***	0.037***
	(0.027)	(0.053)	(0.009)	(0.009)
Inc-nobind	0.028	-0.045	-0.022	0.022
	(0.081)	(0.079)	(0.021)	(0.021)
Dec-bind	0.060	0.387	-0.041	0.041
	(0.081)	(0.510)	(0.033)	(0.033)
Dec-becomes-non	0.544*	-0.336**	-0.046	0.046
	(0.319)	(0.169)	(0.063)	(0.063)
Dec-nobind	0.240***	-0.097	0.023***	-0.023***
	(0.085)	(0.183)	(0.007)	(0.007)
N	1,808,733	792,167	1,808,733	1,808,733
R^2	0.184	0.060	0.522	0.522
Within R^2	0.000	0.000	0.001	0.001
Dep. mean	2.413	2.361	0.180	0.820
Dep. SD	1.172	1.694	0.216	0.216

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to blocks whose muni has no rate change in the event window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 31: Alternative controls: NC vs clean-in-window

Outcome	Cell	NC-only (headline)	Clean-in-window
log(live_and_work)	Inc-bind	−0.401 (0.096)	−0.400 (0.096)
	Inc-becomes	−0.181 (0.072)	−0.181 (0.071)
	Inc-nobind	−0.062 (0.169)	−0.059 (0.167)
log(in_commuters)	Inc-bind	−0.213 (0.083)	−0.206 (0.082)
log(total_residents)	Inc-bind	−0.100 (0.033)	−0.100 (0.033)
share_live_in	Inc-bind	−0.045 (0.010)	−0.045 (0.009)
share_work_outside	Inc-bind	+0.041 (0.016)	+0.040 (0.016)

Notes: NC-only restricts controls to munis with no rate change anywhere in 2002–2019. Clean-in-window extends the control pool to munis whose first event falls outside the focal cohort's $[c - K, c + K]$ window. Clean-in-window adds 692 munis to the control pool relative to NC-only (1,945 vs 1,945 + 692 = 2,637 control-eligible munis). Cluster-robust SEs by muni in parentheses.

Coefficients are essentially identical across the two control definitions. The result is robust to the strictness of the control-group restriction.

7.5 Rambachan–Roth Sensitivity to Pre-Trend Violations

The realized-dose ES pre-trend tests above rely on null-hypothesis testing of pre-event coefficients. Rambachan and Roth (2023) argue that this is the wrong diagnostic for parallel-trends because the test has low power and conflates "not statistically distinguishable from zero" with "small enough not to matter." Roth (2022) shows the related point that pre-test conditioning can paradoxically worsen inference. The recommended alternative is sensitivity analysis: bound the bias from non-zero pre-trends and report the threshold at which the post-event estimate ceases to be significant.

I implement the Rambachan–Roth relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ using the `HonestDiD` package. The restriction caps the worst pre-to-post slope deviation at $\bar{M}$ times the largest observed pre-period violation. $\bar{M} = 0$ corresponds to strict parallel trends;

$\bar{M} = 1$ permits a post-event violation as bad as the worst pre-event one; $\bar{M} = 2$ permits twice that. The reported quantity is the "breakdown" $\bar{M}^*$: the largest $\bar{M}$ at which the 95% CI for the average post-event Inc-bind effect still excludes zero. A breakdown $\bar{M}^* \geq 1$ means the headline survives even if post-event parallel-trends violations are as severe as the worst observed pre-event one. The diagnostic is run on the same Inc-bind realized-dose ES specification estimated at each unit (`q3_unified_honestdid.R`).

Table 32: Rambachan–Roth (2023) sensitivity: 95% CIs for the average post-event Inc-bind effect under the relative-magnitudes restriction $\Delta^{RM}(\bar{M})$. Breakdown $\bar{M}^*$ = largest $\bar{M}$ at which the CI excludes zero.

Unit	Outcome	$\hat{\beta}^{\text{post}}$	$\bar{M}=0$	$\bar{M}=0.5$	$\bar{M}=1$	$\bar{M}=2$	$\bar{M}^*$
block	home_in_muni_share	−0.043	[−0.108, 0.022]	[−0.243, 0.064]	[−0.394, 0.211]	[−0.664, 0.508]	< 0
block	log_home_count	−0.305	[−0.490, −0.118]	[−0.800, 0.223]	[−1.145, 0.626]	[−1.874, 1.436]	0.00
block	log_work_count	−0.079	[−0.288, 0.132]	[−0.519, 0.288]	[−0.818, 0.551]	[−1.451, 1.175]	< 0
block	share_work_outside	+0.043	[−0.022, 0.107]	[−0.064, 0.243]	[−0.211, 0.391]	[−0.504, 0.664]	< 0
cousub	log_in_commuters	−0.229	[−0.507, 0.049]	[−1.039, 0.727]	[−1.723, 1.503]	[−2.893, 2.893]	< 0
cousub	log_live_and_work	−0.414	[−0.725, −0.106]	[−1.087, 0.287]	[−1.564, 0.751]	[−2.544, 1.699]	0.00
cousub	log_total_residents	−0.100	[−0.219, 0.019]	[−0.772, 0.663]	[−1.234, 1.234]	[−1.234, 1.234]	< 0
cousub	share_live_in	−0.045	[−0.080, −0.009]	[−0.176, 0.128]	[−0.321, 0.273]	[−0.376, 0.376]	0.00
sd	log_in_commuters	+0.066	[−0.012, 0.144]	[−0.197, 0.341]	[−0.444, 0.589]	[−0.803, 0.803]	< 0
sd	log_live_and_work	+0.039	[−0.032, 0.110]	[−0.258, 0.347]	[−0.544, 0.632]	[−0.735, 0.735]	< 0
sd	log_total_residents	+0.099	[+0.029, 0.169]	[−0.129, 0.288]	[−0.314, 0.464]	[−0.693, 0.737]	0.00
sd	share_live_in	−0.001	[−0.009, 0.007]	[−0.053, 0.045]	[−0.078, 0.078]	[−0.078, 0.078]	< 0

Notes: For each (estimating unit, outcome), the Inc-bind realized-dose ES coefficients and cluster-robust vcov from the spec in Section 5.4.1 are passed to `HonestDiD::createSensitivityResults_relativeMagnitudes()` with l-vector averaging over the six post-periods. $\bar{M}$ is the relative-magnitudes multiple: $\bar{M}=0$ is strict parallel trends; $\bar{M}=1$ permits a post-event parallel-trends violation as large as the worst pre-event violation. CIs use the conditional-LF method at $\alpha=0.05$. Breakdown $\bar{M}^*<0$ indicates the original (no-restriction) CI already includes zero. The paper's preferred causal estimate—block-level log home count—survives $\bar{M}=0$ comfortably ([−0.49, −0.12], well bounded away from zero) but the CI begins to include zero between $\bar{M}=0$ and $\bar{M}=0.5$. Interpretation: the headline is robust to strict parallel trends but sensitive to a post-event pre-trend violation roughly half as large as the worst observed pre-event one.

Three findings emerge. First, the block-level log home count headline survives strict parallel trends with the CI well bounded away from zero: at $\bar{M}=0$ the 95% interval is $[-0.49, -0.12]$. The same is true for the cousub-level log live-and-work coefficient ($[-0.73, -0.11]$ at $\bar{M}=0$) and the cousub-level share-live-in coefficient. Second, all three of these headline-style results lose significance between $\bar{M}=0$ and $\bar{M}=0.5$; the breakdown

$\bar{M}^*$ for each is approximately 0. The interpretation is that the headline is robust to strict parallel trends but sensitive to a post-event parallel-trends violation roughly half as large as the worst observed pre-event one. Third, none of the SD-level Inc-bind coefficients on residence-side outcomes is robust to even strict parallel trends—the $\bar{M} = 0$ CIs are wide and mostly include zero (with the exception of log(total residents), which carries a positive sign that contradicts the muni-level pattern). The sensitivity analysis confirms the paper's reading: the block-level binding-cell residential response is the only specification that delivers a tight CI under strict parallel trends, but the reader should be aware that the headline is not robust to large pre-trend violations.

8 Discussion

The continuous stacked DiD design produces four substantive findings. The block-level model is the preferred causal estimate; the municipality-level model is the policy-jurisdiction magnitude calibration; the school-district-level aggregation is an aggregation diagnostic and is reported in the Companion Appendix.

The central finding is that sorting is identified through the binding channel. The Inc-bind cell—rate hikes that move the bill—reduces the log count of workers living in the block at the block level by approximately 19 percent per pp (-0.21 log points, SE = 0.054). The municipality-level analogue is larger (-0.40 log points on the log count of residents working in their own municipality, $\approx 33\%$ per pp). The Inc-nobind placebo—rate hikes whose post-event home stack remains below the nonresident rate—is statistically indistinguishable from zero on the block-level count outcomes (-0.056 on the log count of workers living in the block, -0.113 on the log count of workers at the block) and on the municipality-level the log count of residents working in their own municipality (-0.057). Inc-becomes lies between the two. The pattern matches the institutional rule's prediction: the same statutory rate change can produce a strong, moderate, or null behavioral response depending solely on whether

the rate moves the bill. This is direct evidence that workers respond to taxes they actually pay rather than to statutory rates per se.

Block-level identification has cleaner pre-trends than the municipality aggregation, though the contrast is one of degree. At the block level, the realized-dose Sun–Abraham ES has near-zero individual pre-event coefficients (all four with $p > 0.10$) and a joint Wald pre-trend $\beta_{\text{pre}} = -0.027$ ($|t| < 1$). At the municipality level, three of four pre-event coefficients are individually $p > 0.10$, but the joint Wald test returns $\beta_{\text{pre}} = +0.149$ ($|t| = 2.66$), driven by a positive coefficient at $k = -3$. The pattern is consistent with mild high-eventual-dose selection at coarser geography (post-industrial cities on slightly declining trajectories) rather than a sharp parallel-trends failure. The block-level result is the defensible per-pp causal quantity under the strong-parallel-trends assumption required by the recent continuous-DiD literature; the municipality-level estimate remains useful for back-of-envelope welfare calibration but should be cited with the marginal-pre-trend caveat.

The joint regression also separates the two rate-setting channels. The muni resident-rate Inc-bind coefficient on resident outcomes is large and consistent at both the block and municipality levels. The SD-resident-rate cells, estimated jointly in the same regression, produce smaller and direction-mixed effects: $+0.177^{*}$ at the municipality level on the log count of residents working in their own municipality, $+0.097$ on block-level log count of workers living in the block ($p = 0.12$), and $+0.066^{*}$ at the SD level (sign-reversed). The SD rate is not the dominant driver of residential sorting through the binding channel. This is a substantive change from an earlier draft that found a positive SD-rate "attraction" effect, which was an artifact of tract-centroid LODES aggregation; under block-level LODES with the unified two-rate-setter design, SD-rate effects on residents are small or wrong-signed.

The mechanism behind the workplace response appears to be local-services demand spillover rather than direct tax incidence on jobs. Industry-decomposed outcomes at the municipality level show consumer-facing jobs (retail, food, arts, other services) fall by approximately 26 percent in the Inc-bind cell (-0.30 log points, SE = 0.15) while tradeable

jobs (manufacturing, wholesale, finance, professional services) are insulated (point estimate +0.24, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs rising approximately 11 percentage points in the binding cell. The decline in total jobs operates *via* local-services demand spillover from departing residents, not via direct nonresident-tax incidence.

The block-level Inc-bind estimate also has a direct revenue implication. Apply the envelope-theorem decomposition of Saez (2001): for a flat local rate τ on tax base WN with residential semi-elasticity $\beta = \partial \ln N / \partial \tau$, the first-order effect of a small hike is

$$\frac{dR}{d\tau} \;=\; W \cdot N \cdot (1 + \tau\beta),$$

a mechanical gain WN minus the behavioral offset $\tau\beta$ from residents relocating. At the PA muni baseline $\tau \approx 1\%$ and the block-level Inc-bind estimate $\beta \approx -21$ per unit rate (-0.21 log points per pp), $\tau\beta \approx -0.21$: the residential response offsets roughly 21% of the mechanical revenue gain. The hike remains revenue-positive, paralleling the Spanish regional finding in Agrawal and Foremny (2019).

The flat-tax revenue-maximizing rate, obtained by setting $dR/d\tau = 0$, is $\tau^* = 1/|\beta| \approx 4.8\%$. This is the analog, for a flat local tax with only an extensive-margin mobility response, of the top-marginal-rate formula $\tau^* = 1/(1 + a\varepsilon + \eta)$ in the optimal-taxation-with-mobility literature reviewed by Agrawal (2026), where a is the Pareto coefficient of top earners, ε is the intensive-margin elasticity of taxable income, and η is the extensive-margin mobility elasticity. In my flat-tax setting neither a nor ε enters: there is no top tail and no intensive labor-supply channel, only the mobility margin captured by β. PA's statutory EIT range (roughly 0.5–3.4%) sits entirely in the revenue-positive interior; the headline sorting estimate is consistent with a revenue gain to the raising municipality at the muni boundary, with the welfare cost accruing elsewhere through relocation and the local-services demand spillover documented above.

Three limitations merit acknowledgment. First, while the block-level stacked headline on the count outcomes is large, negative, and well-identified, the block-level share outcome (the share of block residents working in their own muni) is null in the binding cell (-0.035, $p = 0.26$) and marginally significant in the Inc-nobind placebo (-0.037, $p = 0.015$). The share outcome at block scale appears to reflect mechanical denominator dynamics on very small populations rather than binding-state-specific behavior, so the share-based block evidence should be read as suggestive rather than identified. The count-based block evidence remains the cleanest test. Second, the school-district aggregation produces direction-mixed coefficients and a marginally non-zero realized-dose pre-event lead; the SD-level results should be read as confirming the directional sign of the muni-rate effect under a coarser aggregation rather than as standalone identification of an SD-channel response. Third, the LODES OD data covers PA municipalities but not cross-state inflows in the intra-state main file used here; the auxiliary file (cross-state inflows) is downloaded but not yet incorporated, which would tighten the estimates of in-commuters for border munis.

9 Conclusion

When do local income taxes cause residential sorting? The institutional answer in Pennsylvania is that they cause sorting when the piecewise max-rule makes the statutory rate increase change the worker's actual bill. At the census block, the level at which both rate-setting jurisdictions enter the regression at their native geography, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by roughly -0.21 log points, approximately 19%, with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The within-design Inc-nobind placebo, where the same statutory increase does not move the bill, is statistically null on the same count outcome. The response is therefore tied to tax liability rather than to the statutory rate change itself, and the binding-state decomposition is what distinguishes the two.

Municipality-level estimates are larger (−0.40 log points on the log count of residents working in their own municipality) but pre-trended, and they enter the paper as policy-jurisdiction magnitude calibration rather than as a second causal headline. School-district aggregations do not robustly detect a sorting response and are reported as a diagnostic non-result. Industry decomposition shows that consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, suggestive of a local-services demand spillover from departing residents rather than direct nonresident-tax incidence on jobs.

The narrow, credible claim is that local income taxes cause residential sorting when they bind. The same design framework—a continuous stacked DiD with binding-state cell decomposition, a realized-dose event study as a strong-parallel-trends diagnostic, and a within-design placebo cell whose theoretical prediction is zero—is portable to other settings where a piecewise tax rule generates predictable cross-sectional variation in whether a statutory rate change moves the worker's bill. Local sales taxes with use-tax-credit rules, county-level income-tax credits across state borders, and special tax districts with overlapping jurisdictions are candidate extensions.

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A Pre-Treatment Balance

Table 33: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
Total Debt ($)	28.6M	3.6M	0.089	
Market Value RE ($)	8.2M	2.1M	0.211	523 / 8,774
Total Index Crime	985	133	0.124	816 / 18,322
Building Permits	19.3	7.7	0.232	644 / 25,066
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. Normalized differences < 0.15 for all fiscal variables except market value of real estate (0.211, limited pre-2012 coverage) and building permits (0.232, treated municipalities are larger and issue more permits in levels). Sources: PA DCED Annual Financial Reports, FBI UCR Offenses Known (93.5% agency match rate), Census Building Permits Survey (93.3% MCD match rate).

B Pre-Trend Tests on Fiscal Observables

I test the parallel trends assumption on eleven observables—six fiscal variables from the PA DCED Annual Financial Reports, real estate market values, total employment from LODES, Part I index crime from the FBI Uniform Crime Reports, and housing units permitted from the Census Building Permits Survey—using the Sun and Abraham (2021) estimator. For each outcome, I estimate Equation 6 with municipality and year fixed effects, clustering

standard errors at the municipality level. Table 34 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for nine of eleven observables: total revenue (max $|t| = 1.57$), EIT revenue (1.40), RE tax revenue (1.69), streets expenditure (0.70), population (1.09), total debt (1.36), total jobs from LODES (0.83), index crime (0.68), and building permits (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level. The jobs, crime, and permits results are particularly important: they confirm that treated and control municipalities were on parallel trajectories in employment, public safety, and housing development before the rate change, ruling out differential economic trends as confounds.

Police expenditure is an exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on the share of workers living in their own muni are robust to controlling for log police expenditure (the robustness section (Section 7)), confirming that this pre-trend does not confound the estimated sorting response.

Market value of real estate also shows a pre-trend at $\ell = -3$ ($t = -2.59$), but this variable has limited coverage (pre-2012 only, $N = 9{,}338$) and the pre-trend may reflect composition effects in the truncated sample rather than a genuine violation of parallel trends.

Table 34: Pre-Trend Tests: Sun-Abraham on Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.57	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.40	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(RE Tax Revenue)	−0.025	−0.060	−0.011	−0.015	1.69	41,683
	(0.044)	(0.036)	(0.029)	(0.017)		
log(Police Exp)	−0.085***	−0.068***	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.70	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		
log(Market Val RE)			−0.403***	−0.026	2.59	9,338
			(0.156)	(0.016)		
log(Total Jobs)	0.006	0.018	−0.001	−0.005	0.83	24,557
	(0.023)	(0.022)	(0.015)	(0.013)		
log(Index Crime)	−0.125	−0.094	−0.084	−0.031	0.65	20,613
	(0.194)	(0.151)	(0.236)	(0.155)		
log(Bldg Permits)	−0.030	−0.057	0.084	0.171	1.35	27,515
	(0.222)	(0.157)	(0.147)	(0.126)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure (lower spending at $\ell = -5$ and $\ell = -4$) and market value of real estate (limited to pre-2012 data, $N = 9{,}338$). Total jobs from LODES; index crime from FBI UCR Offenses Known (93.5% agency-to-municipality match);

C Event-Study Figures

This appendix collects the full set of event-study figures referenced throughout the paper. The figures fall into two families that correspond to the two pre-trend diagnostics described in Section 5.4. The realized-dose Sun–Abraham specification tests strong parallel trends across treatment-intensity magnitudes by interacting relative-event-time indicators with each unit's eventual cumulative dose. The binary Sun–Abraham interaction-weighted specification tests average parallel trends by replacing the dose with a 0/1 cohort indicator and reweighting cohort $\times$ event-time coefficients to deliver a single cohort-robust path. Within each subsection figures are grouped by estimating unit (block, municipality, school district).

C.1 Realized-Dose Sun–Abraham Event Study

Block level, by cell. The block-level realized-dose ES on the log count of workers living in the block (Figure 12 in the main text) is the diagnostic on which the headline causal claim rests; the remaining block-level outcomes are reported here.

Figure 16: Block-level realized-dose ES on the log count of workers working in the block, faceted by binding-state cell; muni-cohort (red) and SD-cohort (blue).

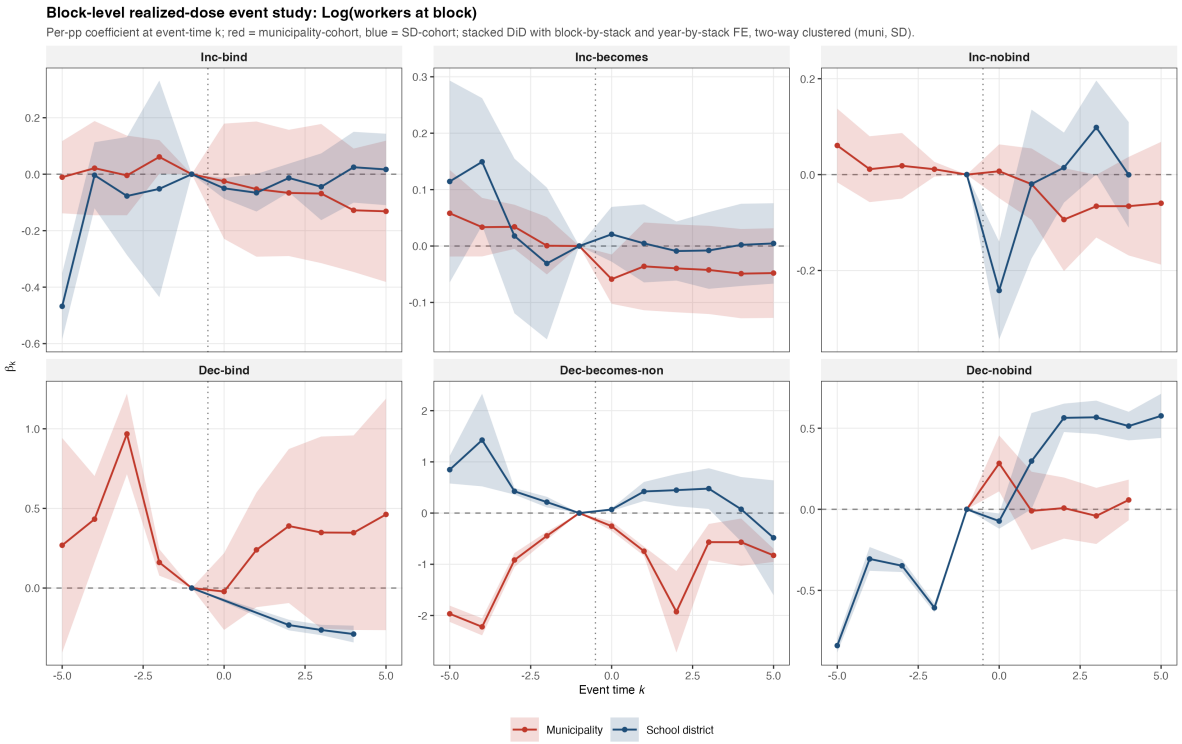


Figure 17: Block-level realized-dose ES on the share of block residents working in their own muni, faceted by binding-state cell.

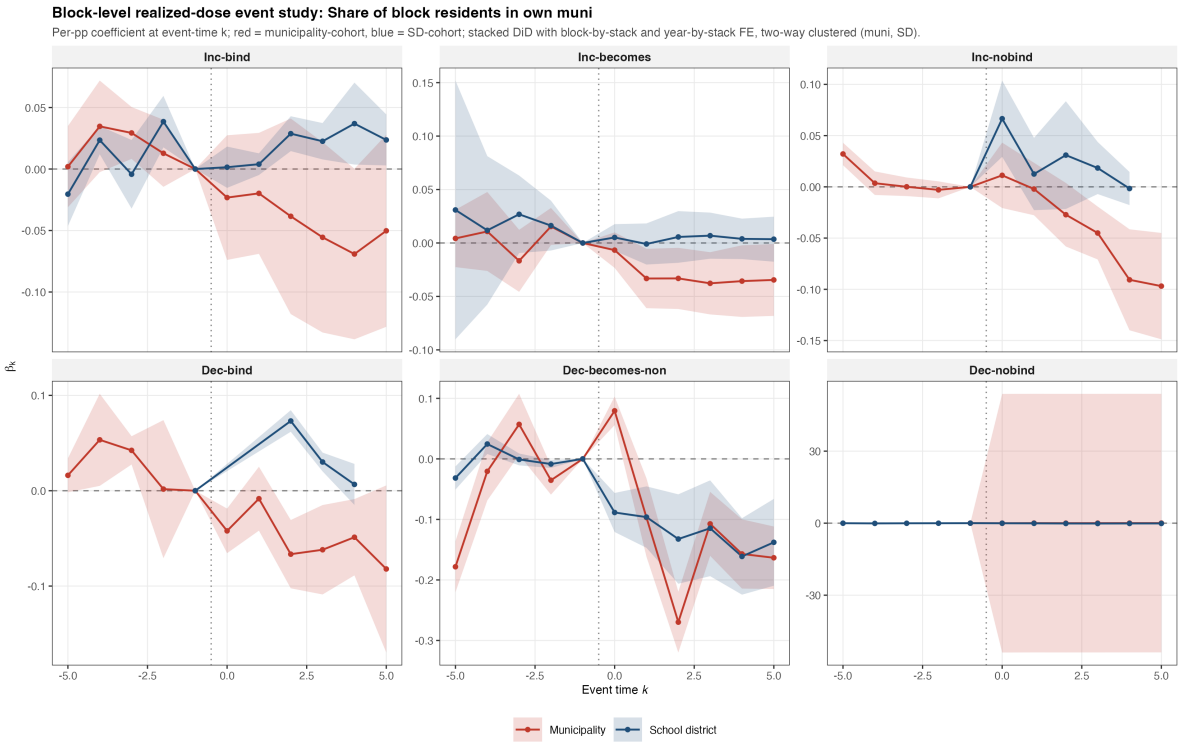
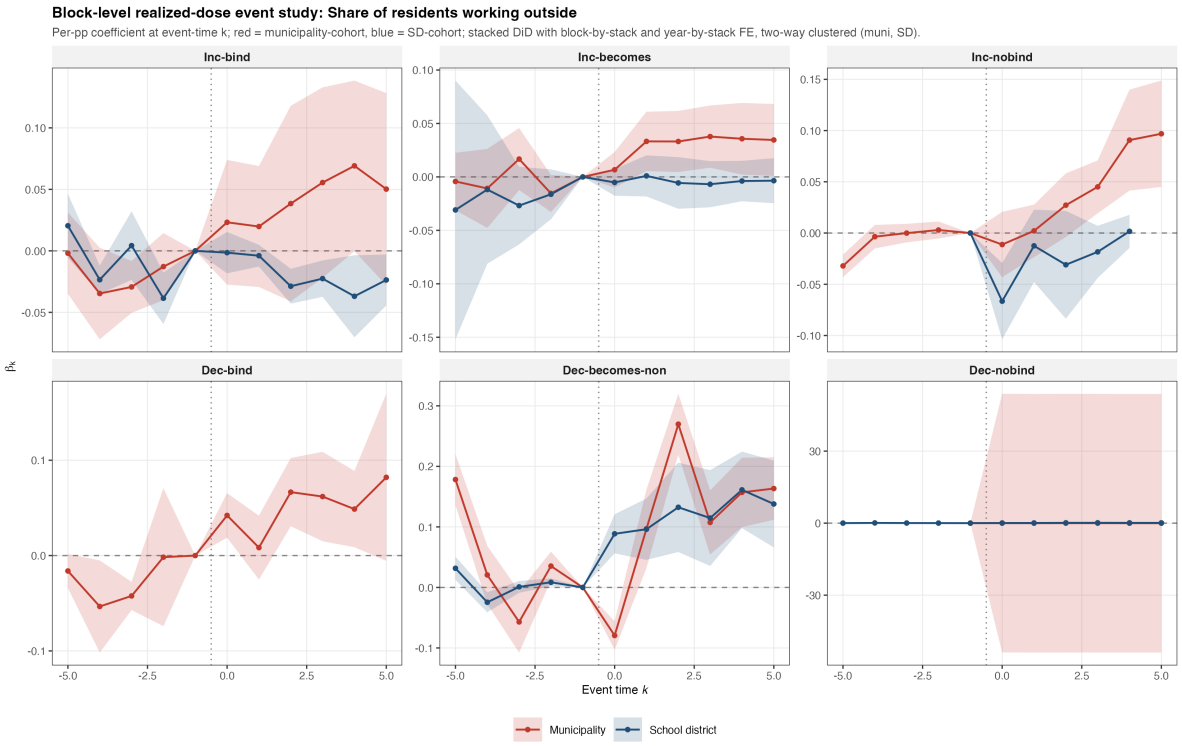


Figure 18: Block-level realized-dose ES on the share of block residents working outside their muni, faceted by binding-state cell.



Municipality level, by cell. The muni-level live-and-work figure (Figure 14) appears in the main text where the joint Wald pre-trend is discussed; the remaining residence-side and commuter outcomes are reported here.

Figure 19: Muni-level realized-dose ES on log(total residents), by cell.

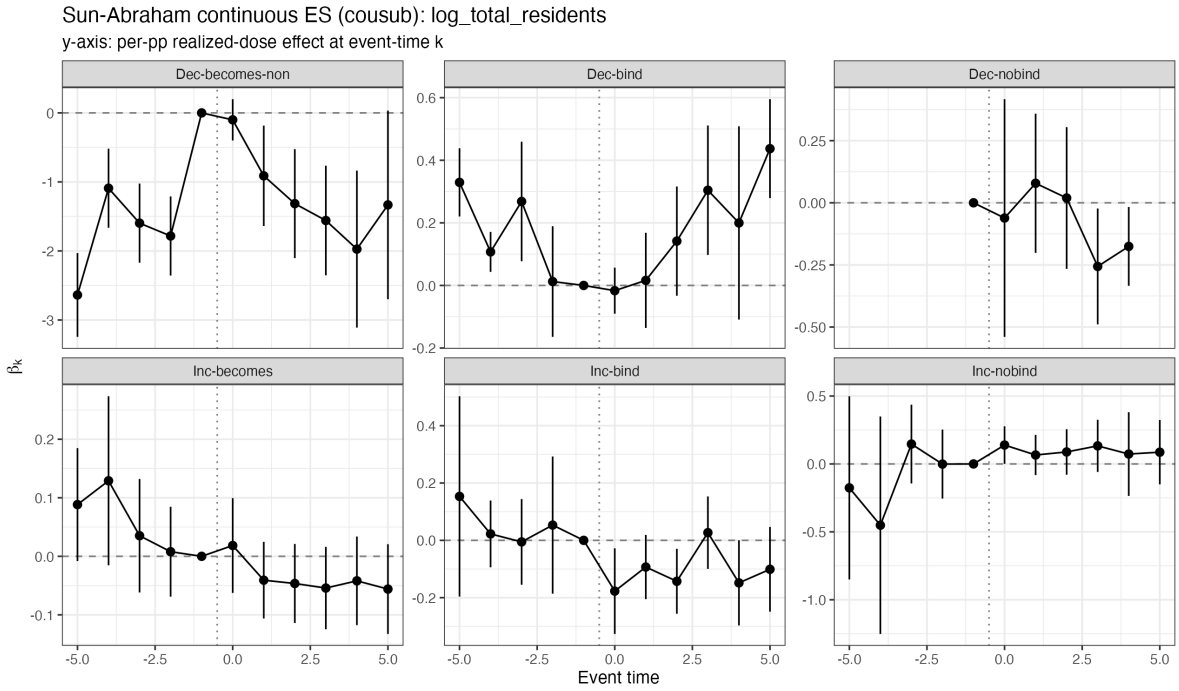


Figure 20: Muni-level realized-dose ES on log(in-commuters), by cell.

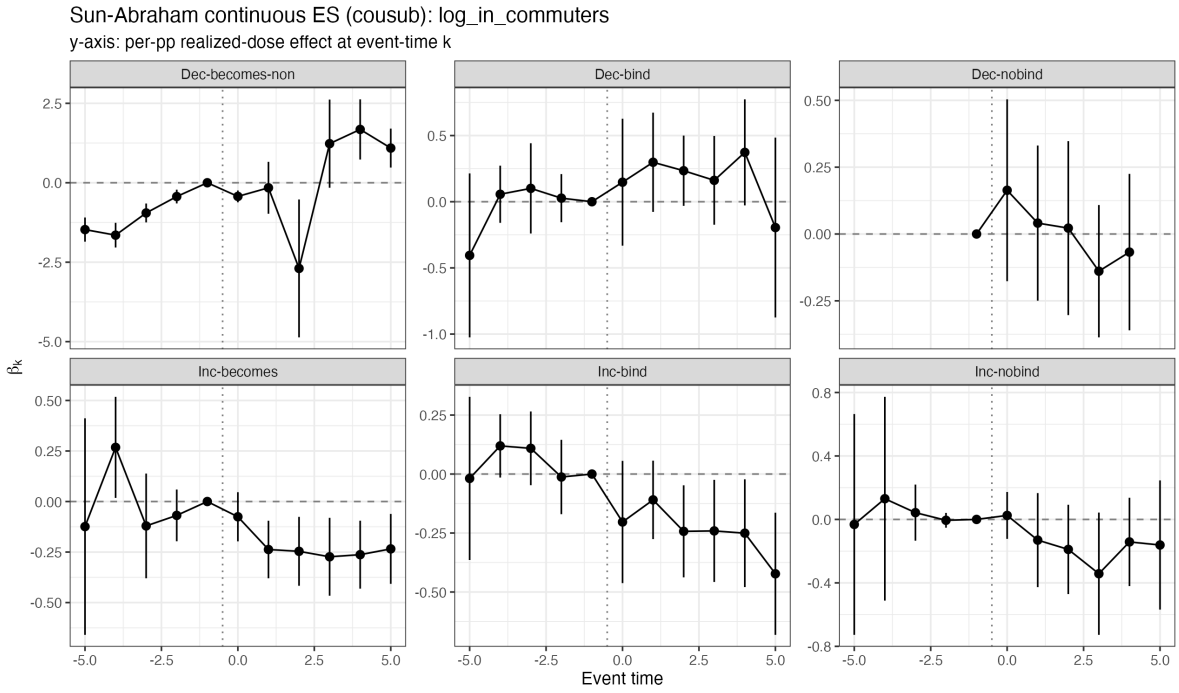


Figure 21: Muni-level realized-dose ES on the share of workers living in their own muni, by cell.

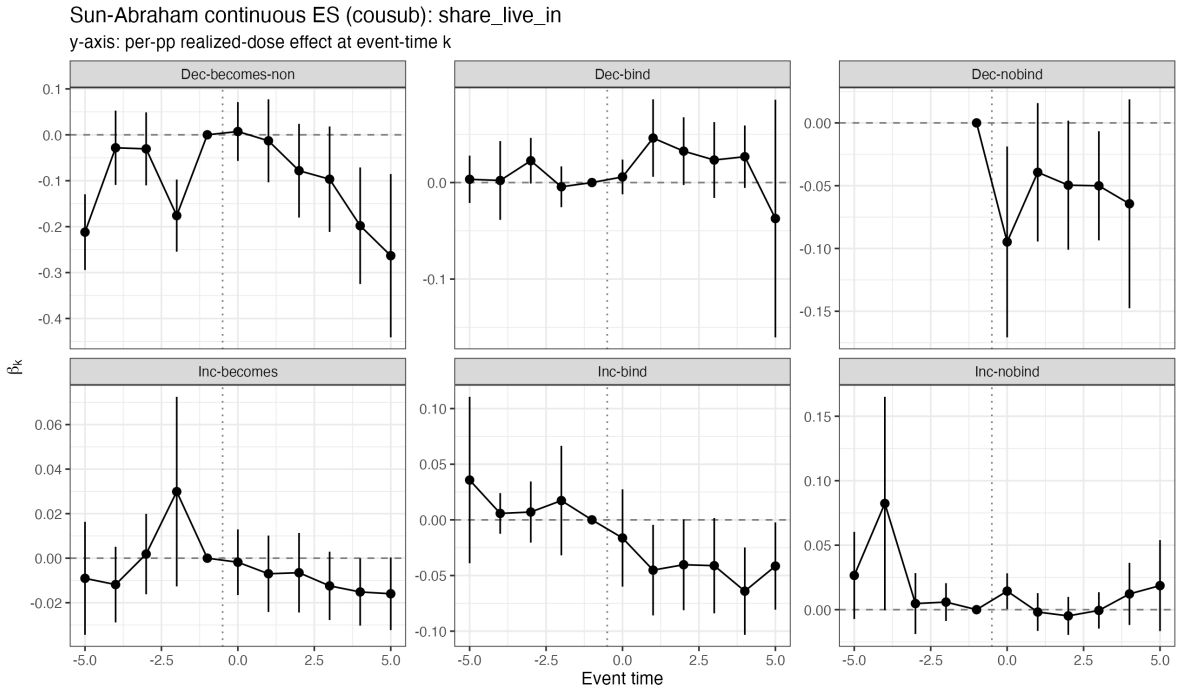
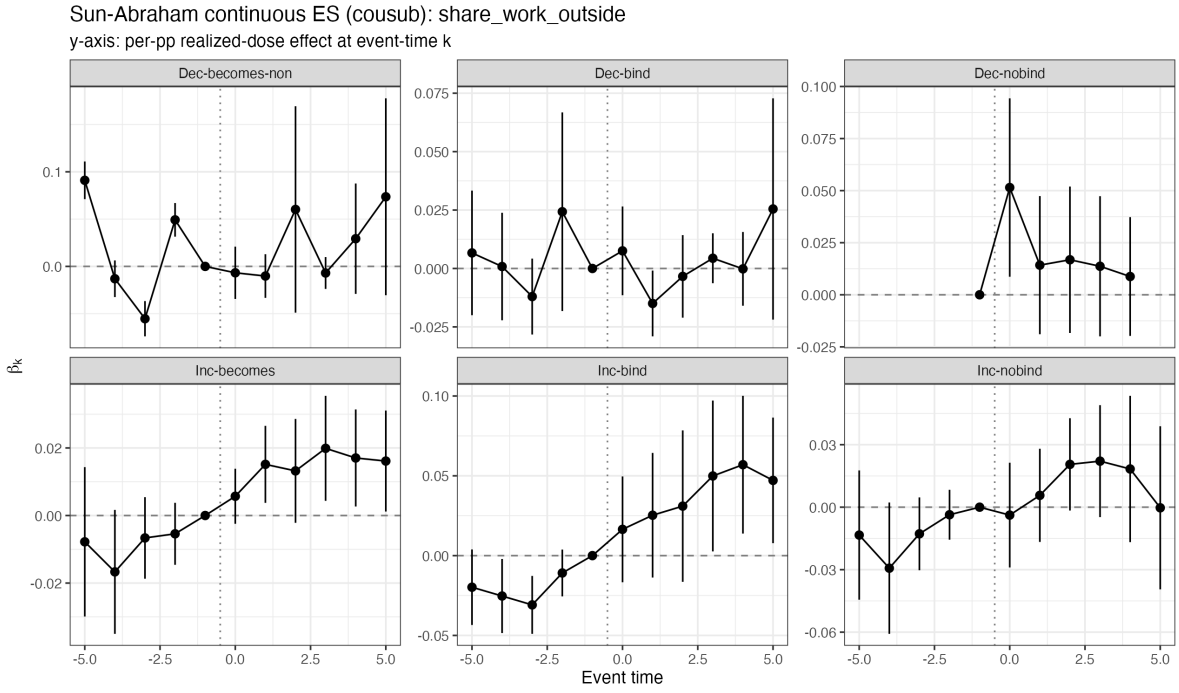


Figure 22: Muni-level realized-dose ES on the share of residents working outside the muni, by cell.



School-district level, by cell. The SD-level realized-dose ES, run on the same residence-side outcomes aggregated to the school district, is the input to the SD aggregation diagnostic in Section 6.6.

Figure 23: SD-level realized-dose ES on log(live_and_work), by cell.

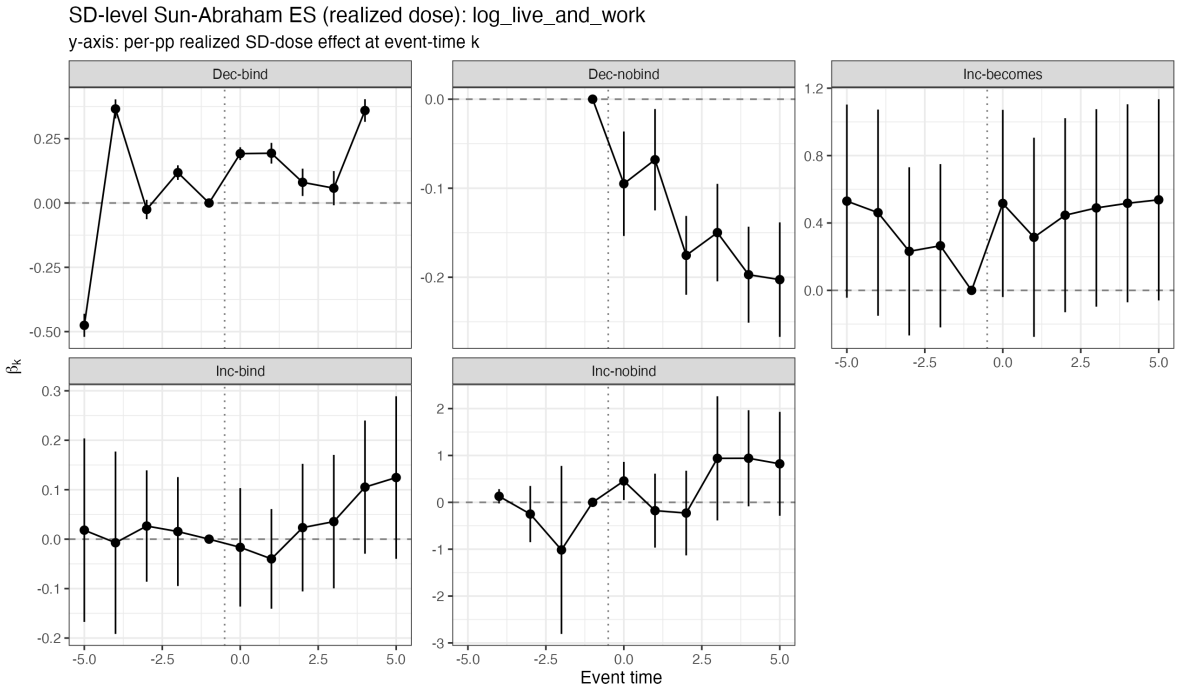


Figure 24: SD-level realized-dose ES on log(total residents), by cell.

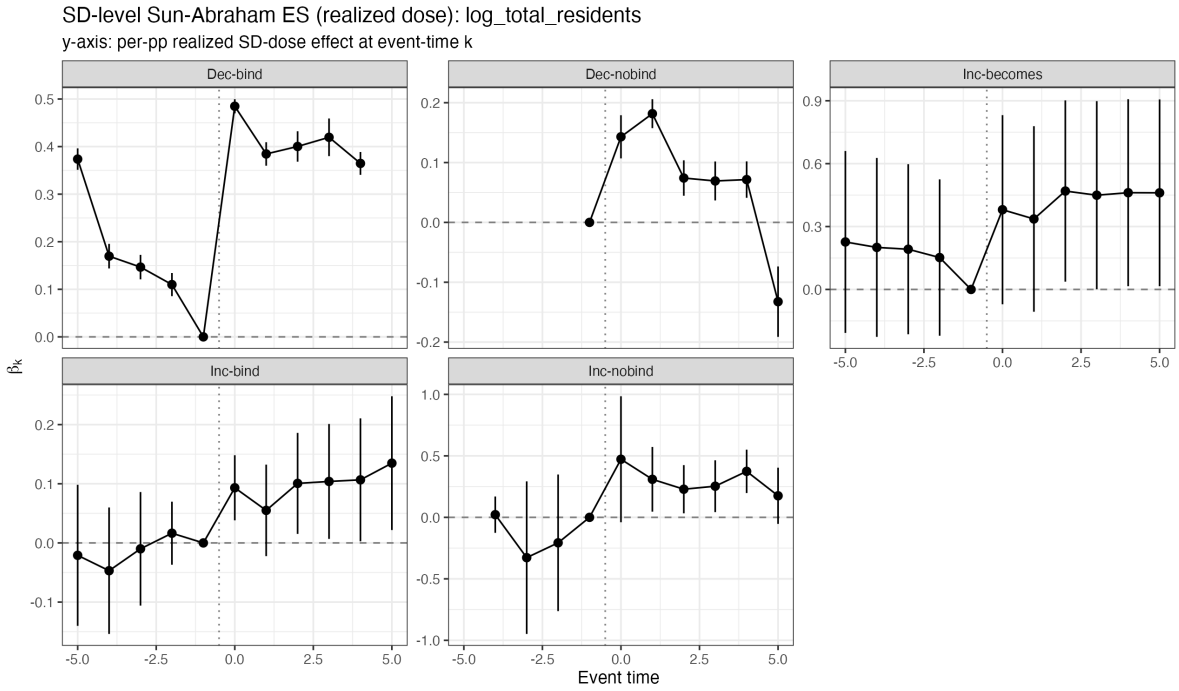


Figure 25: SD-level realized-dose ES on log(in-commuters), by cell.

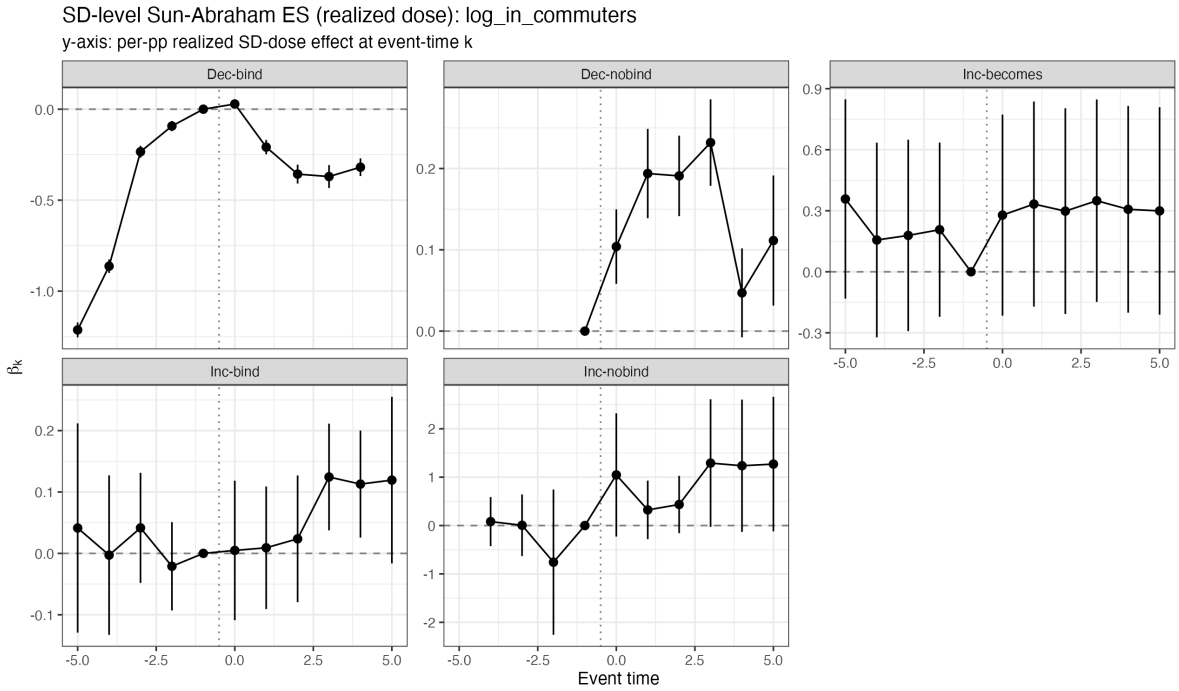


Figure 26: SD-level realized-dose ES on the share of workers living in their own muni, by cell.

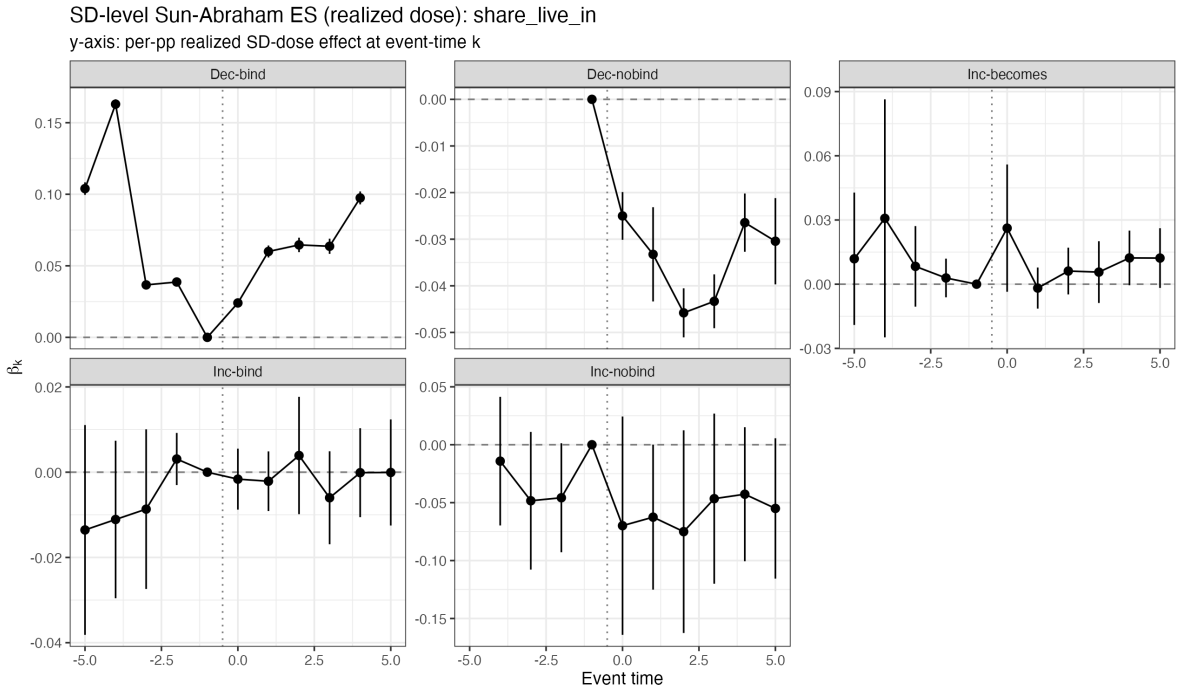
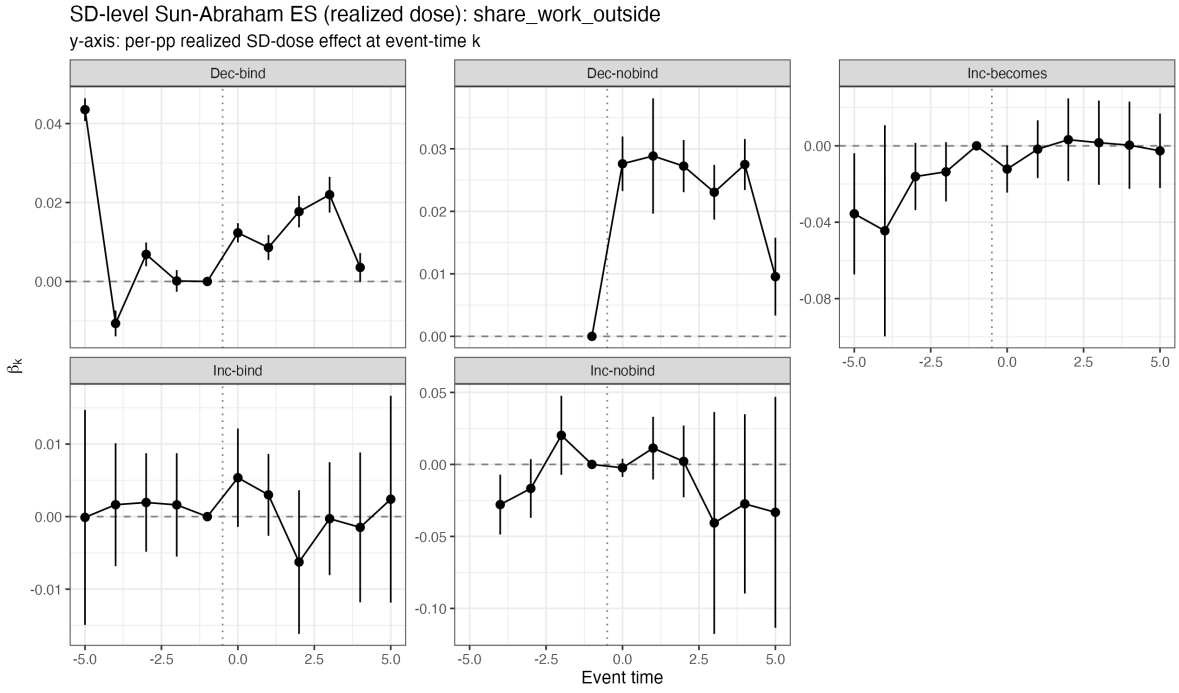


Figure 27: SD-level realized-dose ES on the share of residents working outside the muni, by cell.



C.2 Binary Sun–Abraham Interaction-Weighted Event Study

The binary Sun–Abraham IW specification (Sun & Abraham, 2021) estimates a separate dynamic ATT per cohort × event-time and then interaction-weights across cohorts to produce a single cohort-robust path. The diagnostic complements the realized-dose ES by testing average parallel trends without requiring a continuous dose. Block-level binary SA-IW is unstable in this design (thin cohort × event-time cells under stacking) and is reported with that caveat.

Municipality level. Each figure shows the muni-cohort path on a residence-side outcome.

Figure 28: Muni-level binary SA-IW ES on log(live_and_work).

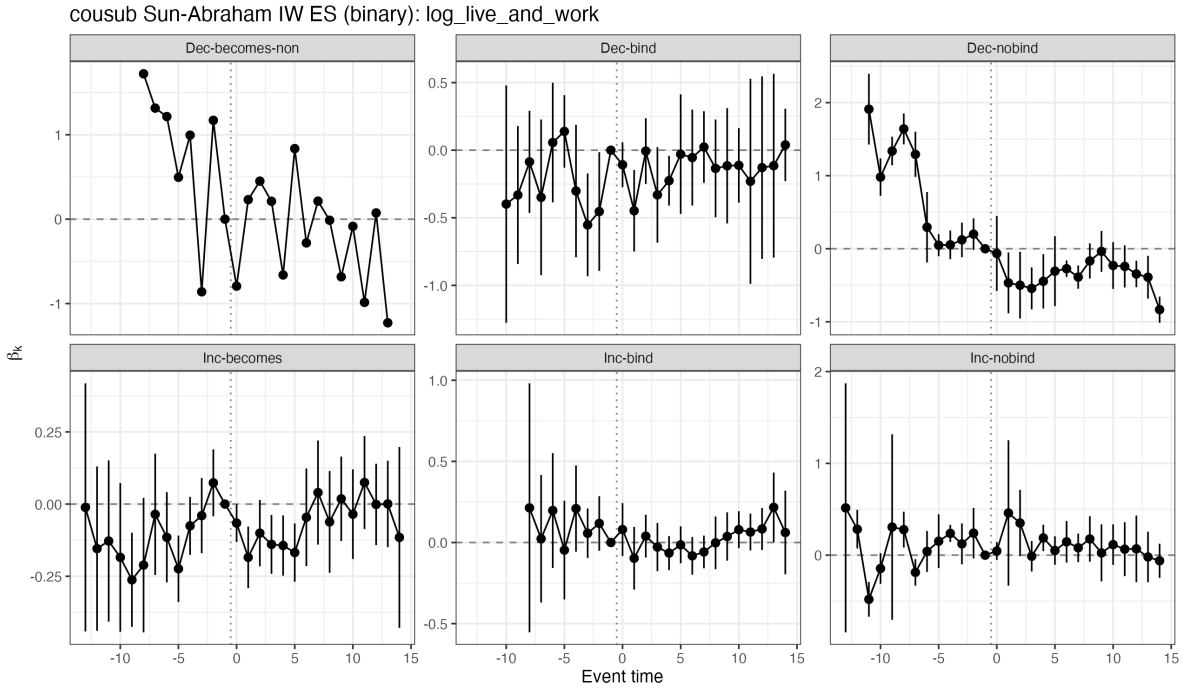


Figure 29: Muni-level binary SA-IW ES on log(total residents).

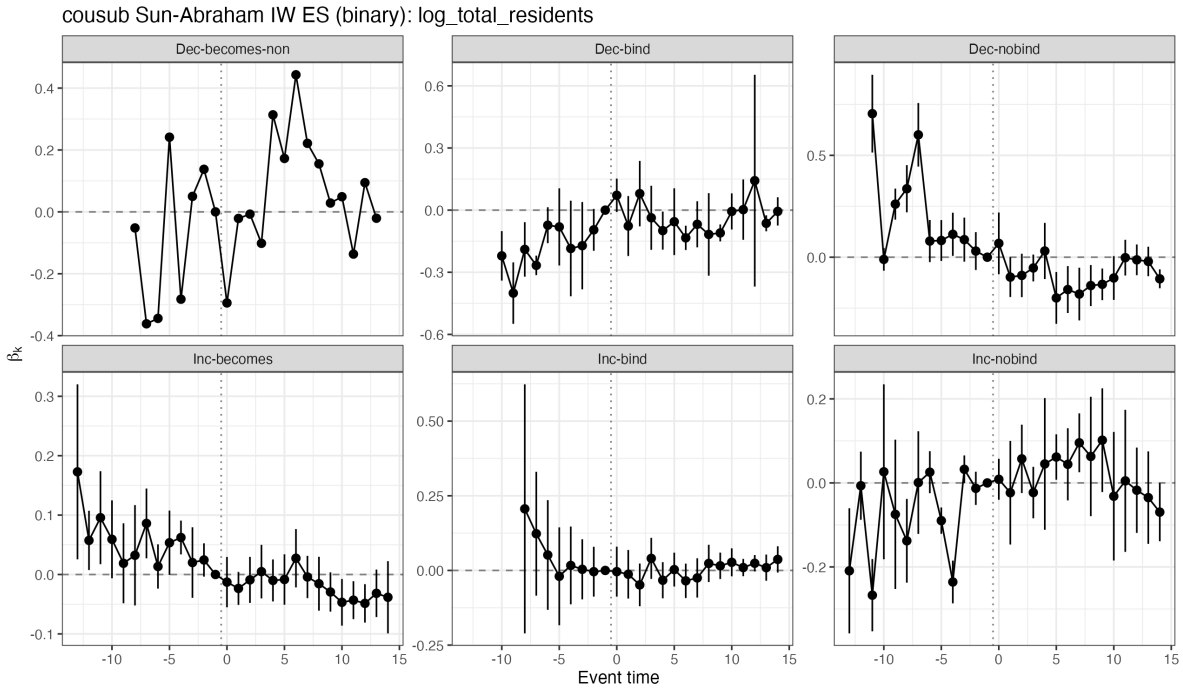


Figure 30: Muni-level binary SA-IW ES on log(in-commuters).

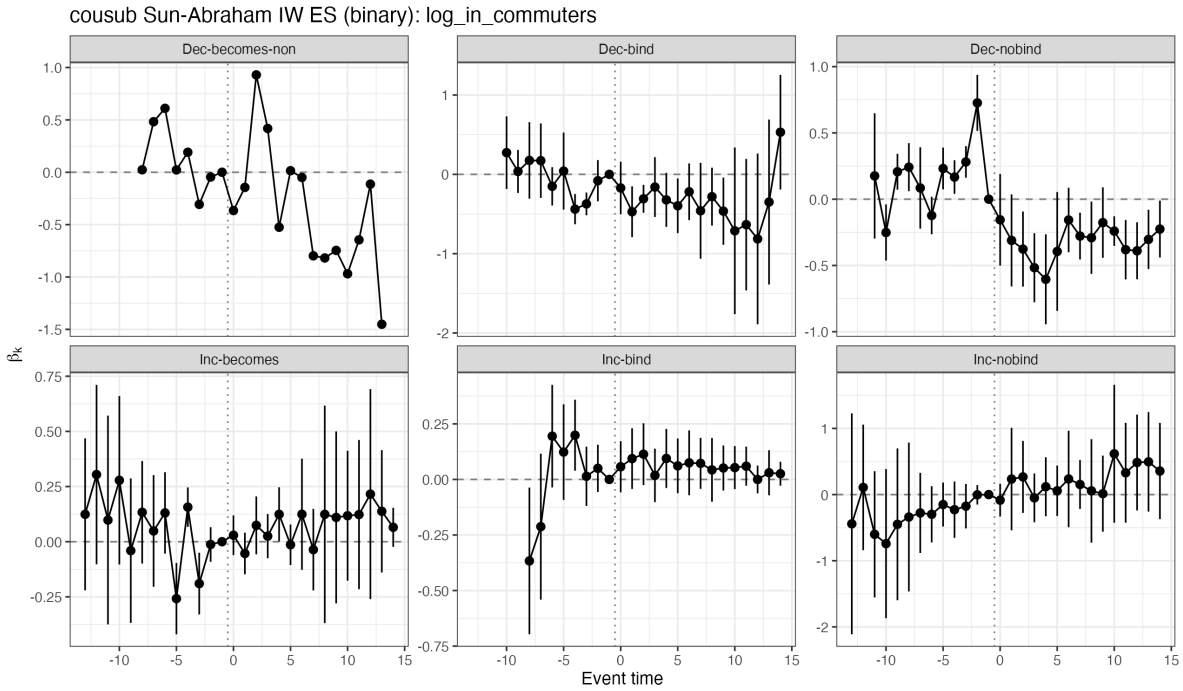


Figure 31: Muni-level binary SA-IW ES on the share of workers living in their own muni.

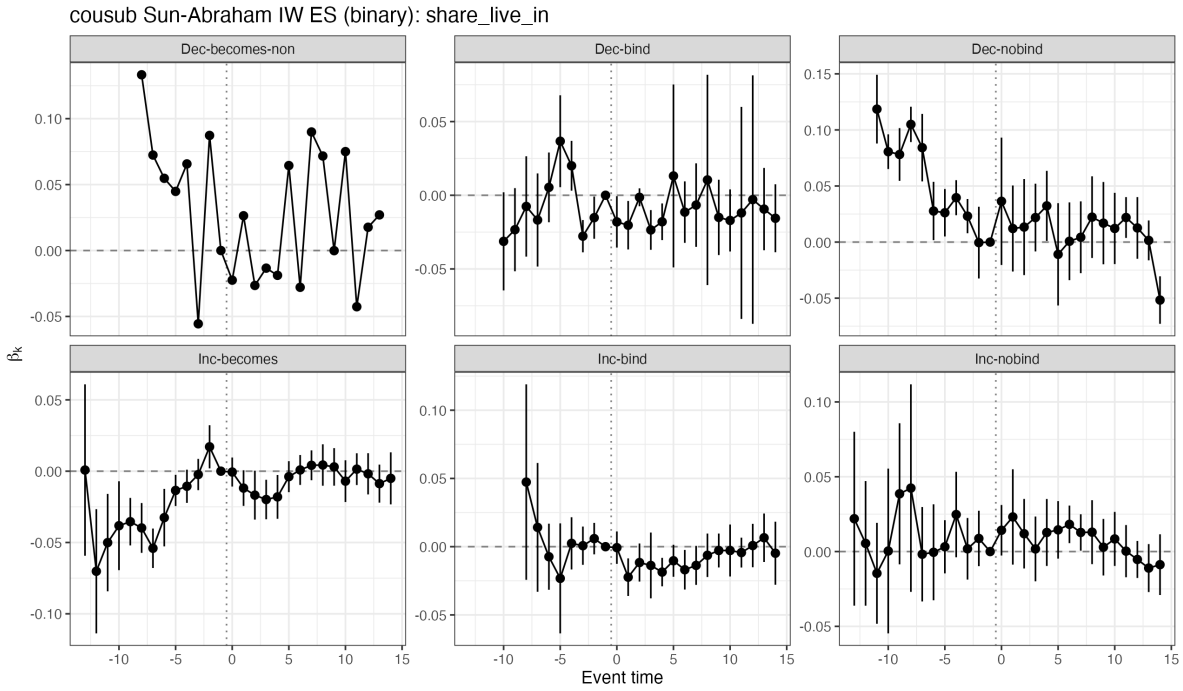


Figure 32: Muni-level binary SA-IW ES on the share of residents working outside the muni.

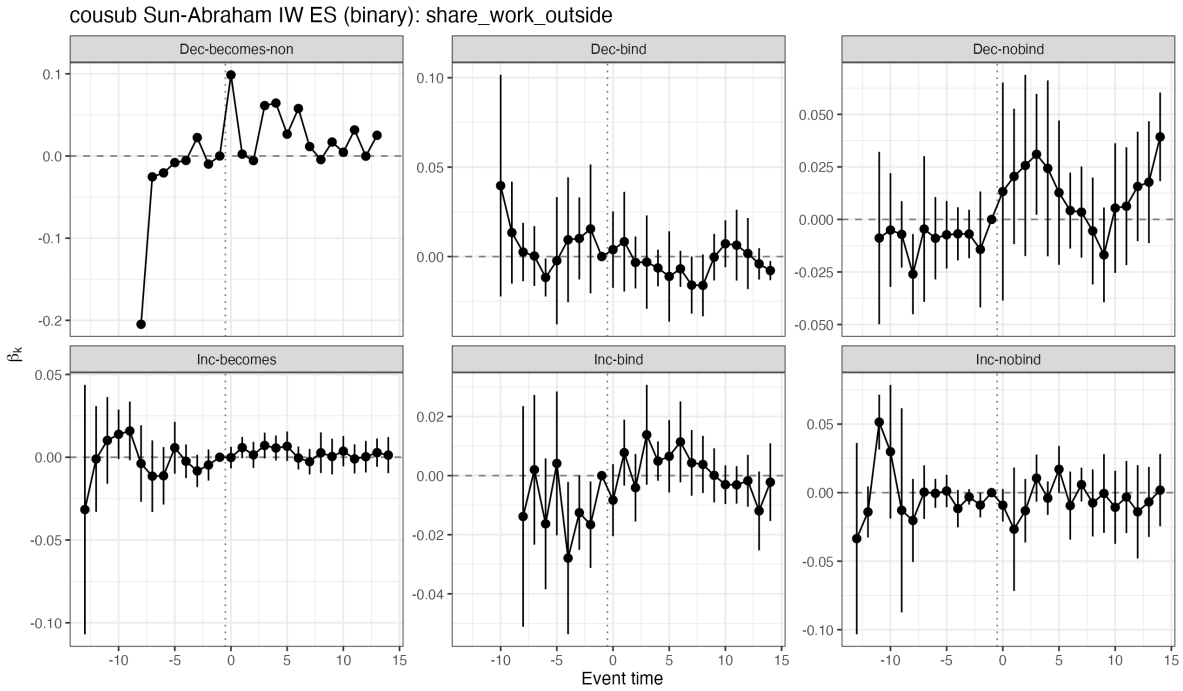


Figure 33: SD-level binary SA-IW ES on log(live_and_work).

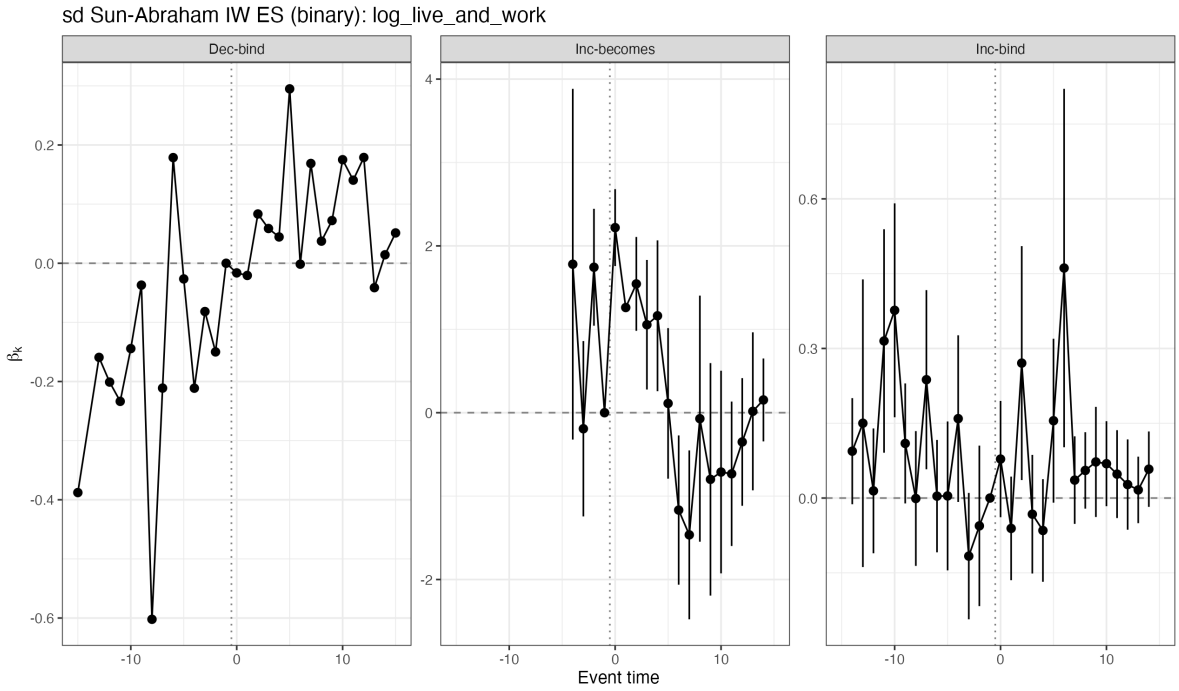


Figure 34: SD-level binary SA-IW ES on log(total residents).

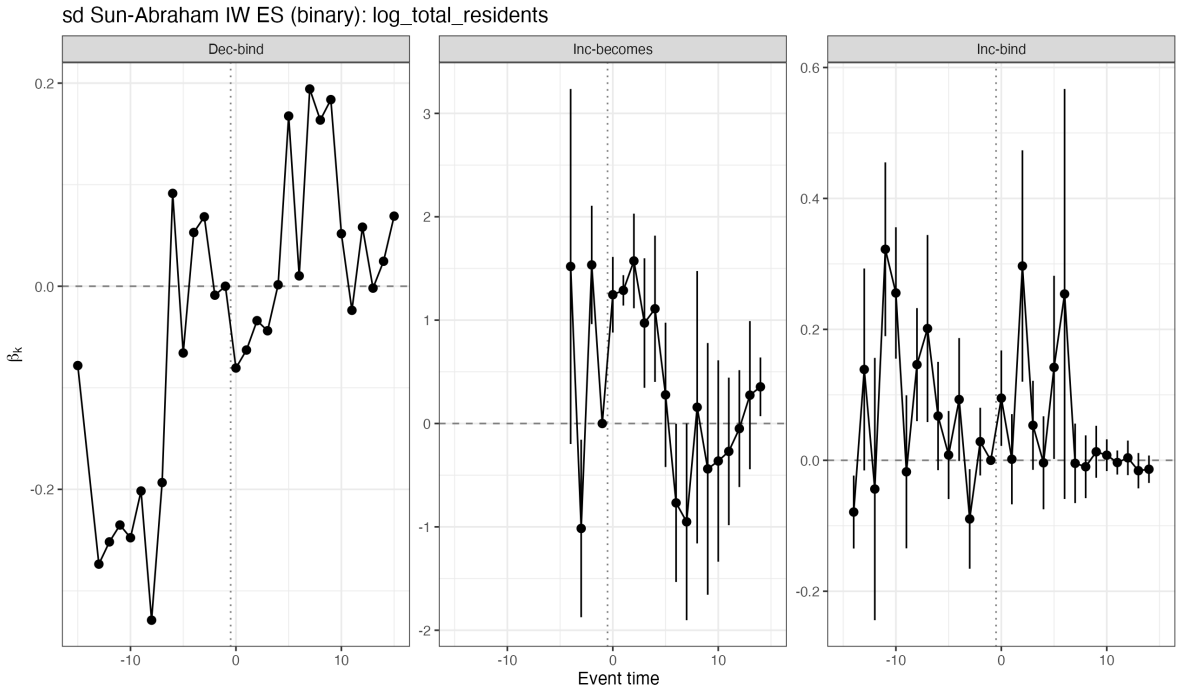


Figure 35: SD-level binary SA-IW ES on log(in-commuters).

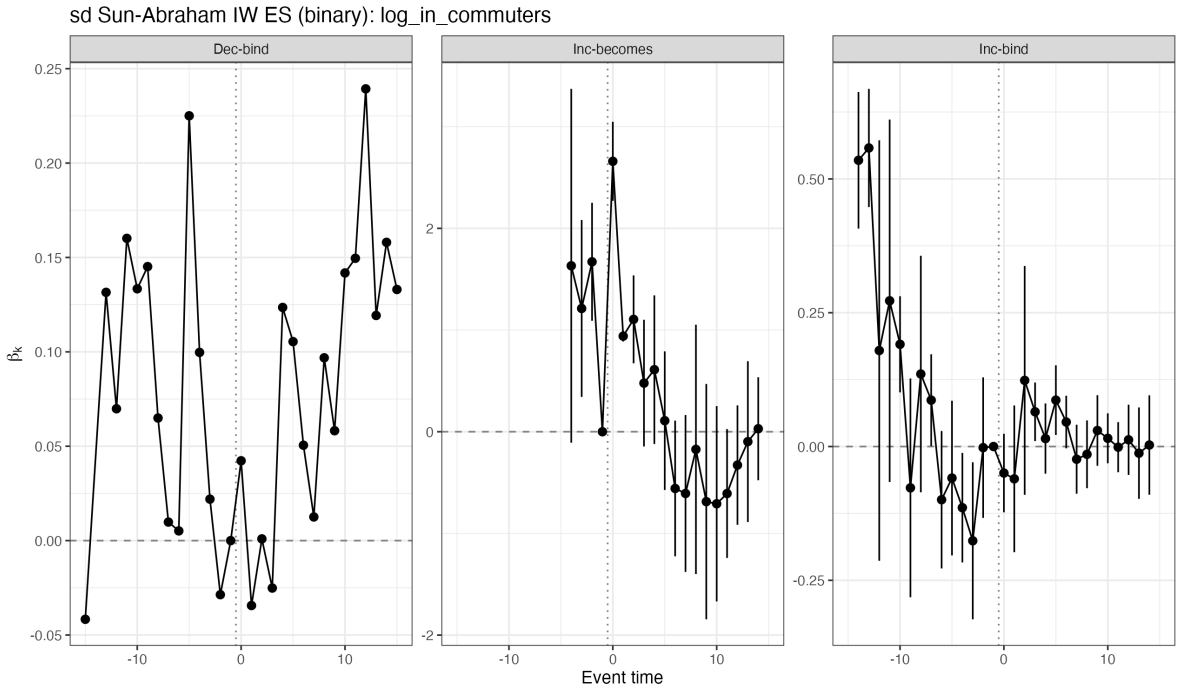


Figure 36: SD-level binary SA-IW ES on the share of workers living in their own muni.

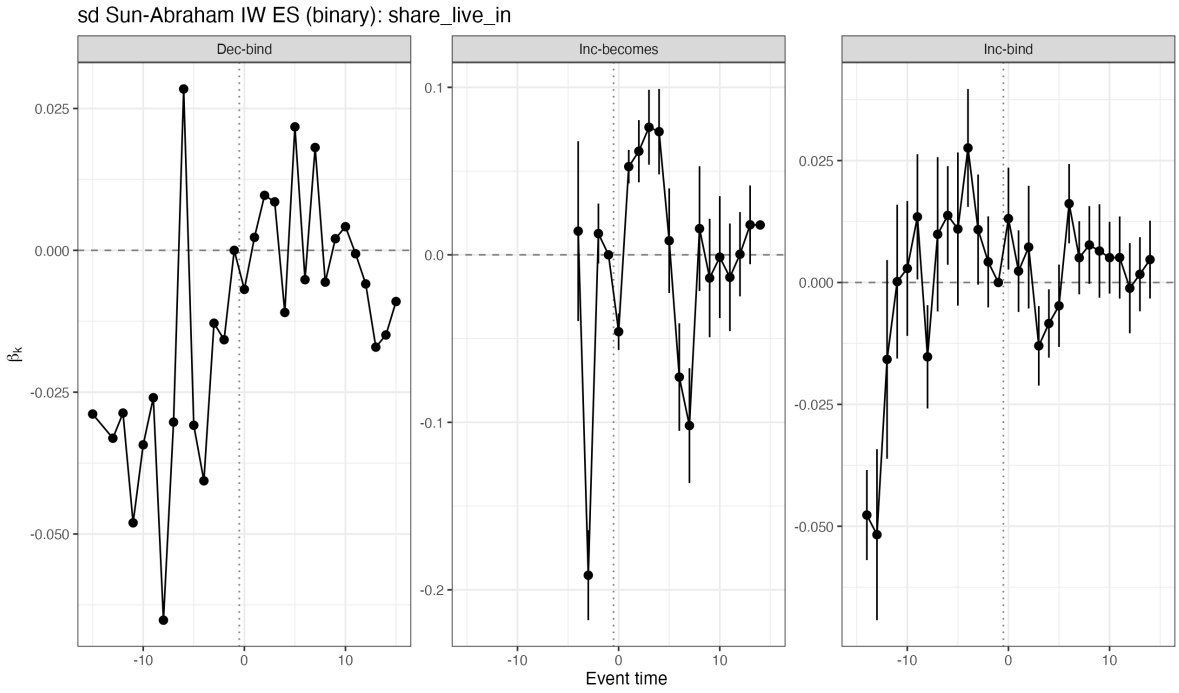
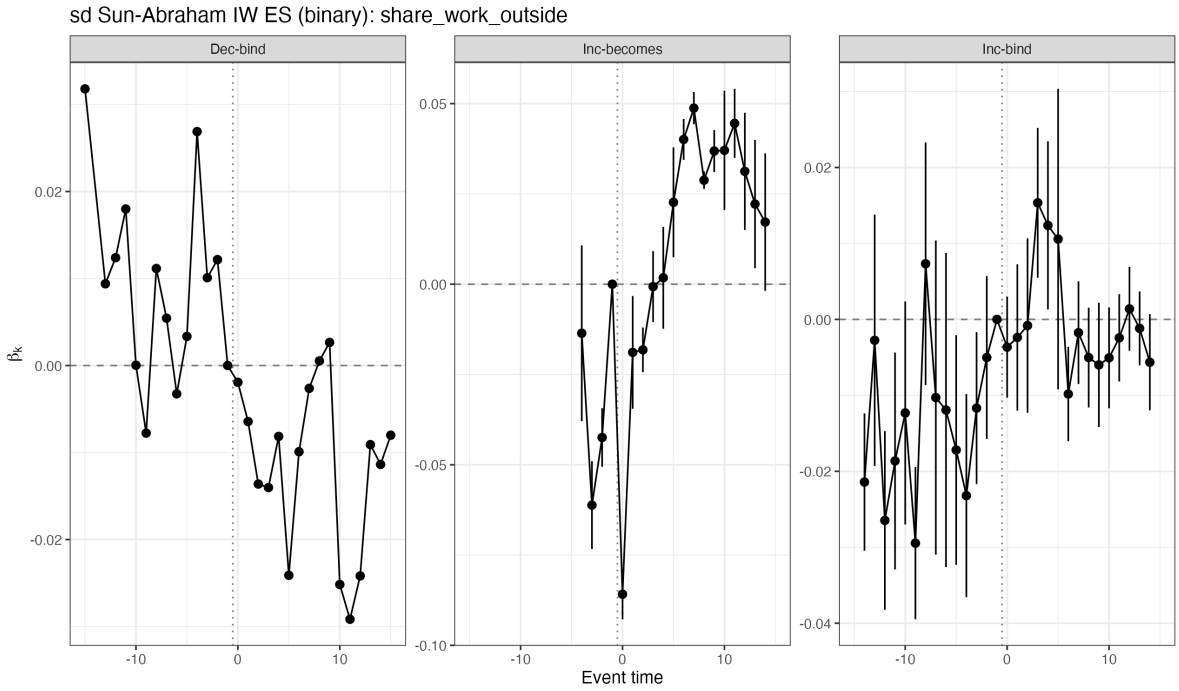


Figure 37: SD-level binary SA-IW ES on the share of residents working outside the muni.



School-district level.

Block level. Reported here for completeness; the thin-cohort caveat applies.

Figure 38: Block-level binary SA-IW ES on the log count of workers living in the block.

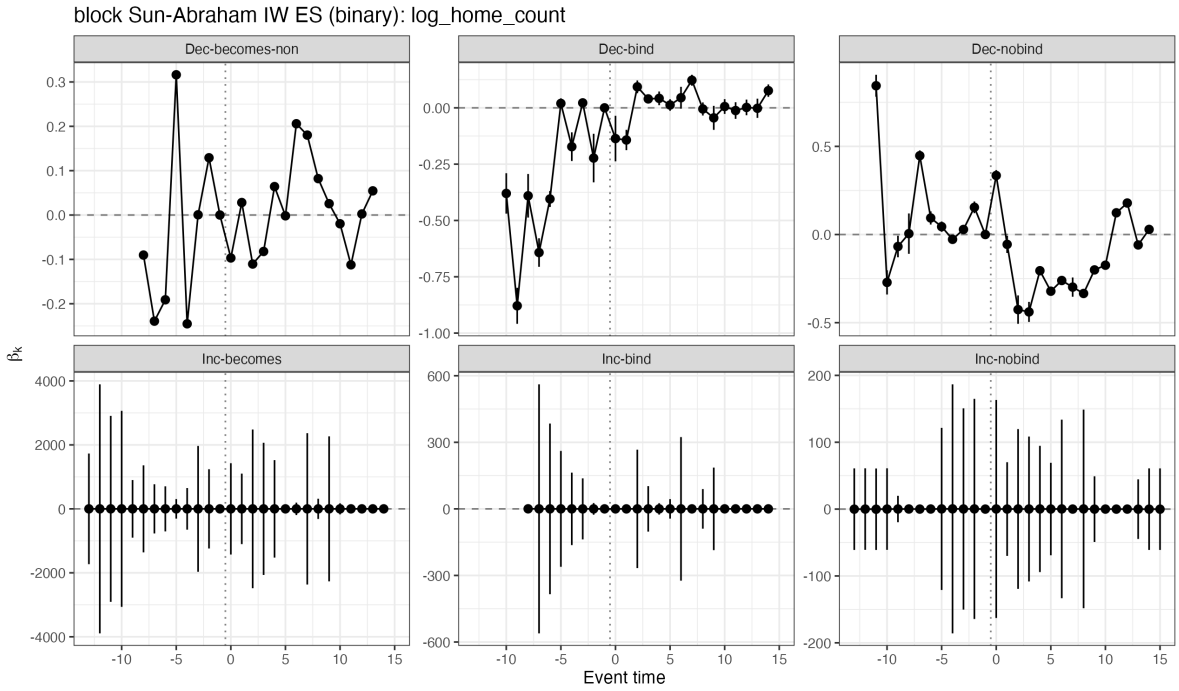


Figure 39: Block-level binary SA-IW ES on the log count of workers working in the block.

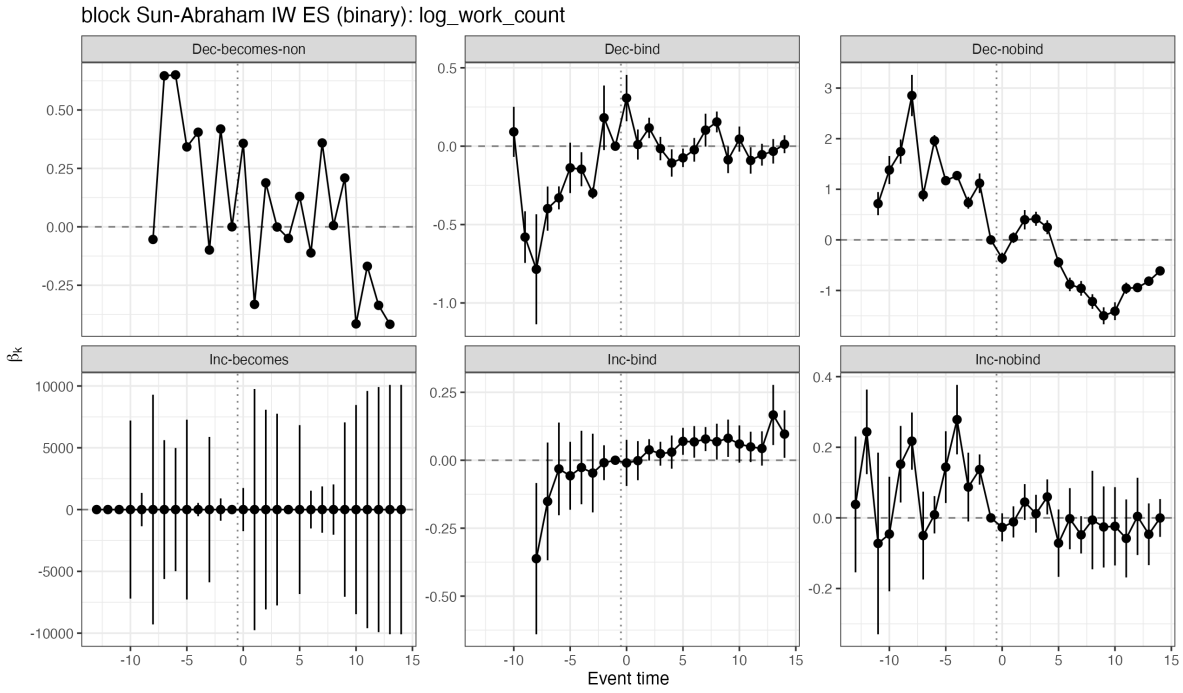


Figure 40: Block-level binary SA-IW ES on the share of block residents working in their own muni.

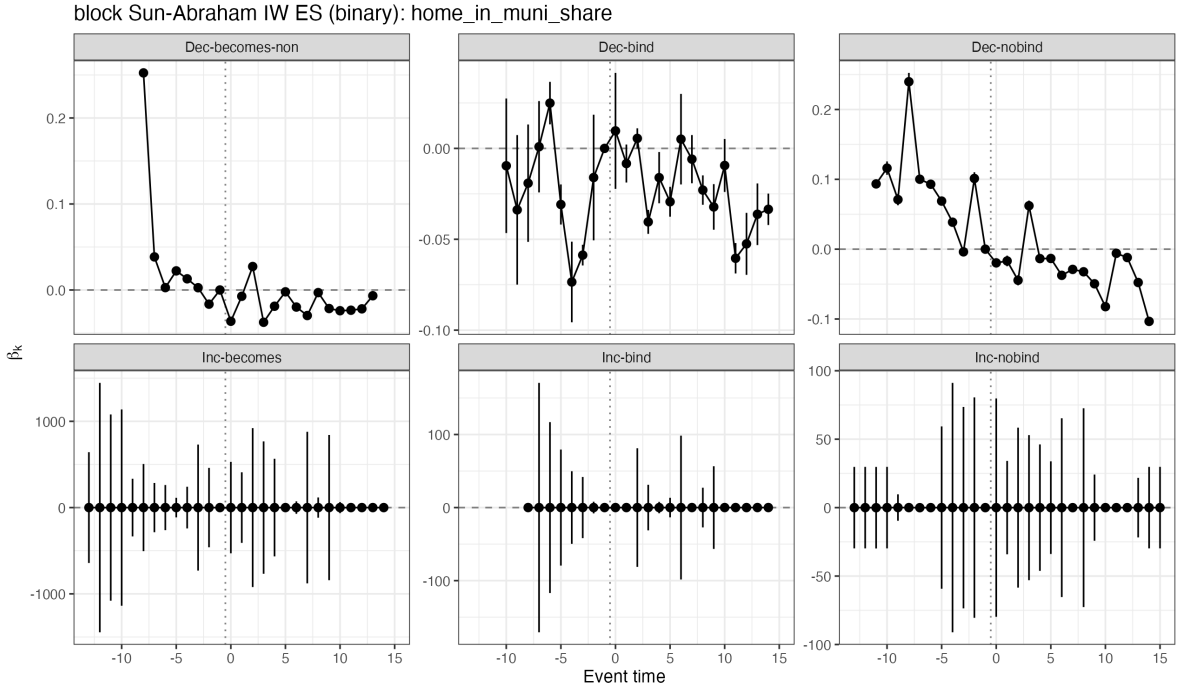


Figure 41: Block-level binary SA-IW ES on the share of block residents working outside the muni.

